

Fortune Favors the Educated

Financial Literacy & Intergenerational Knowledge Transfers in Denmark

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Abstract

Using Danish register data, we show that households with lower educational attainment and non-financial fields of study hold smaller equity shares and earn lower expected excess returns, and that these gaps widen when parental financial knowledge is also low. We build a stochastic heterogeneous-agent life-cycle model calibrated to Danish micro data to quantify the role of financial knowledge in shaping wealth accumulation. Heterogeneity in financial knowledge accounts for 20.7% of the wealth gap at retirement, and we identify a synergy between own and parental knowledge: educated households disproportionately benefit from a financially literate parental background, suggesting that education alone does not fully alleviate children from poor financial decision-making in the parental home. A counterfactual analysis shows that endowing the lowest-literacy households with the financial knowledge of the highest-literacy households would raise their welfare by 3.71%, equivalent to 7,500 DKK per year.

Keywords: household finance, financial literacy, intergenerational transfers, wealth inequality, life-cycle portfolio choice

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1 Introduction

In 2023, a survey by the European Commission found that 40% of the Danes were not confident that they would have enough money to live comfortably throughout their retirement years (European Commission, 2023). Details of the same report reveal that a large fraction of the population remain unfamiliar with fundamental financial concepts such as inflation, compound interest rates, mortgage bonds, and portfolio diversification.

While the Scandinavian countries tend to perform relatively well in financial literacy tests, the literature still documents several areas of improvement. Empirical evidence from Denmark finds that Danish households fail to refinance their mortgages, leading to lower savings at retirement. Moreover, the inaction in the mortgage market is tied to the older and less educated part of the population (Andersen et al., 2020). Similarly, Calvet et al. (2007) finds that their Swedish neighbors hold underdiversified portfolios and that 38 percent of the households don't participate in the financial markets at all, leaving an estimated 2.3 percent return on the table every year.

Importantly, these consequences extend beyond any single generation. A household that takes action on investments, refinancing and avoids high-cost debt accumulates more wealth within a given generation. Moreover, they transfer financial knowledge to their children and as such, an investment into financial knowledge within a household can elevate the financial well-being of their family across generations (Lusardi et al., 2010). Not only is there documented evidence of the persistence of intergenerational knowledge transfers, but also of the degree to which a household translates that into wealth accumulation. Fagereng et al. (2020) finds that the well-educated households, on average, earn a higher return on their wealth, but also that the returns are persistent across generations: if a parent earns a high return on their wealth, so will their child. In this sense, financial literacy acts both as a learnable skill and as an inherited ability. This is particularly interesting in a Danish context, since Denmark consistently ranks among the most equal societies in the world by income, but research shows that this is more likely to stem from government transfers and redistributive efficiency than from genuine educational social mobility (Landersø & Heckman, 2017).

Information, at least, has never been more abundant. Advances in technology and artificial intelligence have made financial data, news, creation of investment accounts and tailored advice available instantaneously and at near-zero cost. Yet abundance may be its own obstacle: when every search returns thousands of results, the capacity to evaluate, filter, and act on information becomes the scarce resource. In that sense, financial literacy becomes more about the ability to filter the good information from the bad than acquiring the information in the first place.

Throughout the last decades, the theoretical literature has made major progress in establishing a mathematical framework for modeling household savings and portfolio choice. Life-cycle models such as those introduced by Carroll (1997) and Cocco et al. (2005) explain optimal savings rates

and portfolio strategies under income uncertainty. Empirical findings have likewise improved the understanding of how financial knowledge develops and how it shapes personal financial decision-making and wealth accumulation over a lifetime (Lusardi & Mitchell, 2007). Recent research has begun to combine these two components: Lusardi et al. (2017) builds a theoretical model incorporating financial knowledge into the quantitative environment of a life-cycle model with a US-based calibration. At the same time, improvements in data availability have enabled findings on the impact of parental background on financial decision-making (Kreiner et al., 2020).

However, to our knowledge, no existing work combines these two strands of literature into a life-cycle model incorporating intergenerational channels of financial literacy transfers. This paper fills that gap and thus, we seek to answer the question: how much of the Danish wealth inequality can be explained by differences in financial literacy and intergenerational knowledge transfers, and what are its welfare implications? To answer the question, we build a stochastic life-cycle model with heterogeneous agents and calibrate it on Danish register data from 2014 to 2024. The novelty of our paper is that we allow agents to differ in their attainable investment returns, depending on their financial knowledge and parental background. We find that financial literacy and intergenerational knowledge transfers account for 20.7% of wealth inequality at retirement, and that endowing the lowest literacy group with the knowledge of the highest literacy group would improve their lifetime consumption by 3.71%.

This paper proceeds as follows. Section 2 reviews the literature on life-cycle modeling and financial literacy to contextualize the paper's contribution. Section 3 outlines the methodology used in our model and estimation procedures. Section 4 describes the dataset and provides descriptive evidence of the effects of education and parental background on financial wealth and decision-making. Section 5 presents our quantitative life-cycle model, which incorporates heterogeneity in income and returns. Section 6 describes and justifies the calibration methods for this model. Section 7 presents our results of optimal decision-making over consumption and portfolio choice and describes their effects on inequality and welfare. Section 8 demonstrates the robustness of these results. Section 9 performs a counterfactual policy analysis of implementing different replacement rates and their effects on inequality and welfare. Section 10 discusses the limitations of the findings and further research opportunities. Section 11 concludes.

2 Literature Review

In this section, we review the literature on life-cycle models and financial literacy, covering both the early findings and the modern developments in these fields. The purpose is to give the reader a clear overview of the main ideas and key contributions, while also providing the background and motivation for our paper. We also discuss the studies that have been most important in shaping our approach.

2.1 Life-Cycle Models

Permanent Income Hypothesis

The study of optimal intertemporal decision-making by households, and the challenge of constructing a normative model that can rationalize their descriptive behavior, traces back to 1936 where Keynes claimed as a fundamental psychological law that households increase consumption when income rises, but by less than the increase in income, so that the saving rate rises with the level of income (Keynes, 2018). Saving, in this view, is a residual that scales with wealth.

This view was challenged in the 1950s by the life-cycle hypothesis of Modigliani and Brumberg (1954) and the permanent income hypothesis of Friedman (1957).¹ Both frameworks propose that forward looking households do not set consumption on the basis of current income, but rather on the basis of lifetime resources. In Modigliani and Brumberg's formulation, consumption equals the annuity value of total wealth, that is, financial assets plus the present discounted value of expected future labor income, so that the consumption-to-wealth ratio is constant and independent of the level of income. In Friedman's formulation, the same result is expressed as consumption being approximately equal to permanent income, defined as the constant-flow equivalent of lifetime resources. In both cases, households save during high-earning years and dissave during retirement in order to smooth consumption over the life cycle, and cross-sectional differences in saving rates reflect transitory income variation rather than a structural relationship between saving and the level of income. A limitation of both frameworks, however, is that income is treated as exogenous and effectively deterministic: the household either knows its future income path with certainty or behaves as if it does, ignoring the role of uncertainty in shaping savings.

Early Empirical Work

Given the prominence of Friedman's and Modigliani's contributions, much of the subsequent empirical literature focused on testing the permanent income hypothesis. Hall (1978) provides the first model: households form rational expectations over uncertain future income and maximize expected lifetime utility, which under quadratic preferences implies that consumption follows a random walk. The household consumes $1/T$ of expected lifetime resources each period. Letting e_t denote a mean-zero innovation with $\mathbb{E}_{t-1}[e_t] = 0$ it follows that $C_t = \mathbb{E}_{t-1}[C_t] + e_t = C_{t-1} + e_t$, which is Hall's random walk result. The intuition is simple: if consumption were expected to rise, the marginal utility of current consumption would exceed expected future marginal utility, and the household would smooth by raising consumption today. As a consequence, consumption only deviates from its previous value by the change in permanent income induced by news arriving in the most recent period. Mathematically, if the change in consumption is regressed on a vector of variables known

¹For a full exposition, see for example Romer (2019) pp. 368-385

at $t - 1$,

$$\Delta C_t = \beta_0 + \beta_1 X_{1,t-1} + \cdots + \beta_K X_{K,t-1} + e_t,$$

then the permanent income hypothesis implies $\beta = \mathbf{0}$. Hall finds predictive power in lagged stock returns, which formally rejects the strict version of the hypothesis, although he argues that because only the most recent quarterly lag matters, a weak version of the hypothesis holds.

Hall's formulation gave the literature an empirical specification, and a substantial body of work followed. Flavin (1981) rejects the permanent income hypothesis more decisively, showing that consumption responds to anticipated movements in current income beyond what the change in permanent income would justify, a phenomenon she labels the excess sensitivity of consumption. Campbell and Mankiw (1989) obtain a similar conclusion using an instrumental variables approach within a hybrid model, in which one fraction of households consume their entire current income each period (so-called rule-of-thumb consumers) while the remainder behave according to Hall's specification.

Buffer Stock Models

While the empirical evidence rejected the strict permanent income hypothesis, early models struggled to provide an alternative that matched the joint life-cycle behavior of consumption, income, and wealth. One proposed solution was that households, particularly the young ones who hold little wealth, face binding liquidity and borrowing constraints, so that their consumption tracks their income almost perfectly. Zeldes (1989) provides one of the earliest such tests, splitting households by total wealth and showing that disposable income predicts consumption only among the constrained households, while having no predictive power among the unconstrained.

A. Deaton (1991) and Carroll et al. (1992) develop what came to be known as the *buffer-stock models*, in which households are simultaneously impatient, in the sense that they prefer consumption today to consumption tomorrow, and prudent, so that uncertainty over future income generates a precautionary saving motive. These two forces pull in opposite directions: impatience pushes the household to run down assets, while prudence pushes it to accumulate them as a buffer against bad realizations. Carroll (1997) formalizes this tension and shows that the model admits a target wealth-to-income ratio. When wealth is below the target, prudence dominates and the household saves. When wealth is above the target, impatience dominates and the household dissaves. These models therefore rationalize why consumption tracks income for impatient liquidity-constrained households, especially early in life. They do not, however, manage to rationalize the lack of savings and hence buffer-stock for some parts of the population.

Extensions & Portfolio Choice

In order to tackle the lack of saving behavior among certain groups of the population, Hubbard et al. (1995) incorporates means-tested social insurance programs that impose an asset-based consumption floor on the households. Because eligibility for such programs is conditioned on having low assets, low-income households face an implicit tax on saving and rationally hold very little wealth, even in the presence of income risk. Their calibration replicates both the wealth-poverty of low-income households and the precautionary holdings of higher-income households within a single framework. Attanasio et al. (1999) address the hump-shaped consumption profile by enriching their model with demographic structure, in particular the arrival and departure of children. They show that once household composition is taken into account, per-equivalent-adult consumption is substantially smoother than per-capita consumption, and that the residual hump can be rationalized by precautionary motives in the presence of income risk that is itself hump-shaped over the life cycle. Gourinchas and Parker (2002) provide what has become the workhorse calibration of the life-cycle model with idiosyncratic income risk. They estimate preference and income parameters by matching age profiles of consumption, and find that the model fits the data well with empirically reasonable values for the discount factor and risk aversion. Cagetti (2003) extends this exercise to wealth profiles, using simulated method of moments on PSID data, and finds that a buffer-stock model with realistic income risk and inheritance motives can match the level and dispersion of wealth across age and education groups.

The portfolio choice extension of the life-cycle model is most prominently articulated by Cocco et al. (2005), which will serve as the benchmark for our model and is therefore described in detail in Section 3.5. They embed a two-asset structure, a risk-free bond and a risky asset, into a life-cycle environment. Their key insight is that human capital, defined as the present discounted value of future labor income, acts as an implicit bond on the household's balance sheet for most workers. Because financial wealth is small relative to human wealth early in life, the optimal portfolio response is to hold close to one hundred percent of financial wealth in equities when young, and to gradually rotate into bonds as human capital is converted into financial wealth through saving. When confronted with micro-data, however, the model over-predicts both equity participation and the risky share, particularly among younger households and households with low financial wealth. This discrepancy motivated much of the subsequent literature.

Remaining Puzzles & Explanations

Standard life-cycle portfolio choice models, calibrated to plausible parameters, predict that essentially all households should participate in the equity market and hold a substantial risky share. The data, however, shows that participation rates and risky shares are lower than these models predict (Calvet et al., 2007; Fagereng et al., 2020). The literature has proposed several mechanisms to explain this observation.

The most widely studied explanation is the presence of fixed costs of participation. Vissing-Jørgensen (2002) provides direct empirical evidence, estimating that a participation cost of fifty dollars is sufficient to rationalize the non-participation of half of US stock market non-participants. The intuition is that the expected utility gain from holding equities is bounded above by the size of the portfolio, and falls below the cost threshold for households with little to invest. Similarly, Haliassos and Michaelides (2003) introduce a fixed cost of participation in an infinite-horizon life-cycle model with liquidity constraints, showing that relatively small fixed entry costs are sufficient to deter stockholding because, for a plausible range of preference parameters, households can achieve desired consumption smoothing with little or no equity exposure. They interpret these costs broadly, attributing them to informational requirements, sign-up fees, and investor inertia. Together, the two studies provide complementary empirical and theoretical support for the view that fixed costs, interpreted to include time costs of monitoring, psychological costs of unfamiliarity, and information acquisition costs, can plausibly account for a substantial share of the participation gap. A more recent study by Filali Adib et al. (2025) provides direct evidence of time costs, utilizing mandated self-isolation of close contacts of infected individuals during the COVID-19 pandemic as an exogenous shock to the time available for managing finances. Quarantined individuals increase their trading propensity by roughly twenty percent during the isolation window, with the largest effects concentrated among the young and the child-free. These are precisely the groups for whom self-isolation represents the largest relative relaxation of time constraints, and they are 3.5 percent more likely to participate in the stock market in the longer run. These results illustrate time costs as a form of fixed participation cost, although the authors also note that the additional trading is associated with lower realized returns.

A second strand of the literature emphasizes the role of income risk and its cointegration with equity returns. Benzoni et al. (2007) argue that the standard calibration, in which labor income is uncorrelated with the stock market in the short run, misses the fact that human capital and aggregate equity values share common long-run sources of variation through their joint exposure to the level of productivity. Their findings rationalize the low risky shares observed early in life and even suggest that young agents should hold short positions in the market to hedge their exposure.

A third explanation appeals to the role of beliefs and experience. Malmendier and Nagel (2011) document that birth cohorts who experienced poor stock market returns in their formative years subsequently invest less in equities, even decades later and after controlling for wealth, income, and standard demographics. They interpret this as evidence that households place excess weight on personally experienced returns when forming subjective beliefs about the equity premium, an effect that naturally rationalizes deviations from the optimal equity share in standard life-cycle models.

A fourth strand departs from the expected-utility framework that underlies the canonical model, and instead modifies preferences or the treatment of probability itself. Cao et al. (2005) consider an equilibrium in which investors face Knightian uncertainty over the distribution of asset returns, and

are heterogeneously averse to that uncertainty. When the dispersion of uncertainty across investors is sufficiently large, those facing the highest perceived ambiguity rationally choose not to participate in the equity market at all, generating limited participation as an equilibrium phenomenon rather than as the consequence of an exogenous cost. Gomes (2005) follows a similar route, modeling investors as loss-averse individuals using the Prospect Theory framework of Kahneman and Tversky (1979). He shows that including loss aversion in a portfolio choice problem yields a discontinuous demand function for risky assets: loss-averse investors hold no equities at all unless the equity premium is sufficiently high to compensate for the disutility of potential losses, and they follow generalized portfolio insurance strategies once they do participate. Both papers illustrate that small departures from expected-utility maximization under rational expectations can generate corner solutions of zero equity holdings without the presence of fixed costs, although in practice the two channels are difficult to separate empirically and likely operate jointly.

The overarching message of this section is that, although we have come a long way since the life-cycle hypothesis of Modigliani and Brumberg (1954), no normative model has yet rationalized the descriptive behavior of households. Still, participation costs, heterogeneous beliefs and preferences, cointegrated income and return processes, and many other plausible explanations exist. In this paper, we draw on model elements from several of the studies above, but our central mechanism is heterogeneous financial knowledge, which we review next.

2.2 Financial Literacy

In line with the proposed solutions above, it seems reasonable to assume that discrepancies between observed and model-predicted risky shares or participation rates may stem from information differences: if a household has more information about the optimal decision in a given context, it seems natural that they will act differently from others.

While the puzzles above revolve around stock market behavior, we do not propose that financial literacy is a one-dimensional character trait. As Lusardi and Mitchell (2023) define it, we consider financial literacy as “people’s knowledge of and ability to use fundamental financial concepts in their economic decision-making”. Hence, we argue that a lack of financial literacy may materialize itself in sub-optimal refinancing of a mortgage, missing opportunities to avoid fees, or, as we shall assume in our paper, a lower return on the stock market. Considering that financial knowledge is unequally distributed across the population, it seems likely that it can rationalize behavior which is, at first glance, irrational. At the same time, quantifying this effect is valuable, since the financial literacy of a population is something a government can intervene on, as opposed to, say, preference heterogeneity.

While defining financial literacy itself can be a difficult task, identifying what leads to an increase in financial knowledge may be even harder: after all, it is difficult to disentangle financial knowledge from education, innate cognitive ability, interest in money and finance, and quantitative

skills more broadly. Therefore, the literature has approached the area from many angles. Lusardi and Mitchell (2007) provide one of the first systematic links between financial literacy, retirement planning, and wealth accumulation, showing that households who fail to plan for retirement, behavior strongly associated with low scores on standard financial literacy questions, accumulate substantially less wealth at retirement, and that the gap remains large after controlling for income, education, and demographics. Clark et al. (2017) extend this evidence into active portfolio choice, showing that within US defined-contribution retirement accounts, financially literate participants earn higher risk-adjusted returns through better diversification and lower fees. The standard approach to measuring financial literacy uses the three questions developed by Lusardi and Mitchell (2011). However, as we shall see below, there are other ways to measure the degree of financial literacy within a household.

Grinblatt et al. (2011) use Finnish military test scores and find that stock market participation is monotonically increasing in IQ, with the difference between the lowest and highest deciles exceeding twenty percentage points, and that high-IQ households are also more diversified conditional on participation. As an alternative measure, Parise and Peijnenburg (2019) find that a lack of non-cognitive abilities such as soft skills and personality traits has a causal effect on the probability of financial distress.

In terms of quantifying financial knowledge, our paper resembles most closely the last common proxy: education. The literature documents that both educational length and field of study play a role in this regard. Fagereng et al. (2020) use Norwegian administrative data to document that returns to wealth are themselves heterogeneous and persistent within households, and that observable measures of human capital, including length and field of study, explain a meaningful share of the dispersion. Altmejd (2025) shows that the field of study chosen at university directly affects portfolio returns, with business and finance graduates outperforming others by economically significant amounts. Our model leans mostly on this approach, as we model financial literacy as a discrepancy in attainable expected excess returns, which depends on the households own education and parental background.

We do so, as the literature documents a causal effect between parental financial literacy and their children. Kreiner et al. (2020) use Danish administrative data to show that the propensity to default on debt is strongly correlated between parents and children, even after controlling for income and education, suggesting that financial behavior is transferred across generations. Moreover, further findings suggest that a lack of social mobility in education creates persistence in education across generations. Landersø and Heckman (2017) argue that while Denmark has a high degree of income mobility, it is not due to educational mobility but rather through redistributive policies. Altmejd (2024) finds that there is not only persistence in the length of one's education, but also a clear persistence in the field of study.

The model that most closely resembles ours is provided by Lusardi et al. (2017), who embed en-

ogenous financial knowledge accumulation in an otherwise standard life-cycle model with stochastic labor income and a portfolio choice between a riskless and a risky asset. In their framework, financial knowledge is a stock that depreciates over time and can be augmented through costly investment, and it raises the return earned on sophisticated assets relative to a basic technology that all households can access without effort. The household therefore faces a joint optimization problem over consumption, portfolio composition, and financial knowledge investment. The optimal knowledge stock is determined by the marginal benefit of higher returns weighed against the marginal cost of acquisition. They find that optimal financial knowledge is strongly increasing in lifetime resources, both because wealthier households earn returns on a larger asset base and because they expect to live in retirement on accumulated wealth rather than on Social Security or other low income sources. Furthermore, they find that the friction of financial knowledge can generate cross-sectional inequality in financial knowledge, with college-educated households investing far more in knowledge than less educated households.

Our framework departs from Lusardi et al. (2017) in treating financial knowledge as exogenous rather than as the outcome of an optimization problem. This is a deliberate modeling choice: we explain distribution consequences not due to heterogeneous income paths leading to different levels of financial knowledge, but heterogeneous starting points of financial knowledge based on own and parental education. Concretely, we endow educational groups with heterogeneous returns that depend on the educational background, allowing us to quantify how much of the observed wealth gap can be attributed to the joint inheritance of financial knowledge and educational attainment.

2.3 Contribution

We consider the contribution of this paper to be original along three dimensions. First, we exploit Danish administrative data to document the joint distribution of returns, equity participation, and portfolio allocation across educational groups. Because the dataset is based on actual observables reported by Danish authorities rather than on survey responses, the precision with which we measure financial outcomes is sharper than in much of the existing literature. The richness of the educational data, which includes length of education, field of study, and grades obtained, additionally allows us to separate the effects of general education, field of study, and innate ability across households, three dimensions that are typically conflated in survey-based work.

Second, we offer a novel theoretical mechanism through which financial knowledge shapes household wealth. Existing structural models of financial knowledge, most prominently Lusardi et al. (2017), rationalize the lack of participation and the low risky shares of less-educated households by arguing that income heterogeneity across educational groups makes participation suboptimal in the first place: low expected lifetime resources lower the marginal benefit of investing in financial knowledge, which in turn reduces participation. We accept that income heterogeneity is part of the explanation, but the novelty of our contribution lies in modeling a lack of financial knowledge as

having a direct effect on the attainable risky return available to the household. Rather than letting low income drive low investment in financial knowledge, we model educated households as earning higher attainable returns directly. We further exploit the richness of the data by incorporating inter-generational knowledge transfers, in which the financial knowledge of a household's parents enters the household's own optimization problem, into a fully structural life-cycle framework. In doing so, we combine three strands of the literature: a life-cycle model with risky assets in the tradition of Cocco et al. (2005), the unequal distribution of financial knowledge studied by Lusardi et al. (2017) and others, and the intergenerational transmission of financial behavior and education documented by Kreiner et al. (2020) and Lusardi et al. (2010).

Third, we calibrate the resulting model to Denmark, an institutional setting that differs in important ways from the United States studied by Lusardi et al. (2017) and from most of the seminal life-cycle literature surveyed in Section 2.1. The most direct difference enters our model through the tax system: marginal income tax rates and the taxation of capital income are substantially higher in Denmark than in the United States, which changes both the after-tax incentive to save and the after-tax return on risky investment that a household with a given level of financial knowledge can attain. Secondly, the Danish public pension system provides a meaningful floor on retirement income, which lowers the incentive to accumulate private wealth and which reduces the wealth-to-income ratios we should expect to see in the data relative to the US benchmark. Although we do not model the pension system explicitly, this institutional feature is useful background for interpreting the wealth distribution our calibrated model is asked to match. Finally, the Danish setting allows us to abstract from features that play a meaningful role in US-calibrated models but that are quantitatively less important here, most notably out-of-pocket medical expenditure in retirement, which has been shown by De Nardi et al. (2010) to be a first-order driver of late-life saving in the United States but which is largely absorbed by universal public health coverage in Denmark. Together, these three contributions allow us to address the central research question of the paper:

- How much of the Danish wealth inequality can be explained by financial education and inter-generational knowledge transfers, and what are the welfare implications for Danish households?

3 Theory

This section outlines the theoretical framework underlying our estimation and modeling methodology. We provide comprehensive replication code for both the empirical analysis and simulations, along with a dummy dataset. The dummy dataset and portions of the simulation code have been generated with the assistance of Generative AI. Materials are available on: [GitHub](#).

Note that the contents of this section are descriptive in nature. They may be skipped without affecting the overall understanding of the paper.

3.1 Econometric Approach

Estimation Strategy

As our dataset is a panel from 2014-2024, it is natural to consider the pooled OLS, fixed effects and random effects estimators. Following Wooldridge (2010), let $t = \{1, \dots, T\}$ and $i = \{1, \dots, N\}$ and suppose the following linear model holds for the i th random draw from the population:

$$y_i = X_i\beta + u_i. \quad (1)$$

The assumptions of contemporaneous exogeneity and full rank,

$$\mathbb{E}(x'_t u_t) = 0, \forall t = \{1, \dots, T\} \quad \text{rank} \sum_{t=1}^T \mathbb{E}(x'_t x_t) = K, \quad (2)$$

are sufficient for consistency of the pooled OLS estimator:

$$\hat{\beta} = (X'X)^{-1}X'y = \left(\sum_{i=1}^N \sum_{t=1}^T x'_{it} x_{it} \right)^{-1} \cdot \sum_{i=1}^N \sum_{t=1}^T x'_{it} y_{it}. \quad (3)$$

As the same individual households are observed over time, however, the errors are typically serially correlated such that:

$$\text{Cov}(u_{it}, u_{is} | X_i) \neq 0, \forall s \neq t. \quad (4)$$

While this leaves consistency intact, it renders pooled OLS inefficient and invalidates default standard errors. To address inference, one can cluster the standard errors at the household level, allowing arbitrary within-household correlation. Alternatively, one can relax the exogeneity assumption altogether: the fixed effects (within) estimator allows for arbitrary correlation between a time-invariant household effect and the regressors, at the cost of requiring the stronger strict exogeneity assumption $\mathbb{E}(u_{it} | x_{i1}, \dots, x_{iT}) = 0$. The time-demeaning transformation, however, removes any time-invariant regressors. As such, if the variable of interest is time-invariant (such as education or skill), it is only possible to estimate the effect of education changing within a certain household, or to interact it with another variable to capture how such a trait moderates time-varying effects. The random effects estimator restores efficiency under serial correlation, but at the price of the additional orthogonality assumption $\mathbb{E}(a_i | x_{it}) = 0$ between the time-invariant household effect and the covariates. We consider this too restrictive in our setting: if unobserved household traits such as cognitive ability or financial sophistication are correlated with educational choices, the RE estimator is inconsistent while the FE estimator remains consistent.

3.2 Utility Functions

We start by introducing utility functions to be able to set up the maximization problem of the household.

To model the economic behavior of agents, it is most common to use utility functions that satisfy three standard conditions: (1) strict monotonicity, $u'(C) > 0$, so that more consumption always raises utility; (2) risk aversion, which under expected utility requires strict concavity $u''(C) < 0$, implying $\mathbb{E}[u(C + \tilde{x})] \leq u(C)$ for any mean-zero risk $\tilde{x}$; and (3) the Inada conditions $\lim_{C \rightarrow 0} u'(C) = \infty$ and $\lim_{C \rightarrow \infty} u'(C) = 0$, which ensure interior solutions. There are several functional forms that satisfy these conditions, including constant absolute risk aversion (CARA) and quadratic utility, each of which is reasonable in particular settings. In this paper, however, we use constant relative risk aversion (CRRA):

$$u(C) = \begin{cases} \frac{C^{1-\theta}}{1-\theta} & \text{if } \theta > 0 \text{ and } \theta \neq 1, \\ \log(C) & \text{if } \theta = 1, \end{cases} \quad (5)$$

where according to L'Hôpital's Rule the function $u(C) = (C^{1-\theta} - 1)/(1 - \theta)$ approaches the logarithmic form in the limit of $\theta \rightarrow 1$, since the $-1/(1 - \theta)$ term as an additive constant does not affect optimization.

The key properties of CRRA can be understood most intuitively in a portfolio choice problem in which the agent allocates wealth between a risky asset and a risk-free bond. The Arrow-Pratt coefficient of absolute risk aversion is $A(W) = -u''(W)/u'(W) = \theta/W$, which is decreasing in W . Decreasing absolute risk aversion implies that the nominal amount of wealth invested in the risky asset rises with wealth, in line with the empirical observation that wealthier households hold larger nominal equity positions. Combined with lognormally distributed returns, the coefficient of relative risk aversion $R(W) = A(W) \cdot W = \theta$ is, by construction, independent of the level of wealth, which also implies that the optimal portfolio share invested in the risky asset is independent of wealth, resulting in the closed-form solution for the optimal risky share introduced by Merton (1969):

$$\alpha^* = \frac{\mu - r}{\theta \sigma^2}, \quad (6)$$

where $\mu - r$ is the equity premium and σ^2 the variance of risky returns. This expression will reappear in our structural model in section 5, where it will provide a practical way to incorporate the role of financial knowledge in determining the different risky returns.

The elasticity of intertemporal substitution (EIS), defined as $-u'(C)/(C u''(C))$, governs how much consumption an agent is willing to shift across periods in response to changes in the interest rate. Under CRRA, the EIS equals $1/\theta$ and is therefore mechanically tied to the coefficient of relative risk aversion. This restriction is a core weakness of CRRA preferences and is most clearly illustrated

by the equity premium puzzle of Mehra and Prescott (1985): within the CRRA framework, matching observed excess returns requires implausibly high values of θ , since risk aversion cannot be raised without simultaneously compressing the EIS. Epstein and Zin (1989) resolve this tension by introducing recursive preferences that disentangle risk aversion from the EIS, but at substantial computational cost since the value function appears explicitly in the optimality conditions and cannot be substituted out.² Despite this limitation, CRRA utility remains attractive on several grounds: it implies stationary risk premia and interest rates even in the presence of long-run economic growth, it is tractable in the presence of multiplicative lognormally distributed risks, and it admits closed-form expressions in many standard portfolio choice problems (Campbell, 2017; Merton, 1969). For these reasons it remains the workhorse utility function of asset pricing, macroeconomics, and life-cycle modeling, and we adopt it as the natural choice for the model developed in the following sections.

3.3 Dynamic Programming

Having introduced utility functions as a way to set up maximization problems for agents, we now move on to dynamic programming to express this problem in an alternative way.

Many economic problems involve an agent's decision-making at different points in time, where previous decisions determine the current optimization problem. Typically, agents maximize their lifetime utility over a finite or infinite horizon, which can be expressed as a sequential problem (SP):

$$w(x_0) = \sup_{\{d_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, d_t) \quad \text{s.t.} \quad d_t \in \Gamma(x_t), \quad x_{t+1} = h(x_t, d_t), \quad x_0 \in X \text{ given,} \quad (\text{SP})$$

where $w(x_0)$ represents the highest achievable lifetime utility given an initial endowment of x_0 , decision variable d_t chosen in every period, feasibility constraint $\Gamma(x_t)$, and the law of motion $h(x_t, d_t)$.

This formulation is a valid way of expressing the optimization problem, but its computational feasibility is very limited. The infinite horizon makes it impossible to maximize directly over the sum, and even in the case of a finite horizon the optimization requires solving over a very high-dimensional object. We therefore look for an alternative expression of the same problem.

The key insight of dynamic programming relies on this alternative expression being a functional equation (FE). Instead of expressing the entire lifetime utility at every single point in time, the optimization problem is narrowed down to the effect of the current decision on today's payoff and all future payoffs, captured by the so-called Bellman equation:

$$v(x) = \max_{d \in \Gamma(x)} \{F(x, d) + \beta v(h(x, d))\}. \quad (\text{FE})$$

²A detailed description of our solution method and why CRRA preferences are critical for its tractability is provided in section 3.6.

In this expression introduced by Bellman (1957), $v(x)$ is the *value function*, which typically represents the highest lifetime utility the agent can achieve given the initial endowment of x , and $F(x, d)$ is the return function which represents the utility received in this period. The variable x is called a *state variable*, since it fully determines the starting point on which the following decision-making is based. Given this state variable, the feasibility constraint $\Gamma(x)$ is the space of all feasible choices for the so-called *control variable* d . This control variable is bounded by the current feasibility constraint, but also determines the upcoming state $h(x, d)$ and the upcoming feasibility constraint $\Gamma(h(x, d))$. This characteristic is essential since it captures the dynamic dependency, where current decisions affect future decisions. It is worth mentioning that the value function is by definition equal to the highest achievable lifetime utility, implying optimal decision-making for all future periods.

The so-called *Principle of Optimality* determines whether the solution of the FE expression equals that of the SP. In other words: does the Bellman equation solve the problem that we are facing? There are two main assumptions required to establish this equivalence formally following Stokey et al. (1989):

Assumption A1. $\Gamma(x)$ is nonempty for all $x \in X$.

Assumption A2. For all initial conditions $x_0 \in X$ and all feasible plans $x \in \tilde{\Pi}(x_0)$

$$\lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t F(x_t, x_{t+1})$$

exists (although it may be $+\infty$ or $-\infty$).

Intuitively, (A1) means that a space of options has to exist from which the agent can choose, while (A2) implies that in the limit, this sum has to converge. Under these two conditions, the Principle of Optimality holds, implying that the Bellman equation represents the same problem as SP. Solving the Bellman equation does not only yield a solution for the value function, but also a so-called *policy function*. A policy function is best described as a rule telling the agent the best decision given every possible starting point. Formally, the policy function expresses the optimal control variable given each possible state:

$$g(x) = \arg \max_{d \in \Gamma(x)} \{F(x, d) + \beta v(h(x, d))\}. \quad (7)$$

The FOC that the policy function must satisfy is given by:

$$F_d(x, g(x)) + \beta v'(h(x, g(x))) h_d(x, g(x)) = 0, \quad (8)$$

which implies that the control variable has to be chosen such that marginal benefits and marginal costs offset. One more unit of the control variable has to balance the direct benefit today, $F_d(x, g(x))$, against the discounted costs of all future periods, $\beta v'(h(x, g(x))) h_d(x, g(x))$.

Computing the policy function directly from equation (8) quickly reaches its limits, since it requires evaluating the derivative of the value function. To ensure computational feasibility, we replace it with a more convenient alternative based on the *Envelope theorem*³, which states that the marginal effect on the value function of one more unit of the state variable has to equal the marginal effect on the return function from optimally allocating that extra unit through the control variable. Intuitively, a small change in the state variable does not change the policy at the margin, since the agent is already acting optimally: we do not need to account for how the policy adjusts, but only for the direct effect of the extra unit on the value function. Formally, this gives

$$v'(x) = F_d(x, g(x)), \quad (9)$$

or, in a life-cycle scenario,

$$v'(x) = u'(c^*(x)), \quad (10)$$

and substituting $v'(x)$ into the original first-order condition (8) yields

$$F_d(x, g(x)) = \beta F_d(x, g(x)) \cdot h_d(x, g(x)), \quad (11)$$

or, in a life-cycle scenario, the Euler equation

$$u'(c_t) = \beta R_{t+1} u'(c_{t+1}). \quad (12)$$

The fact that the policy function is determined by the Euler equation has important implications for efficient solution methods, which we discuss further in section 3.6.

Finally, we want to add stochasticity to the expression. Stokey et al. (1989) show that the Principle of Optimality and the Envelope theorem still hold under uncertainty, so we can apply the expectation operator to the Bellman equation

$$v(x_t) = \max_{d_t \in \Gamma(x_t)} \{F(x_t, d_t) + \beta \mathbb{E}_t[v(x_{t+1})]\}, \quad (13)$$

and to the Euler equation

$$u'(c_t) = \beta \mathbb{E}_t[R_{t+1} u'(c_{t+1})]. \quad (14)$$

In this case, the agent does not maximize over a single future consumption level, but over the possible states of consumption in the upcoming period. A common way to practically implement stochasticity is to use Gauss-Hermite quadrature, where the integral over a continuously distributed random variable is translated into a discrete weighted sum of a finite set of nodes, which is further described in appendix C.

³The full derivation is provided in Appendix C

3.4 The Basic Life-Cycle Model

Having established the recursive formulations, we now use this maximization expression in a life-cycle framework.

The model presented here is mostly based on the buffer-stock model introduced by Carroll (1997). The main assumptions are as follows: prices are given for individual agents, there exists only one (liquid) asset with an exogenous risk-free rate $\bar{r}^f$, a finite horizon, stochastic labor income and a no-borrowing constraint. Every household maximizes expected utility over their life following:

$$\begin{aligned} \max_{\{C_t\}_{t=0}^T} \quad & \mathbb{E} \sum_{t=0}^T \beta^t \frac{C_t^{1-\theta}}{1-\theta} \\ \text{s.t.} \quad & A_{t+1} = \bar{R}^f (A_t + Y_t - C_t), \quad A_t \geq 0, \quad A_0 \text{ given,} \end{aligned} \quad (15)$$

where β is the discount factor, $\bar{R}^f = 1 + \bar{r}^f$ the risk-free return, C_t consumption, A_t end-of-period assets, and Y_t income.

Define cash-on-hand X as $X_t = \bar{R}^f S_{t-1} + Y_t$, which describes the liquid assets every agent can allocate between consumption and savings at the beginning of each period. Using this notation, the maximization problem can be written as

$$\begin{aligned} \max_{\{S_t, C_t\}_{t=0}^T} \quad & \mathbb{E} \sum_{t=0}^T \beta^t \frac{C_t^{1-\theta}}{1-\theta} \\ \text{s.t.} \quad & X_t = C_t + S_t, \quad X_{t+1} = \bar{R}^f S_t + Y_{t+1}, \quad S_t \geq 0, \quad X_0 \text{ given,} \end{aligned} \quad (16)$$

where it is worth discussing the no-borrowing constraint in more detail with two distinct observations:

First, the constraint does not always bind. There exists an implicit borrowing constraint for every household where it is not optimal to take on debt anyway and therefore makes the explicit constraint redundant. It becomes relevant precisely for young households facing an upward-sloping income profile, who would otherwise want to borrow against future earnings to smooth consumption.

Second, the constraint is less restrictive than it may appear. The explicit constraint does not generally rule out taking on debt, but rather taking on a loan beyond current net assets. Taking on debt and acquiring an asset of the exact same nominal value keeps the net assets constant and therefore does not violate this constraint. Intuitively, a potential lender is more willing to lend if the acquired assets serve as collateral, whereas the lender is less willing to lend if the borrowed funds are not covered. Hence, the explicit borrowing constraint introduced here can be interpreted as less strict than not having the chance to take on debt at all.

Having established the constraints on asset accumulation, we now turn to the other key ingredient of the model: the stochastic process governing labor income. There are several alternatives for how to model this stochastic income path. A. S. Deaton (1990) suggests an i.i.d. distribution of

income, but this formulation matches the empirical data in only very specific circumstances (farmers in low-income countries). Tauchen (1986), on the other hand, approximates a continuous AR(1) income process with a finite-state Markov chain. However, the AR(1) process is mean-reverting, which contradicts the empirical evidence of Meghir and Pistaferri (2004), implying that permanent shocks are a substantial component of stochastic income.

The specification of Carroll (1997) addresses this by combining a permanent and a transitory component:

$$\begin{aligned} Y_t &= P_t V_t \\ P_t &= G_t P_{t-1} N_t, \end{aligned} \tag{17}$$

where P_t is permanent income, V_t is a transitory shock, N_t is a permanent shock and G_t is deterministic income growth. Intuitively, permanent income refers to the general direction of the agent's income including the general growth rate based on their age profile and permanent shocks such as long-term illness. Permanent shocks follow a log-normal distribution

$$N_t \sim \log \mathcal{N} \left(-\frac{\sigma_N^2}{2}, \sigma_N^2 \right), \quad \mathbb{E}[N_t] = 1, \tag{18}$$

with an expected value of 1 to keep the general direction of this shock neutral. Transitory shocks follow

$$V_t \sim \begin{cases} 0 & \text{with prob. } p \\ Z_t & \text{with prob. } (1 - p), \end{cases} \tag{19}$$

$$Z_t \sim \log \mathcal{N}(\mu_Z, \sigma_Z^2), \quad \mu_Z \text{ chosen to ensure } \mathbb{E}[V] = 1. \tag{20}$$

Carroll explicitly allows for the possibility of a *disastrous* labor income shock outcome with a probability of p . This is a critical driver for the portfolio choice life-cycle model presented in section 3.5. Note, however, that we do not use this feature in our model in section 5 and provide further discussion regarding this in section 6.

The basic life-cycle model separates lifetime into two different parts: working life and retirement. Income during working life is explained above, while income during retirement is modeled in the following way:

For $t \geq T_{\text{ret}}$, income is given by

$$Y_{it} = \mu P_{i, T_{\text{ret}}-1}, \tag{21}$$

where μ denotes the replacement rate, which is a fixed fraction of the permanent income of the period prior to the entry into retirement. The income path structure of $Y_{it} = P_{it} V_{it}$ and $P_{it} = G_{it} P_{it-1} N_{it}$ can stay even beyond working life since the replacement rate μ can be incorporated into the deterministic growth component G_t .

Typically, one also assumes no income shocks during retirement, so that

$$N_{it} = 1, \quad V_{it} = 1 \quad \forall t \geq T_{\text{ret}}. \quad (22)$$

We express the maximization problem of (16) in the recursive manner explained in section 3.3:

$$\begin{aligned} W(X_t, P_t) &= \max_{S_t} u(C_t) + \beta \mathbb{E} [W(X_{t+1}, P_{t+1})] \\ &= \max_{S_t} u(X_t - S_t) + \beta \mathbb{E} [W(\bar{R}^f S_t + G_{t+1} P_t N_{t+1} V_{t+1}, P_{t+1})]. \end{aligned} \quad (23)$$

Note that we changed the notation for the value function to W to distinguish it from the transitory shocks. The key insight here is that with this definition of the labor income process, permanent income becomes a state variable. One can think of permanent income as an important determinant of current decision-making other than the transitory component, since the transitory shock of the previous period do not add any predictive power for the upcoming period. Having two state variables (X_t, P_t) , however, can become a computational burden. In order to overcome this issue, we proceed with a normalization of the state variables:

$$x_t = \frac{X_t}{P_t}, \quad c_t = \frac{C_t}{P_t}, \quad s_t = \frac{S_t}{P_t}. \quad (24)$$

The economic interpretation of this normalization is that all variables are now measured relative to permanent income, where e.g. $x_t = 2$ means that the current cash-on-hand is double the amount of the current permanent income. Not accounting for the transitory shock in the denominator also means that x_t can fall below 1, while staying within the boundaries of the no-borrowing constraint. This normalization makes use of CRRA utility being homogeneous of degree $(1 - \theta)$:

$$u(C_t) = \frac{(c_t P_t)^{1-\theta}}{1-\theta} = P_t^{1-\theta} u(c_t). \quad (25)$$

The economic interpretation of this normalization is to make households with different levels of income comparable. As discussed in section 3.2, constant relative risk aversion means making decision-making dependent on assets relative to income and independent of the total level of income. Intuitively, two agents make the same decision if their cash-on-hand relative to their income is the same, even if they differ in nominal wealth.

To illustrate the effect of this normalization on the maximization problem, we start with the final period, where it is always optimal to consume all cash-on-hand:

$$\begin{aligned} W_T(X_T, P_T) &= \max_{S_T} U(X_T - S_T) = U(X_T) = P_T^{1-\theta} U(x_t) \\ &= P_T^{1-\theta} W_T(x_T), \end{aligned} \quad (26)$$

and the next-to-last period with

$$\begin{aligned}
W_{T-1}(X_{T-1}, P_{T-1}) &= \max_{s_{T-1}} P_T^{1-\theta} U(x_{T-1} - s_{T-1}) + \beta \mathbb{E} [W_T(X_T, P_T)] \\
&= \max_{s_{T-1}} P_T^{1-\theta} U(x_{T-1} - s_{T-1}) + P_T^{1-\theta} \beta \mathbb{E} [W_T(x_T)] \\
&= P_T^{1-\theta} \cdot \left[\max_{s_{T-1}} U(x_{T-1} - s_{T-1}) + \beta \mathbb{E} [W_T(x_T)] \right] \\
&= P_T^{1-\theta} W_{T-1}(x_{T-1}).
\end{aligned} \tag{27}$$

We can see that the value function $W(x)$ inherits the homogeneity property from the CRRA utility function shown in (25). That is, we can track this characteristic all the way to the initial decision-making period t :

$$\begin{aligned}
P_t^{1-\theta} W(x_t) &= \max_{s_t} P_t^{1-\theta} U(x_t - s_t) + \beta \mathbb{E} [P_{t+1}^{1-\theta} W(x_{t+1})] \\
W(x_t) &= \max_{s_t} U(x_t - s_t) + \beta \mathbb{E} \left[(G_{t+1} N_{t+1})^{1-\theta} W(x_{t+1}) \right].
\end{aligned} \tag{28}$$

As discussed in section 3.3, this formulation provides the base to express the optimality condition using the derivative of the return function (here the utility function) following the Envelope theorem:

$$\begin{aligned}
C_t^{-\theta} &= \beta \bar{R}^f \mathbb{E} [C_{t+1}^{-\theta}] \\
(P_t c_t)^{-\theta} &= \beta \bar{R}^f \mathbb{E} [(P_{t+1} c_{t+1})^{-\theta}] \\
c_t^{-\theta} &= \beta \bar{R}^f \mathbb{E} [(G_{t+1} N_{t+1})^{-\theta} c_{t+1}^{-\theta}].
\end{aligned} \tag{29}$$

This normalized Euler equation will be the central object for our solution method presented in section 3.6, where we exploit this expression by recovering the optimal current consumption over the states of optimal future consumption. This basic model, however, is restricted to one risk-free asset. To go one step further, we need to introduce a portfolio choice problem for the agent with a risky asset, which will be introduced in the following section.

3.5 Portfolio Choice Life-Cycle Model

Having introduced a maximization problem in a basic life-cycle framework, we now want to extend this model by adding risky assets to set a base to make financial knowledge relevant. The model presented here is mainly based on the portfolio choice life-cycle model introduced by Cocco et al. (2005).

This additional feature of a risky asset in the model allows us to understand portfolio choice over the lifetime. Building on the assumptions of the model in section 3.4, we also assume that prices are given, the horizon to be finite, stochastic labor income and the no-borrowing constraint. In contrast, this model does not just assume a risk-free, but also a risky asset the agent can invest in. The maximization problem can be expressed as

$$\begin{aligned}
& \max_{\{C_t, \alpha_t\}_{t=0}^T} \mathbb{E} \left[\sum_{t=0}^T \beta^t \left(\prod_{j=0}^t \phi_j \right) \frac{C_t^{1-\theta}}{1-\theta} \right] \\
& \text{s.t. } C_t + A_t = X_t, \quad X_{t+1} = R_{t+1}^p A_t + Y_{t+1}, \\
& \quad R_{t+1}^p = \bar{R}^f (1 - \alpha_t) + R_{t+1} \alpha_t, \\
& \quad A_t \geq 0, \quad 0 \leq \alpha_t \leq 1, \quad X_0 \text{ given},
\end{aligned} \tag{30}$$

where A_t denotes asset holdings at the end of period t , α_t the portfolio share invested in the risky asset, R_{t+1}^p the portfolio return between t and $t+1$, R_{t+1} the gross return on the risky asset, $\bar{R}^f$ the gross risk-free return, and ϕ_{t+1} the probability of surviving from period t to $t+1$. We can think of the agent deciding every period what fraction of assets to invest in stocks, where this risky asset follows an i.i.d. distribution of

$$R_{t+1} - \bar{R}^f = \tau + \eta_{t+1}, \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2), \tag{31}$$

where τ represents the expected excess return of the risky asset.

We express the maximization problem in a recursive form following section 3.3. Using the normalization of section 3.4, the normalized Bellman equation is

$$\begin{aligned}
W(x_t) &= \max_{a_t, \alpha_t} u(x_t - a_t) + \beta \phi_{t+1} \mathbb{E}[(G_{t+1} N_{t+1})^{1-\theta} W(x_{t+1})] \\
&\text{s.t. } x_{t+1} = \frac{[\bar{R}^f (1 - \alpha_t) + R_{t+1} \alpha_t] a_t}{G_{t+1} N_{t+1}} + V_{t+1}, \quad x_0 \text{ given}.
\end{aligned} \tag{32}$$

This problem has one state variable x_t and two control variables, a_t and α_t , yielding two optimality conditions.

The Envelope theorem introduced in section 3.3, resulting in the Euler equation in section 3.4, is now extended with the portfolio return being a random variable as well and therefore part of the expectation:

$$c_t^{-\theta} = \beta \phi_{t+1} \mathbb{E}[R_{t+1}^p (G_{t+1} N_{t+1})^{-\theta} c_{t+1}^{-\theta}]. \tag{33}$$

This is the first optimality condition that we refer to during the whole dynamic programming process. However, this only pins down the optimal consumption behavior, ignoring the portfolio choice. To get an optimality condition for α_t we differentiate the Bellman equation (32) with respect to α_t :

$$\frac{\partial W(x_t)}{\partial \alpha_t} = \frac{\partial u(c_t)}{\partial \alpha_t} + \beta \phi_{t+1} \mathbb{E} \left[W'(x_{t+1}) \cdot \frac{\partial x_{t+1}}{\partial \alpha_t} \right] = 0, \tag{34}$$

where $\frac{\partial x_{t+1}}{\partial \alpha_t} = \frac{a_t (R_{t+1} - \bar{R}^f)}{G_{t+1} N_{t+1}}$ and since the current consumption does not depend on the portfolio choice but is purely determined by the Euler equation (33) we can also state that $\frac{\partial u(c_t)}{\partial \alpha_t} = 0$. Com-

combined with the Envelope theorem of $W'(x_{t+1}) = u'(c_{t+1}^*)$ and assuming CRRA preferences, we get the following optimality condition:

$$a_t \cdot \beta \phi_{t+1} \mathbb{E}[(G_{t+1} N_{t+1})^{-\theta} (x_t - a_t)^{-\theta} (R_{t+1} - \bar{R}^f)] = 0. \quad (35)$$

Note that the risky share α is hidden in the cash-on-hand of the upcoming period via

$$x_{t+1} = \frac{[\bar{R}^f(1 - \alpha_t) + R_{t+1}\alpha_t] a_t}{G_{t+1} N_{t+1}} + V_{t+1}. \quad (36)$$

This optimality condition is not trivial, which is why we look a bit deeper into it. For the sake of the argument, we simplify (35) by isolating expected marginal utility and expected excess return to

$$\mathbb{E}[u'(c_{t+1}) \cdot (R_{t+1} - \bar{R}^f)] = 0,^4 \quad (37)$$

which can be decomposed into:

$$\mathbb{E}[u'(c_{t+1})] \cdot \mathbb{E}[R_{t+1} - \bar{R}^f] + \text{Cov}(u'(c_{t+1}), R_{t+1} - \bar{R}^f) = 0.^5 \quad (38)$$

Even if labor income and returns are uncorrelated, marginal utility and excess returns will generally be correlated since the realization of investment determines consumption. A low α implies that the state of the stock market does not really matter for consumption since the agent is not really exposed to the stock market anyway. In contrast, a high α means that marginal utility and returns are highly correlated since the high exposure to the stock market makes consumption very dependent on it. Furthermore, the covariance is negative since a high state in the stock market means higher consumption and therefore lower marginal utility. That is, it is the covariance term that drives this optimality condition to zero since it offsets the pushing force of expected marginal utility and expected excess return. If the whole expression is positive, it means that the positive term is dominating, suggesting that the covariance between marginal utility and excess return is not strong enough to be optimal. Hence, the agent should increase α . In contrast, a dominating negative term implies that it is optimal to decrease the exposure to the stock market.

To summarize the most important equations, we have seen that we can describe the optimality conditions for the maximization problem (32) separately for both control variables a_t (33) and α_t (35). These two equations are the central objects for the solution method described in the following section.

⁴Note that we can divide out the strictly positive term $\beta \phi_{t+1}$, unlike end-of-period assets a_t , which could be zero.

⁵according to $\mathbb{E}[A \cdot B] = \mathbb{E}[A] \cdot \mathbb{E}[B] + \text{Cov}(A, B)$

3.6 Solution Method

Endogenous Grid Method

Having introduced a life-cycle model with portfolio choice, we now proceed with an efficient way to solve such a model numerically. The key insight is that we utilize the optimality conditions derived in section 3.5 to recover optimal decision-making rather than iterating through every possible decision and picking the best.

This section is mainly based on the work of Carroll (2006) and gives insights on how to implement the dynamic problem optimization discussed in section 3.4 and section 3.5. Generally, there are multiple approaches to solving a Bellman equation, but in this section we will focus on the most common method used in life-cycle modeling: the Endogenous Grid Method (EGM). Since the focus of this paper is on life-cycle modeling, the following explanation will be closely aligned with the maximization problem of a life-cycle model rather than a general explanation of EGM. The main goal is to recover a policy function for all control variables.

EGM makes use of the optimality conditions derived in section 3.5. An intuitive way of thinking about this method is not to ask: *what is the optimal control variable for each state?*; but rather: *at which state is this decision optimal?*

Following backward induction, we start with the terminal condition, where it is always optimal for any agent to consume everything: $c_T(x_T) = x_T$. That is, there is no optimal α_T^* in that final period, since there are no end-of-period assets.

For the second step, it is useful to distinguish between the mathematical expression and the computational implementation. Mathematically, we fix a candidate pair (a, α) and evaluate the expectation over all possible state realizations (N, V, η) . For each realization, we compute next period's cash-on-hand x' , recover optimal next-period consumption $c'(x')$ from the policy function of the next period, and evaluate the realization-specific product of marginal utility and excess returns weighted by the probability of this state realization occurring. The expectation can therefore be written as

$$\sum_e \sum_n \sum_v w_{e,n,v} u'(c'(\eta_e, N_n, V_v; \alpha, a)) (R_e - \bar{R}^f), \quad (39)$$

where e , n , and v index the discrete realizations of stock return, permanent income, and transitory income shocks, respectively, and $w_{e,n,v}$ denotes the corresponding joint probability weight.

Computationally, however, the procedure is easier to think about in reverse order. The implementation begins by fixing one realization (N, V, η) . Given this realization, one first computes the portfolio return $R^p(\alpha; \eta)$ over the entire α -grid. Using this return realization, one then computes next period's cash-on-hand $x'(\alpha, a)$ for every candidate pair (α, a) . Based on these x' values, the policy function from period $t + 1$ delivers optimal consumption $c'(x')$, which in turn determines

marginal utility $u'(c')$. In the case of the terminal period, it is $c_T(x_T) = x_T$. Finally, the resulting expression is weighted by the probability of the realization and added to the expectation matrix. Repeating this procedure over all realizations fills the expected-value object for every (α, a) candidate pair.

Each (α, a) combination is therefore a candidate for satisfying the portfolio optimality condition

$$0 = a_t \cdot \mathbb{E} \left[(G_{t+1} N_{t+1})^{-\theta} (x_{t+1} - a_{t+1})^{-\theta} (R_{t+1} - \bar{R}^f) \right].$$

For each level of end-of-period assets a , we search over the α -grid for the value that sets the weighted sum to zero. This yields the optimal risky share $\alpha_{T-1}^*(a)$ for every candidate a . The natural next question is therefore: for which state x_t is each level of end-of-period assets a optimal? This is where the endogenous grid logic enters.

$$x'(a) = \frac{R^p(\alpha^*(a)) \cdot a}{G \cdot N_n} + V_v. \quad (40)$$

Equivalently to the process of finding the optimal risky share, we find the optimal consumption of the upcoming period $c'(x')$, get the marginal utility $u'(c')$ and iterate through all possible state realizations to get the expected value of the Euler equation (33):

$$c_t^{-\theta} = \beta \phi_{t+1} \mathbb{E} \left[R_{t+1}^p (G_{t+1} N_{t+1})^{-\theta} c_{t+1}^{-\theta} \right]. \quad (41)$$

Now, one can isolate the optimal current consumption and use the budget constraint to identify the corresponding optimal cash-on-hand of the current period:

$$x^*(a) = c^*(a) + a,$$

leaving a matrix where for every end-of-period assets candidate a , we have the state $x^*(a, t)$ at which this is optimal and the corresponding optimal risky share $\alpha^*(a, t)$. The next stage of iteration is to run through every single period, continuing to start at the oldest age and reiterating until one ends up at the very first period.

This completes the theoretical foundation of the paper. We introduced the econometric approaches required for the calibration in section 6, established a recursive framework for the portfolio choice life-cycle model on which the extension in section 5 is based, and presented the numerical implementation method to solve it.

4 Data

We utilize the Danish registry dataset provided by Statistics Denmark, which we access through our employment at Copenhagen Business School. Our analysis covers the 11-year period from 2014 to

2024, which constitutes the longest unbroken period of data availability across the registries relevant for our analysis. This section describes the data, the sample restrictions we impose, and presents descriptive statistics on the final sample.

4.1 Demographics

The demographics dataset provides individual-level information on age, gender, marital status, immigration status, and place of birth. It also links each individual to their parents and to a unique household identifier. From this dataset we generate indicators for household size, the number of children in the household, and links between an individual and the parents of their spouse. We further merge occupancy status onto the demographics data so that we can track retirement timing and employment status at any given point during the sample period.

4.2 Income and Assets

Our data on financial wealth, including income, asset ownership, and debt, are reported directly by the Danish tax authorities (SKAT). We use disposable income, as this is the most accurate measure to identify a household's consumption-savings space. Statistics Denmark defines this as:

$$\begin{aligned}
 \text{Disposable Income} = & \text{Salary Earnings} + \text{Profits from Business Ownership} \\
 & + \text{Government Transfers} + \text{Private Pension} \\
 & + \text{Capital Gains} + \text{Child Support} \\
 & + \text{Imputed Rental Value of Owner-Occupied Housing} \\
 & - \text{Interest Expenses} - \text{Total Taxes Paid}
 \end{aligned} \tag{42}$$

Household disposable income is not, on its own, an appropriate measure of the utility a household derives from monetary income. A household consisting of two adults and two children does not derive the same utility from a given level of income as a single-individual household. At the same time, a household consisting of two adults can enjoy certain economies of scale, in particular when it comes to expenses for utilities or food. The literature handles this issue in two ways. The first approach scales the utility function based on household demographics, taking the form $U = n_t u(c_t/n_t)$, where n_t is an equivalence scale that accounts for consumption differences across household size and family composition over the life cycle (see for example Scholz et al. (2006) and Lusardi et al. (2017)).

The second approach, which we adopt in this paper, applies the Modified OECD Equivalence Scale directly to income, scaling it according to household composition (see for example Fagereng and Halvorsen (2017)). Following Hagenaars et al. (1994), equivalent disposable income is calculated in three steps:

1. All monetary incomes received from any source by each member of a household are added up. These include income from work, investment, social benefits, and any other household income, while taxes and social contributions paid are deducted from this sum.
2. To reflect differences in household size and composition, total net household income is divided by the number of equivalent adults using the modified OECD scale, which assigns a weight to each member of the household:
 - 1.0 to the first adult
 - 0.5 to the second and each subsequent person aged 14 and over
 - 0.3 to each child aged under 14
3. The resulting figure is the equivalent disposable income, which is attributed equally to each member of the household.

We also obtain information on each individual's financial assets, defined by Statistics Denmark as:

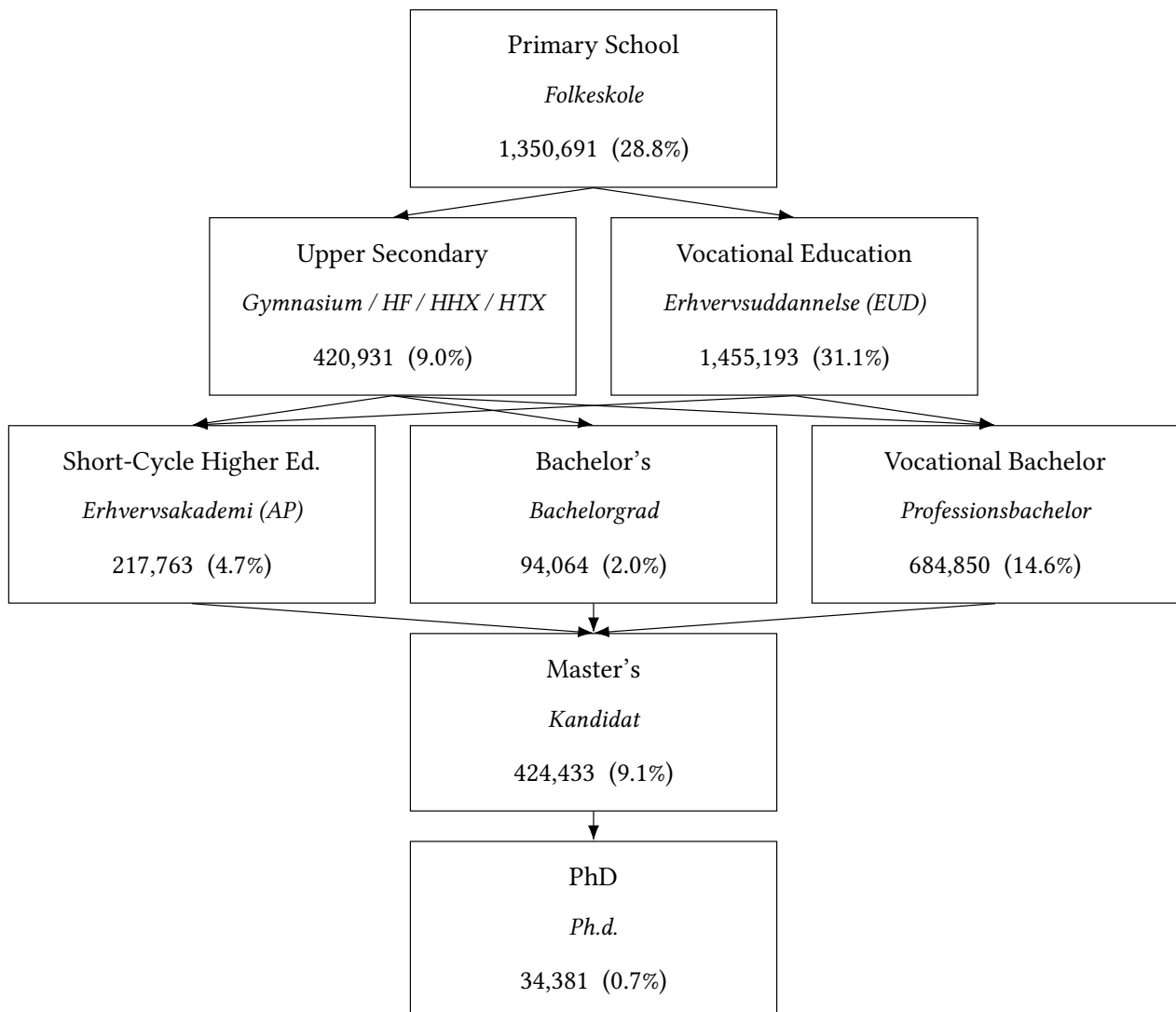
$$\begin{aligned}
 \text{Financial Assets} = & \text{Cash in Financial Institutions} + \text{Bonds} \\
 & + \text{Stocks} + \text{Shares in Investment Funds} \\
 & + \text{Mortgage Bonds} + \text{Cash in Business Account}
 \end{aligned} \tag{43}$$

We subtract cash holdings to obtain financial assets net of cash, which we use to define the extensive and intensive margins of financial market participation. Participation is a binary variable that takes the value of unity if an individual's financial assets net of cash exceed zero. The risky share is defined as the fraction of total financial assets invested in any of the financial instruments above, other than cash.

4.3 Education

Attainment and Field of Study

At the time of writing, Denmark has 5,543 types of education across all areas and levels of the educational system. We construct heterogeneity in education along two dimensions: highest educational attainment and field of study. Each educational classification is mapped to one of 11 categories identifying the highest educational attainment of an individual, ranging from primary school education to PhD programmes. The educational pathways that leads to the 8 most common educational categories in 2019 are illustrated in Figure 1 below. The majority of Danes either complete their formal education after primary school at age 15 (28.8%) or specialize within a trade (31.1%), in occupations such as social and health care assistants and carpenters.

Figure 1: Educational Pathways in the Danish Education System

Notes. Figure illustrating the most natural progression of individuals through the educational system in Denmark in 2019. Progression may require additional specific requirements dependent on the field of study.

Because the literature documents that both educational length and field of study have direct implications for one's financial decision-making (Altmejd, 2025; Fagereng et al., 2020), we proceed to map the educational codes into one of 21 fields of study, as shown in Table 1 below.

Table 1: Distribution of Individuals across Education Area and Category

Field of education	Bach.	Mast.	PhD	Prep.	Prim.	Qual.	Short	U. Sec.	VET	Voc. B.	Total
Agriculture, forestry, fishing	950	9,079	2,246	0	0	0	5,428	0	59,907	2,182	79,792
Arts	8,087	10,036	776	0	0	0	5,414	0	2,673	6,132	33,118
Audio-visual / media	0	0	0	0	0	0	7,861	0	26,468	3,369	37,698
Building / civil engineering	1,795	19,067	1	0	0	0	5,772	0	138,175	28,927	193,737
Business economics, admin., law	27,638	86,984	5	0	0	0	53,887	0	517,547	46,801	732,862
Engineering / industrial production	2,780	27,723	6,097	0	0	1,874	49,397	0	159,414	47,400	294,685
Human sciences	19,141	54,710	2,844	0	0	0	8,874	0	0	11,606	97,175
ICT	2,999	17,441	0	0	0	0	11,331	0	6,936	4,662	43,369
Mechanics / metal	234	2,719	0	0	0	0	11,860	0	228,092	11,687	254,592
Military, police, security	0	4,020	0	0	0	0	19,743	0	3,016	3,630	30,409
Preparatory courses	0	0	0	4,570	0	0	0	0	0	0	4,570
Primary school	0	0	0	0	1,417,349	0	0	0	0	0	1,417,349
Science	6,796	33,793	5,147	0	0	0	321	0	0	861	46,918
Services	30	1,133	0	0	0	0	5,533	0	90,864	2,120	99,680
Social and health	6,825	45,712	7,832	0	0	0	14,267	0	183,458	328,228	586,322
Social sciences	15,936	62,572	3,036	0	0	0	650	0	0	17,071	99,265
Subject not specified	151	35,740	6,607	0	0	0	15,768	0	22,904	37,397	118,567
Teaching and learning	1,154	16,207	10	0	0	0	440	0	386	131,924	150,121
Transport	0	0	0	0	0	0	2,462	0	22,954	3,634	29,050
Unknown	0	0	0	0	15,294	0	0	0	0	0	15,294
Upper secondary, general	0	0	0	0	0	755	0	422,853	0	0	423,608
Total	94,516	426,936	34,601	4,570	1,432,643	2,629	219,008	422,853	1,462,794	687,631	4,788,181

Notes. Counts of individuals classified jointly by field of education (rows) and education category (columns) in 2019. Column abbreviations: Bach. = Bachelor's programmes; Mast. = Master's programmes; PhD = PhD programmes; Prep. = Preparatory courses; Prim. = Primary education; Qual. = Qualifying educational programmes; Short = Short-cycle higher education; U. Sec. = Upper secondary education; VET = Vocational education and training; Voc. B. = Vocational bachelor's educations.

Financial Literacy

Building on the categorization in Table 1, we construct three categories of financial knowledge that aim to capture the two channels through which we hypothesize education affects financial decision-making. First, higher educational attainment may improve an individual's ability to acquire information and learn, allowing them to identify optimal financial decisions more efficiently. Second, certain fields, particularly the quantitative ones, may lead to a higher degree of financial literacy as part of the curriculum.

We define three categories of financial literacy: low (0), medium (1), and high (2). Fagereng et

al. (2020) finds that the length of education has a causal effect on the returns an investor obtains on their wealth. We therefore create our low and medium categories based on educational length, using the two extremes available in the data, Primary School at the bottom and Master's/PhD at the top. As seen in Table 2, PhD programs most cleanly identify the upper extreme, however the number of observations is limited, so we cluster them with Master's programs, which is also the fourth-largest educational category in the dataset. For the third category, we intend to capture the field of study effect on financial literacy. We use the field of *Business Economics, Administration and Law* for two reasons. First, it is the category that most uniquely identifies a group of individuals whose curriculum provides direct exposure to financial decision-making. Second, this closely aligns with the literature's categorization of financial literacy to isolate causal effects on financial decision-making, both in terms of portfolio returns and participation in financial markets (Altmejd, 2025; Fagereng & Halvorsen, 2017).

Table 2: Mean Years of Schooling by Education Category

Education category	Mean years of schooling
1. PhD programmes	21.00
2. Master's programmes	18.07
3. Vocational bachelor's educations	16.28
4. Bachelor's programmes	16.03
5. Short-cycle higher education	15.36
6. Vocational education and training	13.92
7. Upper secondary education	13.02
8. Qualifying educational programmes	13.01
9. Preparatory courses	12.00
10. Primary education	9.49
Mean	13.38

Notes. Mean number of years of formal schooling in each of the ten education categories used to construct the financial-literacy variable.

We define an individual to have a low (0) degree of financial literacy if their highest attained education is Primary School. An individual is assigned a medium (1) degree of financial literacy if their highest attained education is a Master's or PhD programme outside the field of Business Economics, Administration and Law. Finally, an individual is assigned a high (2) degree of financial literacy if their highest attained education is a Master's or PhD programme within the field of Business Economics, Administration and Law.⁶ In summary:

⁶We refer to the field as Business or Economics from here on.

$$\text{Financial Knowledge} = \begin{cases} 0 & \text{if Education} = \text{Primary School} \\ 1 & \text{if Education} = \text{Master/PhD, Field} \neq \text{Business Economics} \\ 2 & \text{if Education} = \text{Master/PhD, Field} = \text{Business Economics} \end{cases} \quad (44)$$

To quantify the effect of having a financially literate parent, we apply the same classification to each individual's parents. For households with two adults, this requires assumptions about how financial knowledge is shared within the household. We assume full spillover of financial knowledge between partners: if at least one adult in a household has a high degree of financial literacy, the entire household is classified as high. By the same logic on the parental side, a household with two adults has up to four parents from whom financial knowledge can be transferred intergenerationally, and the household is assigned the maximum financial literacy across all of them. Under this specification, a household with one business graduate receives the same financial literacy classification as a household with two business graduates, and a household with one parent who graduated in business receives the same parental financial literacy classification as a household with four such parents.

Thus, we classify each household in our dataset with a measure for financial literacy (e, p) , where e refers to the household head's education and p the longest education among the parents. As such, our "low-low" category corresponds to a household with a primary graduate head and a primary graduate parent: $(0,0)$. Meanwhile, our "high-high" category defines a household with a business graduate head and a business graduate parent: $(2,2)$.

4.4 Sample Restrictions

The full sample over the period 2014 to 2024 contains 68,066,158 observations, corresponding to the entire Danish population. We first restrict the dataset to adults above the age of 24 and then collapse observations on the household identifier. Two adults are considered partners, in which case we combine their income and assets, if Statistics Denmark classifies them as married, cohabiting, or registered as an official couple. As described in the previous subsection, we retain only households whose head, defined as the individual with the longest education, falls into one of our three financial literacy categories: Primary School, Master's, or PhD. This leaves us with 10,177,279 households, or roughly 900,000 households per year.

We next remove outliers that may distort the variables on which our analysis depends. Table 3 reports the distribution of equivalent disposable income separately for households whose head has reported business income and those who have not. The data exhibit extreme dispersion in both tails, with both the 1st and 99th pseudo-percentiles being significantly larger in absolute terms for the business income households. We argue that such magnitudes are more likely to reflect non-recurring

events such as large capital gains, business sales, one-time asset disposals, or loss carryforwards, rather than the household's underlying economic resources. Although these households are not measurement errors per se, retaining them would distort estimates of life-cycle income profiles and bias regressions. Furthermore, the standard life-cycle portfolio choice framework (Cocco et al., 2005) treats labor income as exogenous to financial decisions. This assumption is violated by business owners, who jointly determine wages, retained earnings, and portfolio allocation. We therefore exclude individuals with business income (8.6% of the full sample) from the analysis.

Table 3: Income Distribution by Business-Income Status

	Mean	p1 [†]	p10 [†]	p50 [†]	p90 [†]	p99 [†]
No business income	266,490	1,045	123,692	210,825	453,366	889,844
Business income	485,419	−3,106	145,196	343,651	831,190	2,765,516

Notes. Means and pseudo-percentiles of equivalent disposable income (DKK). [†] Pseudo-percentiles (averages of ± 100 adjacent observations).

Excluding business-income households removes the upper-tail problem but leaves the lower tail largely intact. As Table 4 shows, the bottom 1% of equivalent disposable income remains at DKK 1,045 even after the exclusion, which is implausibly low for a Danish working-age household. Our aim is to retain genuinely low-income households without including households with partial residence or under-reporting to the tax authorities. We also require that income be non-negative, since we model the income-age pattern with the natural logarithm.

Table 4: Income Distribution & Trimming

	Mean	p1 [†]	p10 [†]	p50 [†]	p90 [†]	p99 [†]
Income, Pre-trim	260,442	1,045	122,863	208,179	444,422	845,755
Income, Post-trim	258,517	102,705	139,102	213,494	439,097	700,885

Notes. Mean and pseudo-percentiles of equivalent disposable income (DKK) prior to and after trimming the bottom 5% and top 1%. [†] Pseudo-percentiles (averages of ± 100 adjacent observations).

A natural lower bound for household income is the level of government transfers: households with disposable income below this floor are most likely to fall into the categories described above. Denmark's Flexicurity system pairs flexible labor-market regulation with income insurance via two transfer schemes that define the relevant floors (Bjørn, 2023). Members of an A-kasse, the unemployment insurance fund to which the majority of Danes belong, receive up to DKK 19,728 per month at 2023 rates. Uninsured individuals receive support called kontanthjælp, which ranges from DKK 7,095 to DKK 13,060 per month for a single adult. Applying a 40% effective tax rate, these schemes correspond to annual disposable incomes of approximately DKK 142,000 (A-kasse), DKK 94,000 (high

kontanthjælp), and DKK 51,000 (low kontanthjælp). We trim the disposable income distribution at the fifth pseudo-percentile, which corresponds to DKK 96,481 per year and sits between the high kontanthjælp floor and the A-kasse floor.

At the upper tail we trim the top 1%, consistent with the existing literature (see, e.g., Hubbard et al. (1995) and Lusardi et al. (2017)). Our final sample comprises 8,490,899 household-year observations over the 11-year period, or roughly 660,000 households per year.

4.5 Descriptive Statistics

Full descriptive statistics of all relevant variables for the remainder of the analysis are reported in Table A.1 in the Appendix. The average household in our sample contains 1.79 individuals, of which 1.46 are adults; 49% of household heads are male and 31% of households are married. The household head, defined as the individual with the longest education, has on average 12.72 years of schooling, roughly three years longer than the standard duration of primary education in Denmark.

Education, Financial Literacy, and Intergenerational Knowledge Transfer

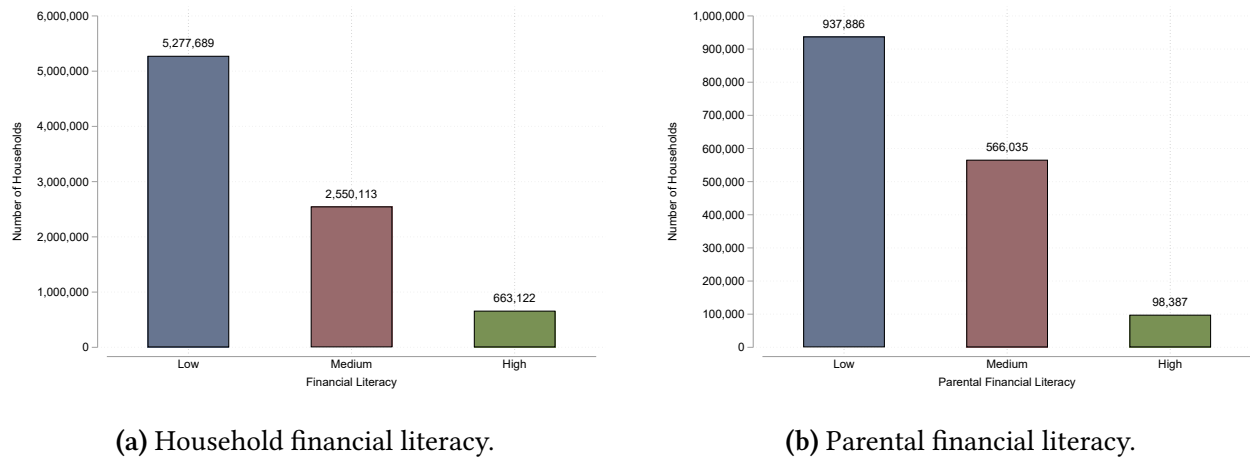
The classification of households into our three financial literacy categories produces a notably uneven distribution. Figure 2 displays the marginal distributions of household financial literacy and parental financial literacy, and the asymmetry is evident: the majority of households fall into the low-literacy category at both the individual and the parental level. Of our 8,490,946 sample observations, 5,277,680 (62.2%) have a household head whose highest education is Primary School, 2,550,134 (30.0%) have a head with a Master's or PhD outside Business Economics, and 663,132 (7.8%) have a head with a Master's or PhD within Business Economics. For parental literacy, the sample is unfortunately rather restrictive as the majority of households with a household head above the age of 50 have no educational data on their parents. Nonetheless, the restricted sample allows us to illustrate the distribution, which is quite similar to that of the households.

The fact that educational data are missing for the parents may also suggest the historical expansion of higher education in Denmark. The cohorts that constitute the parents of our sample completed their education during a period in which high educational attainment was less common than it is today, so the probability of having a parent with a Master's or PhD is low.

The joint distribution of household and parental financial literacy reveals strong intergenerational correlation cf. Table 5. Among households with a low-literacy head, 93.6% have parents classified in the same category while only 0.7% have business economist parents. For the business economists, 18.4% have parents that are business economists themselves and 75% of the parents have at least a master's degree. This pattern is consistent with the substantial intergenerational persistence in education documented for Denmark by Landersø and Heckman (2017), and also with Altmejd (2024), who finds that children are 73% more likely to graduate from the same field of study

as their parent, with the strongest effects in business, engineering, medicine, and the social sciences. Because our high-literacy category is defined by a specific field rather than by attainment alone, the field-inheritance channel is particularly relevant for our setting.

Figure 2: Distribution of Financial Literacy in the Sample



Notes. Panel (a) reports the frequency of households' own education cf. eq. (44). Panel (b) reports the frequency of parental education in the same categories.

Table 5: Joint Distribution of Own and Parental Financial Literacy

Own Financial Literacy	Parental Financial Literacy		
	Primary	Master's	Business
Primary	93.6	5.6	0.7
Master's	21.7	68.7	9.6
Business	24.2	57.4	18.4

Notes. Each cell reports the percentage distribution of parental financial literacy within each household-head financial literacy category. Rows sum to 100 percent. Primary refers to primary education, Master's to Master's/PhD outside of Business Economics, and Business to Master's/PhD in Business Economics.

The unbalanced distribution carries implications for the empirical strategy. The high-literacy category contains roughly 663,000 household-year observations, providing sufficient statistical power for estimating coefficients at the individual literacy margin. The combination of high household and high parental literacy, however, contains only 34,836 observations, so specifications that interact the two will identify their joint effect from this smaller subsample.

Life-Cycle Income and Wealth Profiles

Considering the heuristic argumentation for the parental effect on financial knowledge above, it would be natural to assume that intergenerational knowledge transfer spills over into all facets of a household's life: education, income, returns, and wealth. In this paper, however, we assume that parental education has no direct effect on the income attainable by a household, conditional on the child's own education. Parental education affects child outcomes only through three channels: the educational attainment of the child, their starting wealth, and their attainable returns, the latter two of which we describe in detail in Section 6.1.

We do not claim that parental background does not affect the educational paths of their children. As is evident from Table 5 they do. However, we claim that once we condition on the child's own education, we treat any further parental influence on income as operating through wealth and returns, or through channels that lie outside our model such as professional networks and referrals. Effectively, this assumes that a business graduate household with primary-educated parents has the same income distribution as a business graduate household with business graduate parents, conditional on being business graduates. Or at least, that any remaining differences are absorbed by starting wealth, returns, or unmodeled network effects.

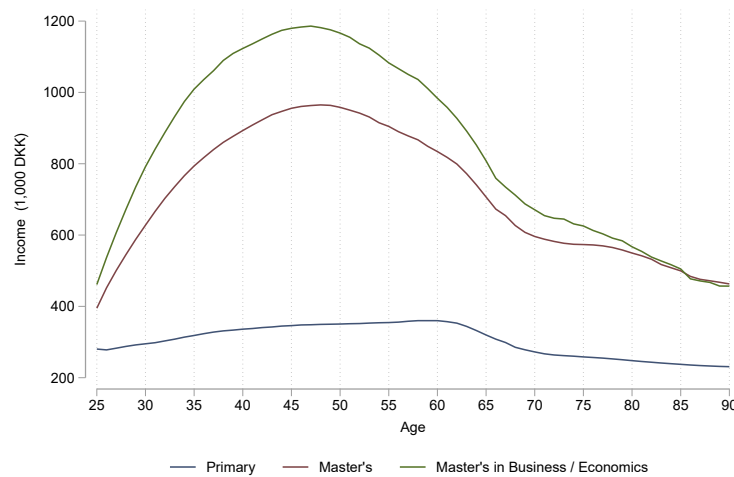
We report the relevant densities in Figure B.1 in the Appendix which supports this restriction. Within each of the three educational categories, the income densities across parental types overlap substantially. The pattern is cleanest for the Master's and Business Economics groups. The Primary School panel exhibits a sharper, narrower distribution, which reflects both the compression of low-end incomes as a result of the Danish transfer system and the trimming we impose at the fifth percentile of the income distribution.

In Figure 3, we report the non-parametric income profile (a) as well as wealth-to-income ratios for financial (b) and net assets (c). The income profiles in Panel (a) display the characteristic hump shape documented widely in the life-cycle literature (Cocco et al., 2005; Fagereng & Halvorsen, 2017; Gourinchas & Parker, 2002). The level differences across educational groups are economically large. Households with a head holding primary school as their highest education exhibit a nearly flat income profile, peaking at around DKK 360,000 at age 60. By contrast, households with a Master's degree reach a peak of approximately DKK 950,000 in their late 40s, and households with a Master's in Business Economics peak at DKK 1.2 million at the same age. Notably, this is household income and as such, it does not account for household composition. As shall become clear later, equivalent disposable household income peaks much later, as children leaving the household push this income measure up as households reach their late 40s. The income difference is present in both the level and the slope: more educated households see steeper income growth in early adulthood and an earlier peak. After retirement, the profiles converge, reflecting that pension benefits and government transfers compress the cross-sectional dispersion in income relative to the working years. Because our model assumes a constant replacement rate after retirement, the analysis that follows

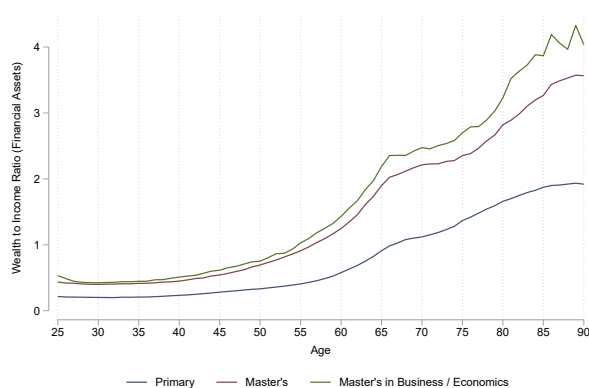
focuses primarily on the working years, where heterogeneity is most pronounced and economically meaningful for portfolio decisions.

The wealth-to-income ratio in Panel (c), which uses total net assets, is consistent with the life-cycle hypothesis of Modigliani and Brumberg (1954): households accumulate wealth through their working years, with the ratio rising steadily and stabilizing in retirement. By age 60, just before the typical retirement age, Master's and Business Economics households hold roughly ten times annual income in net wealth, while Primary School households hold around four times. Initially, it may seem surprising that the wealth-to-income ratio for net assets does not adjust downwards as the households reach the later stages in life. One would assume that households downsize their house, sell off assets and run down savings while they enjoy the last years of their life.

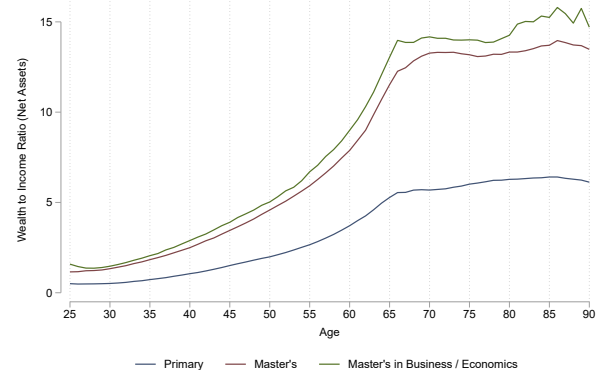
Figure 3: Life-Cycle Income and Wealth-to-Income Profiles by Education



(a) Household income.



(b) Wealth-to-income ratio: financial assets.



(c) Wealth-to-income ratio: total net assets.

Notes. Panel (a) plots household income (in 1,000 DKK) by age. Panel (b) plots the ratio of financial assets to household income. Panel (c) plots the ratio of total net assets, including housing equity, to household income.

As we see in Panel (a), however, all educational groups exhibit decreasing income which perfectly offsets the decrease in total net assets of the households.⁷ The discrete jump around retirement across all three groups reflects the drop in the denominator at retirement. However, the steady increase in the wealth-to-income ratio for financial assets indicates that households are indeed selling off their illiquid assets. The financial wealth-to-income ratio is also significantly lower across all types of households. Even business graduate households hold less than one year of income in financial assets until their mid-50s, and by age 65 they hold approximately two and a half years. Master's households hold approximately two years at the same age, while Primary School households hold roughly one year. These low levels are consistent with the wealthy-hand-to-mouth phenomenon described by Kaplan et al. (2014), in which households exhibit very low wealth-to-income ratios when illiquid wealth is excluded.

The contrast between the two panels also reflects the fact that the bulk of Danish household wealth sits outside liquid financial markets, primarily in housing and in the occupational pension system. The Danish retirement system rests on a universal public pension (*folkepension*), an occupational component effectively made mandatory through industry-wide collective bargaining agreements, and voluntary private savings. The occupational pillar is well organized with defined-contribution schemes covering 76% of the working population in 2006, with typical contribution rates of 9% for blue-collar workers, 15% for white-collar workers, and 12% for public-sector employees (OECD, 2009). Because contributions to the occupational pillar accumulate inside pension accounts that are not captured by our financial-asset measure, households that appear to hold modest financial wealth in the registry data may be holding considerably more total retirement wealth once pension claims are included.

As noted by Campbell et al. (2003), the choice of asset measure is an empirical difficulty in the literature. In this paper, we define wealth as *financial assets* and as such, the figures in 3b and B.2a are the appropriate measures. We do this for two reasons: first, our theoretical framework concerns the choice between risky and safe financial instruments, for which housing is not a relevant comparison asset. Second, using housing as a calibration measure would necessitate an inclusion of housing choice in our model. As we seek to isolate the effect of financial literacy on income and financial market behavior, we believe including this in our model would make our findings less clear. We acknowledge, however, that financial assets understate true retirement wealth because the occupational pension pillar is not observed, and we return to the implications of this measurement choice in Sections 7 and 10

Financial Market Participation and Portfolio Composition

The heterogeneity across the educational groups is not limited to the income and wealth distributions across time. As seen in Figure 4, there is also an evident discrepancy in the participation rates

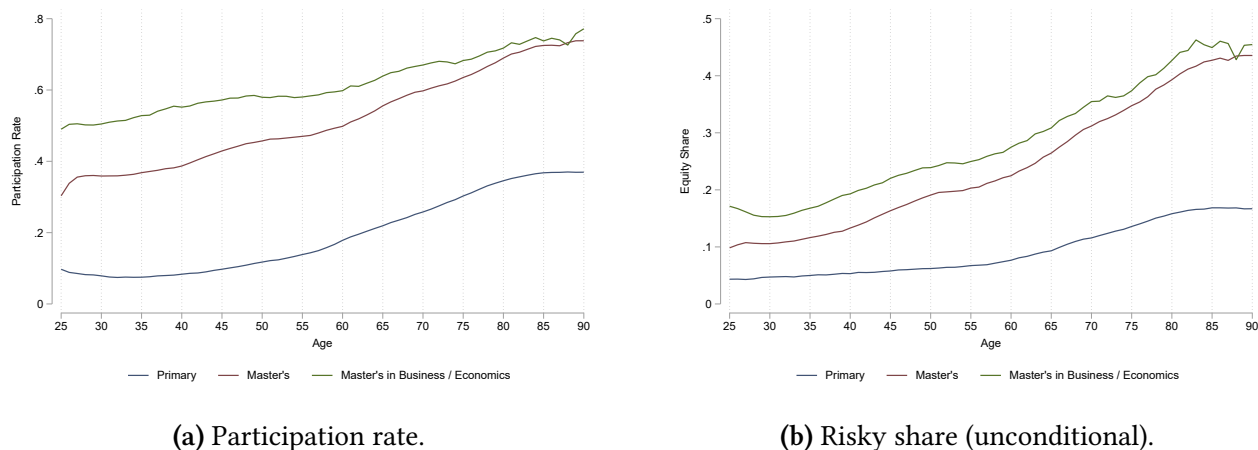
⁷See Figure B.2 for documentation of this behavior.

and the unconditional risky share of the households.

The discrepancy is most apparent along the extensive margin. Among Primary School households, participation begins below 10% in early adulthood and rises only gradually to roughly 25% by retirement. Master's households start at roughly 30% participation and reach 45% by retirement, while Business Economics households start at 45% participation already at age 25 and exceed 60% by retirement. At age 35, a household head with a Master's in Business Economics is roughly six times more likely to participate in financial markets than one with only Primary School.

The unconditional risky share in Panel (b) reveals a similar pattern. Over the life cycle, the average risky share rises from roughly 5% to 17% for Primary School households, from 10% to 44% for Master's households, and from 17% to 46% for Business Economics households. The vertical distance between Master's and Business Economics households is smaller in Panel (b) than in Panel (a), suggesting that the field-specific component of financial literacy operates primarily on the extensive margin: business economists are more likely to enter financial markets at all, but conditional on participating, the share they allocate to risky assets is similar to that of other Master's holders. This is consistent with the broader finding in the literature that education discrepancies in portfolio choice are stronger at the participation margin than at the conditional risky share (Calvet et al., 2007).

Figure 4: Financial Market Participation and Risky Share by Education and Age



Notes. Panel (a) plots the participation rates of the households over the life cycle. Panel (b) reports the unconditional equity share.

Taken together, the descriptive evidence in Figures 3 and 4 establishes three facts that motivate the empirical analysis in the next section. First, there are pronounced differences in the level and slope of income profiles across educational groups, with Business Economist households exhibiting the steepest growth and highest peaks. Second, these income differentials translate into even larger wealth differentials in net terms, although they are substantially lower when measured against financial assets alone, reflecting the importance of housing and occupational pensions in the Danish

wealth distribution. Third, the educational difference in financial-market participation is substantially stronger than the difference in the conditional risky share, indicating that the primary margin of adjustment is whether households enter financial markets at all. In the sections that follow, we attempt to rationalize this behavior and model its welfare implications for Danish households.

5 Model

Building on the theoretical framework of section 3, we now specify the life-cycle model used to create our benchmark.

Households receive labor income of Y each year. They choose consumption C and a risky portfolio share α , they hold an according amount of end-of-period assets A , which is bounded by a no-borrowing constraint. Lifetime is assumed to start at the age of 25 ($t = 0$) with certain death at the age of $T = 100$, while there is a survival probability each year of ϕ_{t+1} to survive to the next year.

As described in section 4.3, we assign households a level of financial knowledge based on their own level of education:

$$\text{Financial Knowledge} = \begin{cases} 0 & \text{if Education} = \text{Primary School} \\ 1 & \text{if Education} = \text{Master/PhD, Field} \neq \text{Business Economics} \\ 2 & \text{if Education} = \text{Master/PhD, Field} = \text{Business Economics.} \end{cases} \quad (45)$$

Households are ex-ante heterogeneous in two dimensions: across the three own-education groups they differ in income growth G_t , income risk $V[Y_t]$, and replacement rates μ ; across another three parental-education groups, we have nine groups that differ in excess returns τ and starting wealth A_0 . All of them, however, share identical CRRA preferences to isolate the effect of educational heterogeneity.

The labor income during working life follows the permanent-transitory specification of Carroll (1997). For an agent with education level e , income is described by

$$\begin{aligned} Y_t^e &= P_t^e V_t^e, \\ P_t^e &= G_t^e P_{t-1}^e N_t^e, \end{aligned} \quad (46)$$

where permanent income P_t^e evolves through a deterministic age profile G_t^e and a permanent shock N_t^e , while V_t^e is a transitory shock. Both shocks are log-normally distributed with unit mean,

$$N_t^e \sim \log \mathcal{N}\left(-\frac{1}{2}\sigma_N^{2,e}, \sigma_N^{2,e}\right), \quad V_t^e \sim \log \mathcal{N}\left(-\frac{1}{2}\sigma_V^{2,e}, \sigma_V^{2,e}\right), \quad (47)$$

and are independent of each other and across time. Groups differ in both the deterministic profile G_t^e and the shock variances $\sigma_N^{2,e}$ and $\sigma_V^{2,e}$, capturing the fact that higher-educated households face

steeper but more volatile income paths.

Upon retirement at age T_{ret} , labor income is replaced by a fixed fraction of pre-retirement permanent income,

$$Y_t^e = \mu^e P_{T_{\text{ret}}-1}^e, \quad t \geq T_{\text{ret}}, \quad (48)$$

where the replacement rate μ^e varies by own education level e only. This is implemented by absorbing μ^e into G_t^e and setting $N_t^e = V_t^e = 1$ for all $t \geq T_{\text{ret}}$, which guarantees that households do not face income risk during retirement.

The excess return on the risky asset is defined as $R_{t+1} - \bar{R}^f = \tau + \eta_{t+1}$, where $\eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$ is a return shock common to all groups. Just as in the theory section 3, x_t denotes normalized cash-on-hand, with lowercase variables normalized by permanent income. The maximization problem can then be expressed as a Bellman equation, where $\gamma = (e, p)$ represents the group categorization:

$$\begin{aligned} W^\gamma(x_t) &= \max_{a_t, \alpha_t} \left\{ \frac{c_t^{1-\theta}}{1-\theta} \right. \\ &\quad \left. + \beta \phi_{t+1} \int_\eta \int_N \int_V (G_{t+1}^e N_{t+1})^{1-\theta} W^\gamma(x_{t+1}) dF_e(V) dF_e(N) dF(\eta) \right\} \\ x_{t+1} &= \frac{R_{t+1}^{p,\gamma} a_t}{G_{t+1}^e N_{t+1}} + V_{t+1} \\ R_{t+1}^{p,\gamma} &= \bar{R}^f (1 - \alpha_t) + (\bar{R}^f + \tau^\gamma + \eta_{t+1}) \alpha_t \\ a_t &= x_t - c_t \geq 0 \\ a_0^\gamma &\text{ given} \\ 0 &\leq \alpha_t \leq 1, \end{aligned} \quad (49)$$

where lowercase variables denote quantities normalized by permanent income, following the normalization introduced in section 3.5. That is, a household starts with initial assets a_0^γ (lowercase since normalized by permanent income) and receives labor income, which it can either consume or invest. By choosing consumption c_t , the residual $a_t = x_t - c_t$ is invested in the risky or the risk-free asset. Hence, the household chooses the fraction to invest in the risky asset, α , and the fraction to invest in the risk-free asset $(1 - \alpha)$, which yields the portfolio return at any period t : $R_{t+1}^{p,\gamma}$, where the no-short-sales condition $0 \leq \alpha \leq 1$ is satisfied. The joint decision of α and a_t thus determines the cash-on-hand in the next period, x_{t+1} .

We solve the model by backward induction using the Endogenous Grid Method (EGM).⁸ The asset grid contains 330 points and the risky share grid 210 points, both curved to concentrate nodes at low levels where policy functions are most nonlinear due to the binding borrowing constraint. Income shocks are approximated via Gauss-Hermite quadrature with 9 nodes each and the return

⁸See Section 3.6 for a full exposition.

shock with 5 nodes.⁹

6 Calibration and Parametrization

In this section we outline our approach for the calibration of the heterogeneous variables: labor income process, return process, replacement rates and initial wealth of the households. Furthermore, we justify the choices of parameters for risk aversion, discount factors, risk-free rates and mortality rates.

6.1 Labor Income Process

A naive approach to calibrating the income process would be to use the non-parametric age profile of income shown in Figure 3a. The interpretation of such a profile, however, is ambiguous. Consider the income growth observed between ages 49 and 50 in our data. This growth could reflect a genuine age effect, in which households earn more as they accumulate experience, a cohort effect, in which the specific cohort turning 50 during our sample window had systematically different conditions throughout their lifetime, or a time effect, in which macroeconomic conditions raised earnings for all 50-year-olds in a particular year. Disentangling these three components matters for the results of the analysis. As we shall see however, that process is not trivial.

The ACT Identification Problem

The fundamental obstacle, as established by A. Deaton (1985), is that age, time, and cohort effects cannot be jointly identified from the data. The reason is the linear identity: $\text{birth year} = \text{year} - \text{age}$, which means that knowing any two of the three variables determines the third. Identification therefore requires the researcher to impose a restriction on at least one of the three components, and Ameriks and Zeldes (2004) document that the choice of restriction can produce qualitatively different life-cycle profiles from identical data.

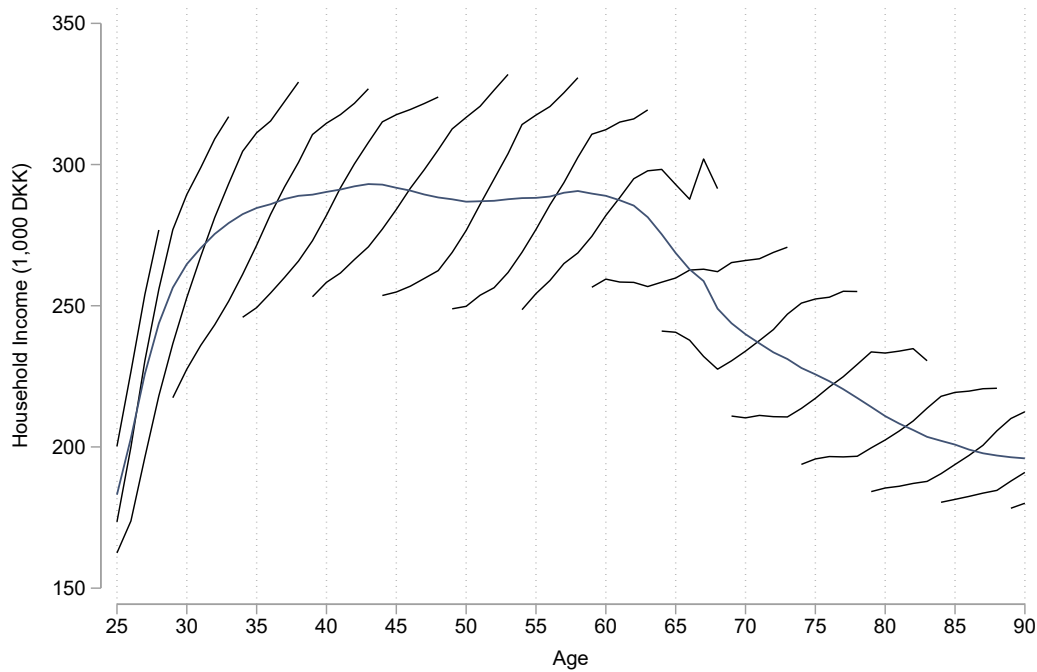
Figure 5 illustrates the empirical magnitude of the cohort problem in our sample. Each black line plots mean household disposable equivalent income by age for a five-year birth cohort observed over the 2014–2024 period, while the blue line shows the pooled cross-sectional age profile across all cohorts. The cross-section appears relatively flat between ages 35 and 60, but the within-cohort lines reveal income growth across these ages within any given cohort. The apparent flatness of the cross-section arises because older ages are populated by cohorts with systematically lower lifetime incomes. At post-retirement ages, the opposite bias emerges: the cross-section overstates the income decline relative to the within-cohort profiles. The slope of the cross-sectional age profile

⁹See Appendix C for details.

therefore combines true age effects with cohort effects and cannot be interpreted as the life-cycle income path that any particular household experiences.

Omitting time effects from the regression specification, on the other hand, would imply that the increase in income from turning 50 in 2014 is identical to that of turning 50 in 2024. The literature handles the issue in different ways: while Campbell et al. (2003) argue that the most natural assumption for empirical work is that cohort effects are absent, Lusardi et al. (2017) excludes time effects, at the cost of attributing macroeconomic variation over the sample period to the age-cohort profile. Given Ameriks and Zeldes (2004)'s findings on the importance of this specification choice, we estimate both versions and compare the results.

Figure 5: Income Heterogeneity by Birth Cohort



Notes. The figure depicts the presence of cohort effects in the sample.

Estimation of Income-Age Profiles

We follow the established literature in regressing the log of household income on a polynomial in age while controlling for cohort and time effects, respectively (Gourinchas & Parker, 2002; Hubbard et al., 1995; Lusardi et al., 2017). We estimate the regression separately for each financial literacy group, and we consider both a third-degree and a fifth-degree specification of the age polynomial:

$$\log y_{i,t} = \alpha + \sum_{k=1}^K \beta_k \cdot \text{age}_{i,t}^k + \sum_c \gamma_c \cdot \mathbb{1}\{\text{cohort}_i = c\} + \varepsilon_{i,t} \quad (50)$$

$$\log y_{i,t} = \alpha + \sum_{k=1}^K \beta_k \cdot \text{age}_{i,t}^k + \sum_{t=2014}^{2024} \delta_t \cdot \mathbb{1}\{\text{year}_t = t\} + \varepsilon_{i,t} \quad (51)$$

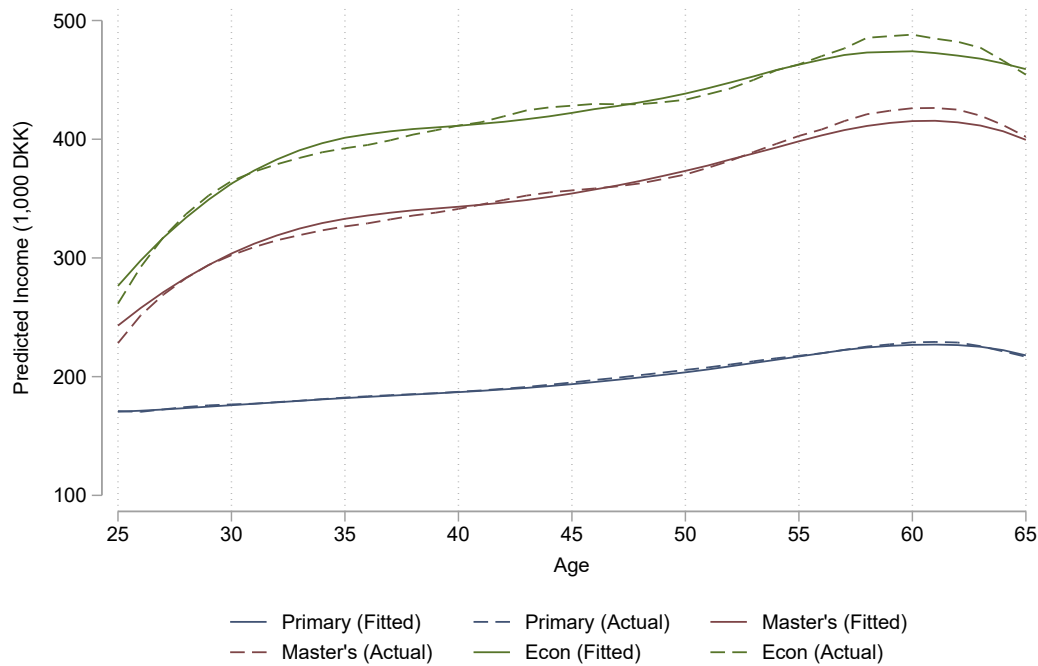
where $K \in \{3, 5\}$ denotes the polynomial degree, c indexes five-year birth cohort bins, and t indexes years. All four specifications have relatively strong explanatory power across age coefficients, are highly significant for both time and cohort effects, and yield empirical fits that are close to the data.

We adopt the fifth-degree time-effects specification as our benchmark for two reasons. First, while we do not wish to take a stance on the relative magnitude of cohort and time effects in our sample period, it seems natural to include time effects given that the period contains the Covid-19 pandemic. Second, the fifth-degree polynomial captures the hump shape just before retirement that the third-degree polynomials miss. The coefficients of our preferred model and its empirical fit are presented in Table 6 and Figure 6 below while the cohort specification and the remaining empirical fits are reported in Appendix Table A.2 and Figures B.3, B.4, and B.5.

Table 6: Income-Age Profiles by Financial Literacy: Time-Effects Specification

	Primary		Master's		Business	
	(1) Third	(2) Fifth	(3) Third	(4) Fifth	(5) Third	(6) Fifth
<i>Panel A: Age polynomial</i>						
Age	−0.046*** (0.001)	−0.378*** (0.025)	0.120*** (0.001)	0.860*** (0.041)	0.184*** (0.002)	1.576*** (0.069)
Age ² × 10 ²	0.125*** (0.002)	2.050*** (0.117)	−0.231*** (0.003)	−2.940*** (0.196)	−0.371*** (0.006)	−6.240*** (0.330)
Age ³ × 10 ⁴	−0.092*** (0.001)	−5.370*** (0.271)	0.156*** (0.002)	4.560*** (0.457)	0.252*** (0.004)	12.100*** (0.777)
Age ⁴ × 10 ⁶	—	6.860*** (0.306)	—	−2.900*** (0.522)	—	−11.400*** (0.895)
Age ⁵ × 10 ⁸	—	−3.410*** (0.136)	—	0.386* (0.234)	—	4.150*** (0.404)
Constant	12.510*** (0.012)	14.641*** (0.201)	10.575*** (0.020)	3.119*** (0.338)	9.829*** (0.033)	−2.830*** (0.560)
<i>Panel B: Specification and fit</i>						
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Cohort effects	No	No	No	No	No	No
Observations	2,861,609	2,861,609	2,160,761	2,160,761	609,816	609,816
Adj. R ²	0.107	0.108	0.097	0.100	0.121	0.126

Notes. The dependent variable is log labor income. Each column reports pooled OLS estimates from a separate regression of log income on a polynomial in age and a full set of year dummies (2014–2024, with 2019 as the omitted base). Errors are clustered. Year-dummy coefficients are estimated but suppressed for compactness. “Third” columns use a third-order polynomial in age; “Fifth” columns use a fifth-order polynomial. Higher-order age coefficients and their standard errors are reported on the scale indicated in the row label. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Predicted Income-Age Profile, Fifth-Degree Time-Effects Specification

Notes. Predicted income-age profiles from using coefficients presented in Table 6 with a fifth-degree polynomial and time effects against their actual empirical observations.

Having estimated the deterministic age profile, we decompose the residual variation into permanent and transitory income shocks using the framework developed by Carroll and Samwick (1997), derived in appendix C. The estimated variance components are reported in Table 7. Importantly, the permanent and transitory variance components are virtually identical across the four specifications, with each moving by less than 0.001 in any direction. The income-process moments that feed into our structural model are therefore robust to both the choice of polynomial degree and the choice between cohort and time effects.

A perhaps counterintuitive finding is that the permanent and transitory income variances are strictly greater for the well-educated groups than they are for the primary graduates. For the Master's graduates, we argue that this may stem from the fact that our classification groups all Master's graduates in the same cohort. Hence, variances increase because the income discrepancy between fields of study can be large. Furthermore, we acknowledge that the variances for the primary graduates may be too low, as we have excluded business owners. One could argue that it is a difficult path for a primary graduate to get a salary in the top 1%, unless they start their own business. This finding is, however, consistent with the pattern that Cocco et al. (2005) finds for the permanent income variances.¹⁰

¹⁰In fact, Cocco et al. (2005) results are more extreme in the variance difference between the least and most educated group implying a 61% difference.

Table 7: Permanent and Transitory Income Shocks Across Specifications

	<i>Time Effects</i>		<i>Cohort Effects</i>	
	Third	Fifth	Third	Fifth
<i>Permanent</i>				
Primary	0.0140	0.0139	0.0142	0.0139
Master	0.0181	0.0176	0.0183	0.0178
Econ	0.0168	0.0163	0.0172	0.0162
<i>Transitory</i>				
Primary	0.0051	0.0052	0.0051	0.0050
Master	0.0060	0.0061	0.0060	0.0060
Econ	0.0067	0.0069	0.0063	0.0062

Notes. Estimated variances of permanent and transitory income shocks. The **fifth-order** time-effects specification is our preferred model. Estimates are highly similar across specifications.

6.2 Return Heterogeneity

The labor income process pins down one source of heterogeneity across educational groups, but our analysis relies on a second channel: differences in financial knowledge accumulated over the life cycle, which we model as different attainable expected excess returns. We think of education as a lever that elevates an individual's ability to make better decisions in the stock market, either through an improved skill of acquiring relevant knowledge more efficiently (education length) or through studying in finance directly (field of study). We consider three potential approaches to quantifying this effect.

The first approach is to draw on existing estimates from the literature. As discussed in Section 2.2, it is well established that education, especially in quantitative fields, improves household participation rates in financial markets. More recent work, demonstrates that education length and field of study, in particular in business and economics, are causal explanations for higher attained returns to wealth and portfolios (Altmejd, 2025; Fagereng et al., 2020). We use calibrations from these papers as a sanity check for our own estimates below.

The second approach would be to estimate return heterogeneity directly as a function of education, portfolio holdings, and household parameters. Unfortunately, our dataset only reports realized gains on portfolio holdings, which is an imperfect measure of true portfolio returns. Realized returns capture only the gains that households choose to realize, often for tax-incentive purposes, to fund major birthdays or vacations, or to cover unexpected life events such as a car breaking down. The resulting estimates would be both noisy, because realization timing is endogenous to household-specific shocks, and biased upwards, because our dataset does not distinguish whether an increase in financial assets is due to portfolio returns or to shifting liquidity by acquiring new assets.

Risky Shares & Merton's Formulation

Obtaining excess returns directly from realized portfolio gains is, as discussed in the previous subsection, infeasible in our data. We therefore adopt an indirect approach. Under the classical portfolio-choice framework with CRRA utility, the optimal risky share is a known function of the excess return, the variance of the risky asset, and the household's risk aversion. Inverting this mapping at observed risky shares yields the implied excess return that rationalizes household behavior. The starting point is the Merton optimum (Merton, 1969); see Campbell et al. (2003) for the discrete-time analog:

$$\alpha^* = \frac{\mathbb{E}(\tilde{R}^e) - R_f}{\theta \cdot \sigma^2}, \quad (52)$$

which assumes lognormal returns, deterministic labor income, and a coefficient of relative risk aversion θ that is constant across the population. The deterministic income assumption, however, is at odds with our calibrated income process, in which labor income is subject to permanent and transitory shocks. We therefore use the formulation of Viceira (2001), who derives the optimal portfolio rule under stochastic labor income separately for retired and employed investors:

$$\alpha^* = \frac{\mu + \sigma_u^2/2}{\theta \cdot \bar{b}_1^r \cdot \sigma_u^2}, \quad (53)$$

$$\alpha^* = \frac{\mu + \sigma_u^2/2}{\theta \cdot \bar{b}_1^e \cdot \sigma_u^2} - \frac{\pi^e(1 - b_1)}{\bar{b}_1^e} \cdot \frac{\sigma_{\xi u}}{\sigma_u^2}, \quad (54)$$

where b_1 denotes the elasticity of consumption with respect to financial wealth and the second term in equation (54) captures the hedging demand induced by the covariance between labor-income innovations and equity returns, $\sigma_{\xi u}$.

Two assumptions allow us to use the Viceira framework while retaining a tractable inversion. First, we set $\sigma_{\xi u} = 0$, so the income-return covariance term vanishes. This simplifying assumption is backed up by the empirical evidence of Heaton and Lucas (1997), who find a very low correlation between income and return.

Second, we set $b_1^r = b_1^e = 1$. This collapses the Viceira correction to the classical Merton form, but does so under the explicit assumption that the elasticity of consumption with respect to financial wealth is unity on average over the life cycle and is the same during employment as in retirement. This is a strong assumption: Viceira's central insight is precisely that b_1 is generally below unity for working-age households, and setting it to one therefore understates the precautionary demand for the safe asset and, which raises the implied excess return. We retain $b_1 = 1$ for tractability but report a comprehensive robustness check in section 8 in which we vary b_1 across its empirically relevant range. Under these two simplifications, equations (53) and (54) both collapse to equation (52), which we invert to recover the implied excess returns:

$$\mathbb{E}(\widehat{\tilde{R}^e}) - R_f = \hat{\tau} = \hat{\alpha}_e^* \cdot \theta \cdot \sigma^2. \quad (55)$$

The remaining identifying assumption is that the coefficient of relative risk aversion θ does not vary systematically across our financial literacy groups. We adopt the standard assumption of homogeneous θ as our baseline and revisit it in the discussion section. Importantly, any group-level variation in true risk aversion that we miss will be loaded onto the implied excess return, biasing the estimated return difference in the same direction as the risk-aversion difference.

It is also worth being explicit about the interpretation of $\hat{\tau}$. The Merton inversion does not measure the excess return that households actually receive in equity markets, but rather the excess return that would rationalize the observed risky share under the assumed model. The two coincide only if households are rationally following the Merton optimum. Departures from the optimum, for instance due to fixed costs of participation, ambiguity aversion, or behavioral biases, will be absorbed into $\hat{\tau}$ rather than identified separately. We interpret $\hat{\tau}$ accordingly as the implied excess return under the model, and note that it may be biased in the presence of the effects described above.

Estimating Excess Returns

The Merton inversion takes the optimum α_e^* as input. We obtain it from a pooled OLS regression of the observed household risky share on financial literacy, parental financial literacy, their interaction, and a set of demographic and financial controls:

$$\alpha_{it} = \mathbf{X}_{it}'\boldsymbol{\beta} + \sum_{e=0}^2 \sum_{p=0}^2 \theta_{ep} \cdot \mathbb{1}\{\text{finlit}_i = e\} \times \mathbb{1}\{\text{finlitparent}_i = p\} + \epsilon_{it}, \quad (56)$$

where $\mathbf{X}_{it}$ contains gender, age, marital status, number of children, household income, financial assets, and the household head's primary school math GPA, along with year fixed effects. The inclusion of the primary school math GPA is the central identifying restriction: it absorbs the component of risky-share variation that is driven by inherent quantitative ability or interest, rather than by educational attainment or field of study effects. The estimated θ_{ep} coefficients therefore measure the residual effect of education itself on portfolio choice, holding both observable demographics and a proxy for innate quantitative ability fixed. We use pooled OLS rather than household fixed effects because financial literacy is time-invariant within our 11-year sample for almost all households. Identifying its effect from within-household variation would discard the comparison we care about. Coefficient estimates are reported in Table 8.¹¹

As expected, the coefficients on both household and parental financial literacy are positive and statistically and economically significant, and the interaction terms reveal a synergy between the two. The coefficient on the 9th-grade math test is also positive: the difference between an individual

¹¹See Table A.3 in the Appendix for the full table.

who received a 02 and a 12 in their 9th-grade math exam, the lowest and highest passing grades, corresponds to a predicted increase in the household risky share of five percentage points.

Table 8: Determinants of the Household Risky Asset Share

	(1)
<i>Panel A: Literacy and Innate Ability</i>	
Finlit = 1	0.0162*** (0.0018)
Finlit = 2	0.0410*** (0.0028)
Parental finlit = 1	0.0076*** (0.0016)
Parental finlit = 2	0.0241*** (0.0042)
9th Grade Math GPA	0.0050*** (0.0002)
<i>Panel B: Controls and Specification</i>	
Demographic controls	Y
Financial controls	Y
Interaction terms	Y
Year fixed effects	Y
Observations	362,750
Adj. R^2	0.192

Notes. The dependent variable is the household risky asset share. Column (1) reports OLS estimates with robust standard errors in parentheses. Demographic controls include gender, marital status, number of children, and age. Year fixed effects (2014–2024) are included with 2019 as the base year. Note that Finlit = 1 refers to Master’s/PhD outside of Business Economics and Finlit = 2 to Master’s/PhD in Business Economics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The full regression output can be found in Table A.3.

To make the magnitudes interpretable, Figure 7 and Table 9 report adjusted predictions of the risky share for each combination of household and parental financial literacy, evaluated for a 45-year-old at the sample mean of all other covariates. The predicted risky share rises monotonically along both dimensions. A household with low literacy and low-literacy parents is predicted to hold a risky share of 7.73%, while a household with high literacy and high-literacy parents is predicted to hold 16.24%, more than double. The synergistic effect noted above is also evident: increasing a household’s financial literacy from low to high raises the predicted risky share by 4.5 percentage points for households with primary-educated parents, but 6.09 percentage points for households with business economist parents. As Figure 7 shows, the lines are roughly parallel, indicating that own and parental financial literacy operate as additive rather than complementary determinants of

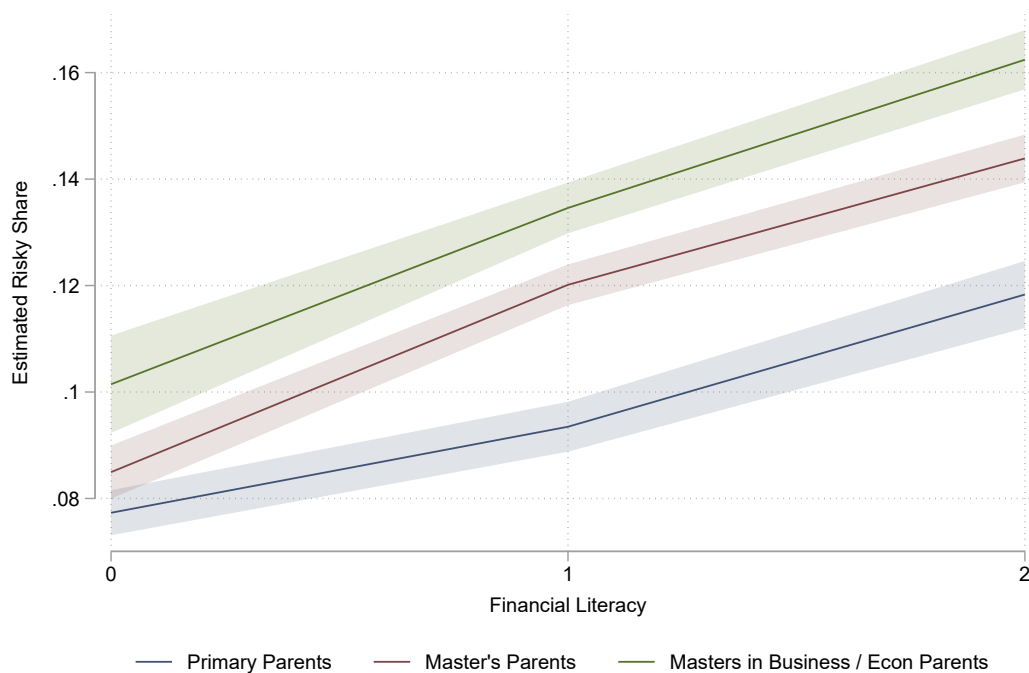
the risky share.

Table 9: Predicted Risky Share by Own and Parental Financial Literacy

Own Financial Literacy	Parental Financial Literacy			Average
	Primary	Master's	Business	
Primary	0.0773 (0.0022)	0.0850 (0.0025)	0.1015 (0.0047)	0.0845 (0.0023)
Master's	0.0935 (0.0024)	0.1201 (0.0019)	0.1346 (0.0024)	0.1133 (0.0020)
Business	0.1183 (0.0032)	0.1439 (0.0023)	0.1624 (0.0028)	0.1379 (0.0022)

Notes. Predicted household risky asset share from the `margins` command (linear model) on regression output in Table A.3, evaluated at age = 45 and at the mean values of all other covariates. Standard errors (delta method) in parentheses. The rightmost column shows the marginal (average) prediction for own financial literacy, averaging over parental financial literacy.

Figure 7: Predicted Risky Share by Financial Literacy and Parental Financial Literacy



Notes. Predicted risky share by financial literacy and parental financial literacy, evaluated at age = 45 and at the mean values of all other covariates.

Equipped with the estimated risky shares for each financial literacy combination, we now invert the Merton formula in equation (55) to recover the implied excess returns. As described in Section 6.3, we calibrate the standard deviation of risky assets at 15.7% and the coefficient of relative risk aversion at $\theta = 7$, which yields the excess returns reported in Table 10.

Table 10: Predicted Excess Returns by Own and Parental Financial Literacy

Own Financial Literacy	Parental Financial Literacy			Average
	Primary	Master's	Business	
Primary	1.33%	1.47%	1.75%	1.46%
Master's	1.61%	2.07%	2.32%	1.95%
Business	2.04%	2.48%	2.81%	2.38%

Notes. Predicted excess returns (in percentage points) implied by inverting the Merton formula at the predicted risky shares from Table 9, evaluated at age = 45 and at the mean values of the other covariates. Number of observations = 362,750.

The predicted returns are broadly consistent with the existing literature. Both educational length and field of study raise expected excess returns, in line with Fagereng et al. (2020) and Altmejd (2025). Fagereng et al. (2020) estimates an increase in returns to wealth of 0.1492 percentage points per extra year of study and an additional 0.1296 percentage points from a Business and Economics degree. Using the average years of education reported in Table 2, this implies a $(18.07 - 9.40) \cdot 0.1492 = 1.29$ percentage-point gap between the low and medium categories and a further $1.29 + 0.1296 = 1.41$ percentage-point gap between low and high. Our model predicts a $2.81\% - 1.33\% = 1.48$ percentage-point spread between the high-high and low-low groups, and a $2.07\% - 1.33\% = 0.74$ percentage-point spread between medium-medium and low-low. Averaging over parental literacy, the corresponding differentials are 0.92 and 0.49 percentage points, respectively.

Altmejd (2025) finds a slightly higher excess-return gap of 1.85 percentage points between business and non-business graduates, using a difference-in-discontinuities design. He notes, however, that this effect is most pronounced in volatile and downturn markets, which may explain why our estimated gap is slightly smaller: the 2014–2024 period contained only one major downturn in global equity markets, in 2022.

6.3 Other Parameters

As described above, we set the discount factor to $\beta = 0.96$ and the coefficient of relative risk aversion to $\theta = 7$, where the latter is subject to the largest variation in the literature. Mehra and Prescott (1985) set the upper bound of reasonable values at 10, a convention later adopted by Cocco et al. (2005). To avoid benchmarking edge cases, we choose the lower value of 7 and back it up with robustness checks across all proposed variations in the literature.

Replacement Rates

We calibrate heterogeneous replacement rates for the three educational groups, defined as the household's average income over the retirement period relative to income in the last full working year:

$$\text{Replacement Rate} = \frac{\frac{1}{T-\tau+1} \sum_{t=\tau}^T \text{Income}_t}{\text{Income}_{t=\tau-1}} \quad (57)$$

A household is classified as retired when the household head's labor-market attachment in the Statistics Denmark registers changes to "pensioner" and their age exceeds 65. The resulting replacement rates, reported in Table 11, are relatively stable across educational groups. This finding may come as a surprise, as the literature documents that better-educated households often face lower replacement rates, driven by the progressivity of public safety-net programs (Lusardi et al., 2017). In a Danish context, we do not find this result unreasonable given the high, often mandatory, pension contribution schemes in the Danish labor market, described in section 4.5. Furthermore, this is consistent with the findings of Cocco et al. (2005), who document similar replacement rates for their high and low categories.

Table 11: Replacement Rates by Financial Literacy Group

	Primary	Master's	Business
Replacement rate	0.914	0.893	0.892

Notes. Replacement rates as defined in equation (57).

Initial Wealth

While we do not calibrate our model with the wealth-to-income paths of households, we endow households with heterogeneous starting wealth according to their empirical median wealth-to-income ratios at age 25, reported in Table 12. The parental effect is evident among primary and master's-degree holders, but surprisingly negligible among business economists. We also see that own education has a significant and consistent effect across all parental types. This result is puzzling, since primary-school graduates have had ten years to accumulate wealth since completing their education, while master's graduates and business economists have had at most one to two. One would also expect the starting salary of a business economist to exceed that of a primary-school graduate, even early in the career. Both forces should push the wealth-to-income ratio downward for master's and business economists and upward for primary-school graduates, yet we observe the opposite. This suggests that, despite the head start, the consumption of primary-school graduates has roughly tracked their income over the first ten working years.

Table 12: Wealth-to-Income Ratios at Age 25 by Own and Parental Financial Literacy

Own Financial Literacy	Parental Financial Literacy					
	Primary		Master's		Business	
	Mean	Median [†]	Mean	Median [†]	Mean	Median [†]
Primary	0.237	0.088	0.535	0.153	0.698	0.166
Master's	0.665	0.409	1.043	0.606	1.164	0.703
Business	0.707	0.707	0.972	0.616	1.258	0.654

Notes. Wealth-to-income ratios are computed as total net assets divided by equivalised disposable income at age 25.

[†] Median values are pseudo-medians (average of ± 100 adjacent observations).

Risk-Free Rate

The standard papers of life-cycle modeling, mainly referring back to Cocco et al. (2005), typically assume a risk-free rate of 2%. However, we argue that the market framework has changed significantly since then with a long period of low interest rates. It is not just the US that faced this environment, but also Denmark, which faced a long-term government 10-year bond monthly yield average of approximately 0.829% (Federal Reserve Bank of St. Louis, 2026). We therefore assume a risk-free rate of 1% in our benchmark model and examine the sensitivity of our results to alternative higher and lower risk-free rates.

Death Rates

Finally, we obtain age-specific death and population statistics from Statistics Denmark's StatBank, tables DOD and BEF5. Using these tables, we compute the survival probability as the empirical probability of surviving for a given age group in a given year, then average across our 2014–2024 sample period (Statistics Denmark, 2026a, 2026b). The resulting death rates are plotted in Figure B.6.

7 Results

This section presents the simulation results of the model based on 100,000 agents over the life cycle. We analyze and discuss shapes and cross-sectional differences between educational groups. First, we present the benchmark model, followed by a decomposition analysis of inequality and welfare as well as a counterfactual analysis.

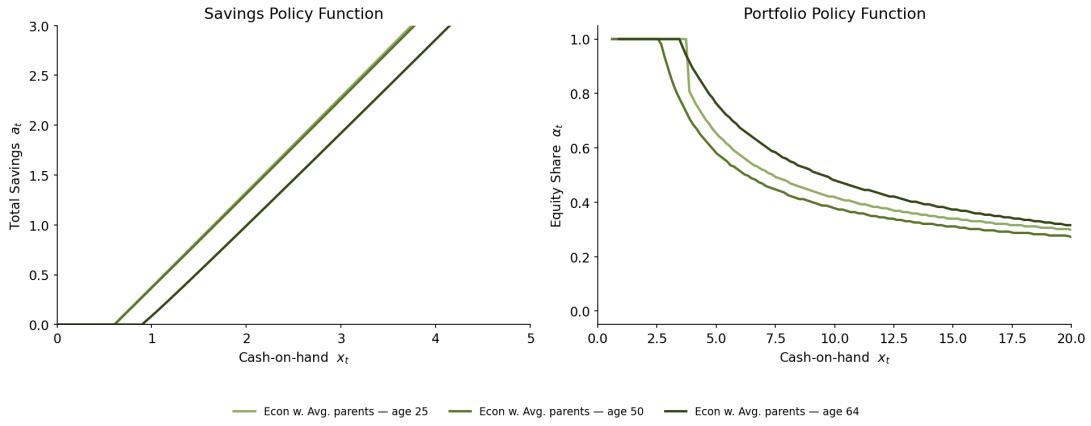
7.1 Benchmark

In the benchmark case, we solve the model for households with average parental literacy across our three educational groups. As such, we assume that all groups have the same parents and that heterogeneities are entirely determined by the household's level of financial literacy.

Policy Functions

Solving the model yields the policy functions for savings (a) and portfolio choice (b) presented in Figure 8, shown for a business economist household with average parents at three different ages.

Figure 8: Benchmark Model Policy Functions for Economists



Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for economists with average parents.

In Panel (a), we see the driving forces of consumption-smoothing behavior across the life cycle. The savings rate is equal to zero for low cash-on-hand, which is due to the borrowing constraint: the agent would prefer to borrow in order to smooth consumption, but the no-borrowing constraint in equation (49) binds. The sharp kink in the policy function occurs where it is no longer optimal for the agent to borrow. At this point, the constraint is no longer binding, creating a non-continuous increase. Furthermore, we see that the optimal consumption policy changes across age. Given the same cash-on-hand, an old person faces a different optimization problem compared to a person peaking in labor income, since the latter has to offset upcoming low-consumption periods by saving more compared to those who are already in these low-consumption periods.

The portfolio policy in Panel (b) is determined by the optimal risky share condition derived by Cocco et al. (2005):

$$\alpha_t = \frac{\tau}{\theta \sigma_\eta^2} \cdot \frac{W_t + HC_t}{W_t}, \quad (58)$$

where τ is the excess return, θ the coefficient of relative risk aversion, σ_η^2 the variance of the risky

asset, W_t the financial wealth at time t , and HC_t the human capital, defined as the present value of all future non-financial income. It is the ratio of financial wealth to human capital W_t/HC_t that determines the optimal risky share. Consider an asset-wealthy and an asset-poor agent who have the same labor income along their lifetime. It is optimal for the wealthy agent to invest a lower fraction of those assets in the risky asset, since a bad state in the returns of the risky asset is followed by low consumption and an according increase in marginal utility. The poor agent, in contrast, does not suffer as much from a bad state of the risky asset, since the main driver of current consumption is labor income, not financial wealth.

Two forces drive the development of financial wealth W : the life-cycle motive and the precautionary savings motive. First, the life-cycle motive induces a hump-shaped savings profile: agents accumulate wealth during high-income periods to finance consumption in low-income periods. Second, our agents are prudent and receive uncertain labor income, meaning that they exhibit precautionary savings behavior in order to prepare for bad states of labor income. Thus, agents save to prepare for bad states and then invest all these savings in the most risky asset, since the bad state of labor income might overlap with a state of low (or even negative) returns. That is, the more uncertainty lies ahead of an agent, the more cautious the investment strategy, by lowering the risky share in the portfolio.

As proposed by Jagannathan and Kocherlakota (1996), one can consider a deterministic labor income process as a large risk-free bond. This shifts the portfolio weights heavily towards the risk-free asset, which makes it optimal to invest more in the risky asset to keep the balance of an optimal risky share. Although our model has a stochastic income process, the intuition still holds, since labor income is still closer to a risk-free asset than to a risky asset. In fact, it is the no-shorting constraint $\alpha \leq 1$ which prevents the agent from making the portfolio even riskier when cash-on-hand is low.

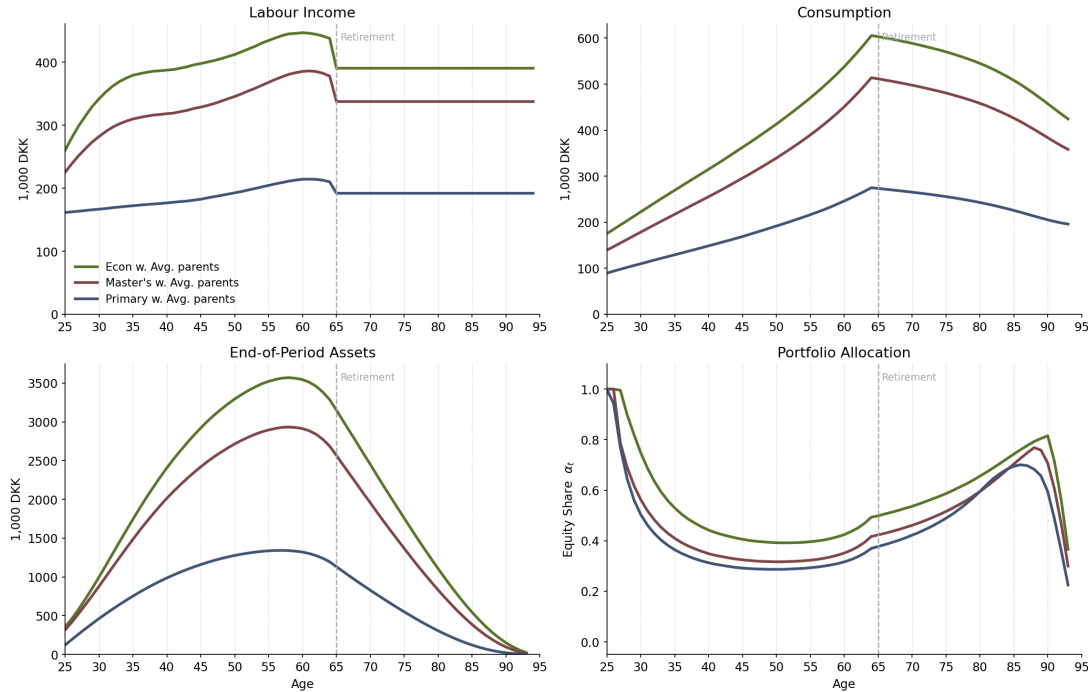
Furthermore, it is worth noting the non-monotonic difference in the portfolio policy function across age. Given a certain cash-on-hand x_t , a 64-year-old has the highest optimal risky share, a 50-year-old the lowest, and a 25-year-old in between the two. When the agent is young, there are low savings and high human capital, which drives the optimal equity share up. The remaining labor income stochasticity, however, strengthens the precautionary savings motive, making financial wealth increase. Conversely, when an agent approaches retirement, they face a low amount of human capital relative to financial wealth, but with significantly less labor income uncertainty. Since income is modeled to be constant during retirement, the optimal share of risky assets increases again due to the removed uncertainty in labor income.

Life-Cycle Profiles

Figure 9 shows the dynamics across the lifetime of the simulated agents, with the labor income path calibrated as described in Section 6.1. The labor income panel reproduces the hump-shaped income profile that motivates the calibration, and the consumption panel clearly shows the smoothing mo-

tive. The end-of-period assets illustrate the compounded savings behavior and portfolio allocation decisions.

Figure 9: Benchmark Life-Cycle Profiles



Notes. The figure depicts the labor income, consumption, end-of-period assets, and portfolio allocation paths of primary, master's, and economist groups.

The shape of the end-of-period assets captures the core dynamics of life-cycle models. Around the age of 60, a_t peaks, and agents start running down their assets as the low-income period of retirement approaches. Furthermore, the fact that retirement income is certain removes the precautionary savings motive, which slows down asset accumulation in the years leading up to retirement.

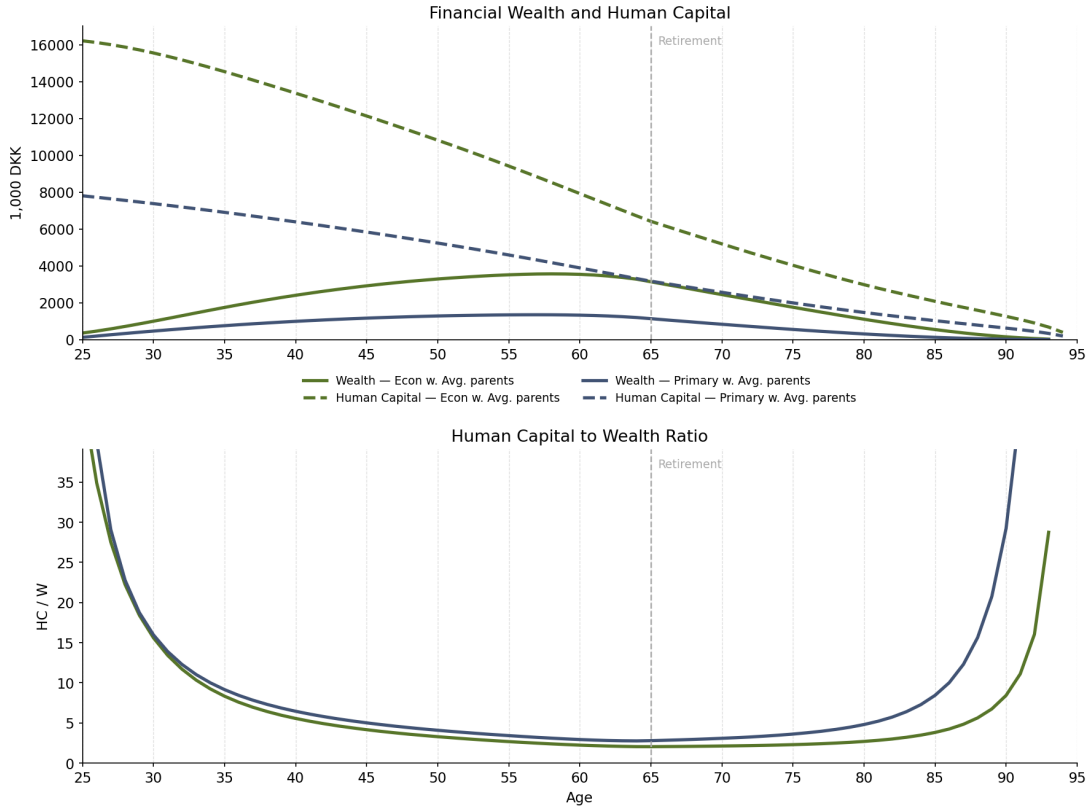
The Drivers of the Risky Share

In order to illustrate the two opposing effects that determine the optimal equity share α , we plot the financial wealth and remaining human capital as a function of age in Figure 10.

Human capital declines monotonically over the lifetime, since there are fewer remaining periods of labor income in later years. Financial wealth, in contrast, follows an inverted U-shape, which leads to a sharp decrease in the human-capital-to-wealth ratio in early adulthood. That is, the risky share declines steeply between ages 25 and 35 across all educational groups. Between ages 40 and 70, the HC/W ratio is relatively flat, which means that the main driver of the increase in α_t^* during this period is the gradual reduction in labor income uncertainty rather than further changes in the wealth composition. Towards the end of life the human-capital-to-wealth ratio increases sharply again, as financial wealth is drawn down faster than the present value of remaining labor income falls. This

last pattern reflects the fact that the certain retirement income makes the remaining human capital effectively bond-like, while uncertain mortality pushes agents to draw down financial wealth more quickly. The same pattern is documented in Cocco et al. (2005).

Figure 10: Financial Wealth and Human Capital over the Life Cycle



Notes. The figure depicts the financial wealth and human capital paths over the life cycle, illustrating the drivers of eq. (58).

7.2 Inequality Implications

Following Lusardi et al. (2017), we express the degree of wealth inequality as the ratio of wealth-to-income ratios (WIR ratio) at retirement. As discussed in Section 4.5, the definition of wealth will necessarily bias the evaluation of our model against the empirical data. As such, we do not intend to evaluate the model against the absolute value of the wealth-to-income ratios of our educational categories, but rather the ratio $\frac{WIR(2,2)}{WIR(0,0)}$ at the time of retirement.

Inequality at the Benchmark

Our model predicts that a business economist household with business economist parents (2,2) has a wealth-to-income ratio of 10.62, while a primary household with primary parents (0,0) has a wealth-to-income ratio of 7.29. Comparing this with the data, we observe much lower wealth-to-income

ratios of 2.80 and 1.30, respectively. While the absolute differences may seem large, this is a natural implication stemming from the fact that our model has no pension system nor any illiquid assets. In the absence of these components, households are more incentivized to save for retirement, as their financial assets alone have to sustain them until the end of life. The observed liquid hand-to-mouth behavior of households in the data is consistent with the findings of Kaplan et al. (2014).

Table 13: Simulated and Observed Wealth-to-Income Ratios at Retirement (Age 65)

Own Education	Parental Education		
	Primary	Master's	Business
<i>Panel A: Simulation (Mean WIR)</i>			
Primary	7.29	7.40	7.62
Master's	9.41	9.91	10.19
Business	9.63	10.19	10.62
<i>Panel B: Data (Mean WIR)</i>			
Primary	1.30	1.57	0.80
Master's	2.60	2.59	2.10
Business	2.77	2.73	2.80
<i>Panel C: WIR Ratios</i>			
	(1, 1)/(0, 0)	(2, 2)/(1, 1)	(2, 2)/(0, 0)
Data	1.99	1.08	2.15
Simulation	1.36	1.07	1.46
Model fit (%)	36.25	91.50	39.81

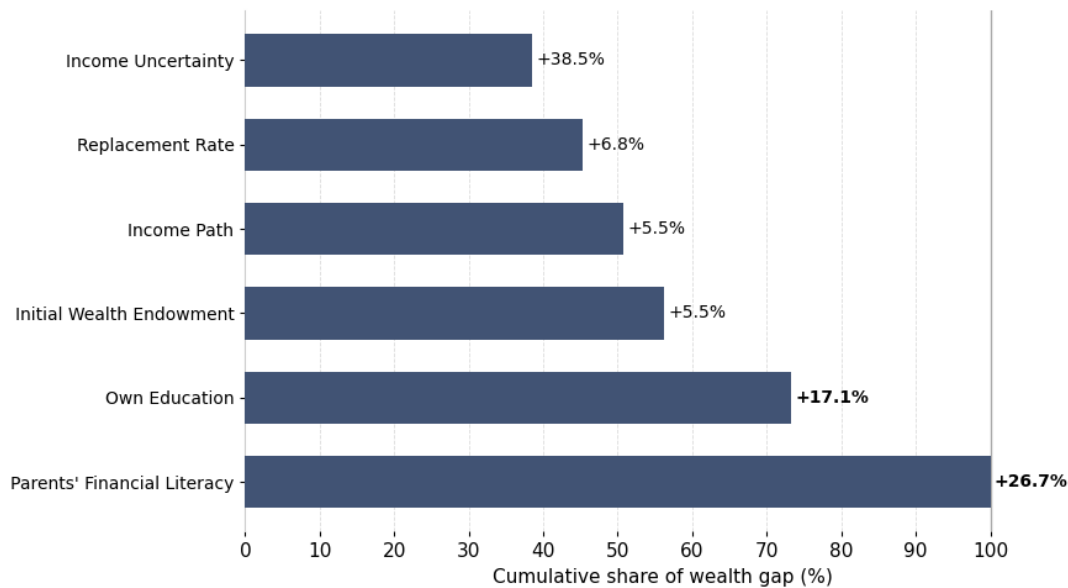
Notes. The table compares simulated and observed wealth-to-income ratios at retirement across combinations of own and parental education levels. Panel C reports relative wealth ratios and model fit statistics.

The ratio of wealth-to-income ratios of our highest and lowest literacy groups is 2.15 in the data and 1.46 in our simulation. Our model is therefore able to account for 39.81% of the inequality observed in the data at retirement. This fraction must be explained by the collection of ex-ante imposed heterogeneity that we feed into the model: (1) income variances, (2) replacement rates, (3) income path growth rates, (4) initial wealth endowment, and (5) excess returns.

Decomposition

In Figure 11 we present a decomposition of the simulated inequality at the time of retirement. That is, the relative magnitude of each of the above ex-ante sources of heterogeneity that move the ratio of wealth-to-income ratios from 1.00 to 1.46 in our model.¹²

¹²Full life-cycle profiles as well as policy functions for the decompositions can be found in the appendix from Figure B.23 onward.

Figure 11: Decomposition of the Simulated (2,2)/(0,0) Wealth-to-Income Ratio at Retirement

Notes. The figure decomposes the drivers behind the wealth-to-income ratio discrepancies between a (0,0) and a (2,2) household at retirement.

The largest single contributor is income uncertainty, which on its face is surprising. The mechanism is the following: as we argued in Section 6.1, the business economists exhibit larger permanent and transitory income shocks, and the observed variance discrepancies are robust to our model specifications. Given the assumption that the coefficient of relative risk aversion, and hence the degree of prudence, is homogeneous across the educational groups, larger shocks strengthen the precautionary savings motive of the business economists. Combined with their higher attainable returns, they therefore accumulate a relatively higher wealth-to-income ratio at retirement.

The income path itself plays a relatively small role at 5.5%, which is somewhat counterintuitive given how visually different the income paths in Figure 9 are. This stems from the fact that we use a CRRA utility function and a relative inequality measure at retirement. The inequality is therefore independent of the level of the income path, and only depends on its shape.

Replacement rates and the initial wealth endowment collectively explain about 12% of the simulated inequality. As we saw in Table 11, the replacement rate of the primary group is slightly higher than for the business economist graduates, which allows them to consume more of their labor income throughout their working life. Comparatively, the business economist graduates have to make up for a relatively larger drop in income when they retire, which pulls up their wealth at retirement and hence their wealth-to-income ratio. Similarly, the higher initial wealth of the business economist trivially translates into a higher wealth-to-income ratio at retirement.

Finally, we see that the household's own education and their parents' education play the second and third largest roles, respectively, in explaining the simulated wealth inequality at retirement. These are our measures for financial literacy and are thus driven by the higher attainable excess

returns of the groups: “own education” refers to a vertical move in Table 10, while the effect of parents’ financial literacy corresponds to a horizontal move in the same table. The accumulated effect of the two channels corresponds to moving from a 1.33% excess return to a 2.81% excess return. Our model therefore finds that the wealth-to-income inequality would be 43.8% smaller at retirement (the sum of the 17.1% from own education and the 26.7% from parental financial literacy) if the primary graduates with primary parents and the business graduates with business graduate parents faced the same expected excess returns.

7.3 Welfare Analysis

Next, we illustrate the impact on wealth at retirement and on welfare if households had homogeneous financial knowledge and thus homogeneous excess returns in the financial markets. To be specific, we replicate the simulations of the households from Table 13 with the same income shocks and initial conditions, and compare the outcomes to a counterfactual world in which all households are endowed with the excess returns of a high literacy household at age 25. We quantify the welfare gains using a consumption equivalent, derived in Appendix C.

Welfare & Inequality

Panel B shows that the inequality metric $\frac{WIR(2,2)}{WIR(0,0)}$ falls from 1.46 in the baseline to $10.62/8.66 = 1.22$ in the counterfactual, a reduction of $(1.46 - 1.22)/(1.46 - 1) = 52.2\%$ of the simulated gap. Since our model accounts for 39.81% of the observed inequality in the data, this corresponds to closing roughly $52.2\% \times 39.81\% = 20.7\%$ of the empirical wealth gap at retirement. Notably, all Master’s category households end up with a strictly greater counterfactual wealth-to-income ratio than the highest literacy reference household. This reflects the precautionary saving channel that we identified in the decomposition: Master’s households face higher income variance than business economists, so when given the same excess return profile they accumulate even more financial wealth as a buffer against income risk.

Panel C reports the welfare gains. The consumption equivalents are strictly positive and rise monotonically as own and parental literacy fall. A (0,0) household is indifferent between obtaining the (2,2) household’s excess returns at age 25 ongoing and increasing consumption in every period over the rest of their life by 3.71%. This corresponds to approximately 7,500 DKK per year or 514,000 DKK over the whole lifetime. The corresponding values for (2,0) and (0,2) households are 2.16% and 2.89%. These magnitudes are comparable to those of Lusardi et al. (2017), who report consumption equivalents of 1.9% and 2.7% for primary and master’s groups, respectively, in a US calibration that does not include a parental literacy dimension.

Table 14: Wealth-to-Income Ratios at Retirement and the Welfare Cost of Financial Illiteracy

Own Financial Literacy	Parental Financial Literacy		
	Primary	Master's	Business
<i>Panel A: Baseline mean WIR at age 65</i>			
Primary	7.29	7.40	7.62
Master's	9.41	9.91	10.19
Business	9.63	10.19	10.62
<i>Panel B: Counterfactual mean WIR at age 65, endowed with (2,2) excess returns</i>			
Primary	8.66	8.69	8.69
Master's	10.65	10.75	10.78
Business	10.50	10.61	10.62
<i>Panel C: Consumption-equivalent gain from being endowed with (2,2) excess returns</i>			
Primary	3.71%	3.47%	2.89%
Master's	3.18%	2.11%	1.46%
Business	2.16%	0.98%	0.00%

Notes. Panel A reports the mean simulated WIR at age 65. Panel B reports the counterfactual WIR when each household is endowed with the excess return profile of a (2,2) household holding all other ex-ante heterogeneities fixed. Panel C reports the consumption equivalent of moving from the household's baseline to the counterfactual return profile.

Synergy Effects

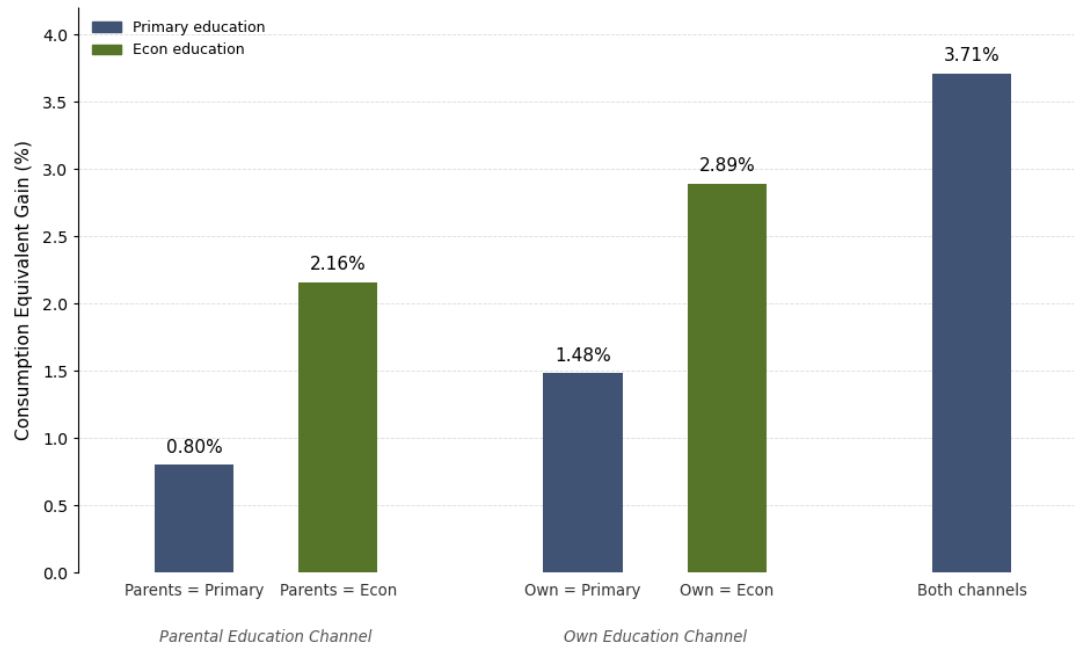
The diagonal of Panel C decomposes naturally into a parental channel and an own-education channel, and Figure 12 plots this decomposition for the (0,0) household.¹³ Two patterns are worth highlighting. First, the synergy effect between own and parental literacy that we documented in Section 6.2 analysis carries through into welfare. Upgrading parents from Primary to Business Economist is worth 0.80% to a household that is itself Primary, but 2.16% to a household that is itself a Business Economist. Symmetrically, upgrading own education from Primary to Business Economist is worth 1.48% to a household with Primary parents, but 2.89% to a household with Business Economist parents. The benefits of financial knowledge are larger when the household's parents are also knowledgeable, and the same holds in reverse.

Second, these welfare gains operate primarily through the extensive margin documented in Section 4.5. As we showed there, the educational difference in financial-market participation is substantially steeper than the difference in the conditional risky share. The 0.80% consumption equivalent for upgrading the parents of a Primary household therefore mainly measures, in welfare terms, the value of moving that household from one participation regime to another while holding the rest of the life-cycle environment constant. A reasonable interpretation is that the welfare cost of low

¹³The decomposition for (0,0) to (1,1) and (1,1) to (2,2) can be found in Figures B.33 and B.34 in the appendix.

parental financial literacy is, in large part, the welfare cost of being less likely to enter financial markets at all.

Figure 12: Benchmark Model Consumption Equivalent Decomposition: (0,0) Household



Notes. Panel (A) illustrates the consumption equivalent of improving parental education. Panel (B) illustrates the consumption equivalent of improving household education. Panel (C) illustrates the joint effect.

These results have a simple intuitive reading. If one were to sample two siblings, who are children to business economist, and one chooses to study Business Economics while the other does not go beyond primary school, the educated child will benefit more from the fact that the parents are financially literate. In fact, our model predicts a consumption equivalent of the business graduate of 2.16%, while that of the Primary is 0.8%. Conversely, sampling two classmates in a Business Economist program, where one is a child of business economist and the other a child of primary graduates, the first will benefit more from pursuing this degree. That is, the consumption equivalent of the former is 2.89%, while the latter one's increase in financial knowledge is equivalent to an annual consumption increase of only 1.48%.

8 Robustness

Next, we provide a sensitivity analysis to assess whether our results are robust to the chosen parameters. In Table 15 below, we restate our benchmark results along with alternative parameters for the coefficient of relative risk aversion θ , discount factor β , risk-free rate r_f , and elasticity of consumption with respect to financial assets b_1 . All simulated life-cycle profiles and policy functions for the

sensitivity analysis can be found in the appendix from Figure B.35 onward. Column 1 reports the consumption equivalent of a (0,0) household improving both their own and parental education to a (2,2) household. Columns 2 and 3 report the wealth-to-income ratios at retirement while column 4 reports the ratio of the two.

Table 15: Sensitivity Analysis for Preference Parameters

	CE	Mean WIR		WIR Ratio
		WIR (0,0)	WIR (2,2)	
Benchmark	3.71%	7.29	10.62	1.457
$\theta = 4$	1.52%	2.95	5.03	1.706
$\theta = 10$	5.74%	10.17	14.37	1.412
$\beta = 0.95$	3.58%	6.66	9.86	1.482
$\beta = 0.97$	3.84%	7.94	11.39	1.435
$r_f = 0.5\%$	3.92%	6.94	10.26	1.480
$r_f = 2\%$	3.32%	7.95	11.27	1.418
$b_1 = 0.9$	3.04%	7.21	10.26	1.422
$b_1 = 0.75$	2.14%	7.11	9.79	1.377

Notes. This table reports the consumption-equivalent welfare gain (CE) and wealth-to-income ratios for different values of preference and interest rate parameters. WIR Ratio = WIR(2,2) / WIR(0,0).

Benchmark has parameters $\theta = 7$, $\beta = 0.96$, $r_f = 0.01$, $b_1 = 1$.

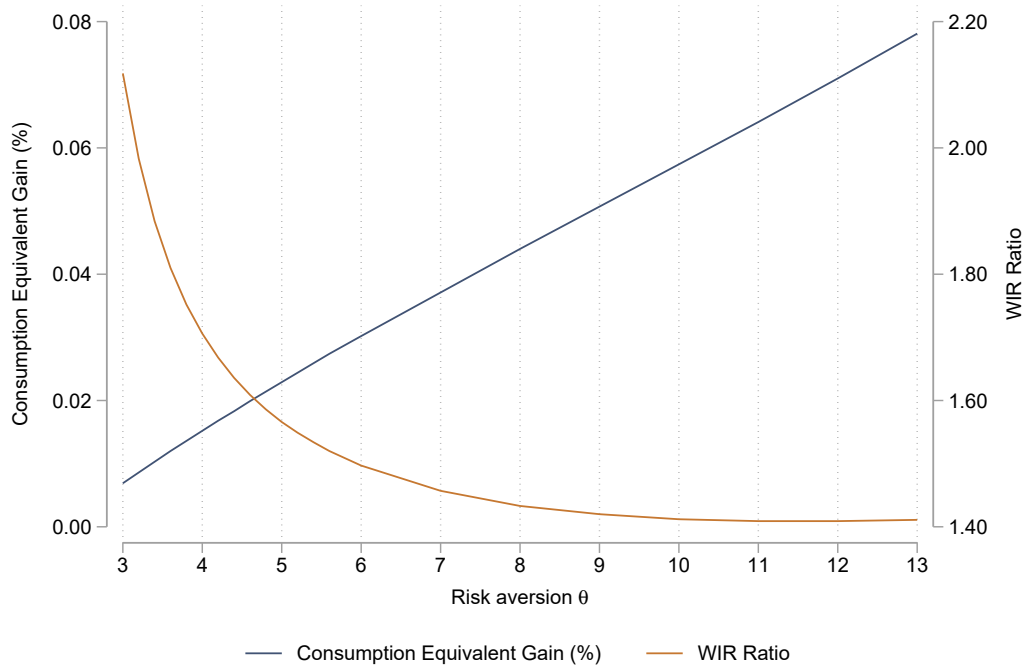
Risk Aversion

First, we vary the coefficient of relative risk aversion θ from 3 to 13. We consider these cases rather extreme given the coefficients that are most pronounced in the literature (Cocco et al., 2005; Mehra & Prescott, 1985). We see that the consumption equivalents are relatively sensitive to the choice of this coefficient: a higher degree of risk aversion leads to a lower fraction of risky investment or, in our model, a higher excess return τ through the Merton inversion in eq. (55). Therefore, the discrepancy in excess returns between the lowest literacy and highest literacy households increases. This means that the consumption equivalent is increasing in θ , as the increasing discrepancy in excess returns makes it more attractive for the less-educated households to obtain a higher level of financial knowledge.

Conversely, the ratio of wealth-to-income ratios at retirement decreases in θ . A higher degree of relative risk aversion implies a lower elasticity of intertemporal substitution (EIS) with CRRA preferences, as $EIS = 1/\theta$. When the EIS is low, it makes consumers less willing to substitute across periods and therefore, despite the business economist graduates getting a higher excess return, the income effect dominates and they spend more today rather than save for retirement, which puts downward pressure on their wealth-to-income ratio. We illustrate the trade-off in fig. 13 below. Notably, the ratio of wealth-to-income ratios converges at 1.4 while the consumption equivalent is

linear in θ .

Figure 13: WIR Ratio & Consumption Equivalent



Notes. The figure depicts the consumption equivalent gain of a (0,0) household obtaining the excess returns of a (2,2) household against their relative wealth-to-income ratios at retirement for varying values of θ .

Discount Factor

The choice of discount factor β is rarely a topic of discussion in the literature and, as such, our robustness check is tight compared to the preference parameter above. A lower β implies that the households are more impatient which, all else being equal, makes the impatience incentive dominate the precautionary savings (prudence) incentive. As such, a lower β makes the economist graduates less likely to utilize the higher returns. In fact, since the primaries also accumulate fewer assets, yielding lower wealth-to-income ratios, they would benefit less from the higher excess return of the business economist graduates, which lowers the consumption equivalent in this scenario. By inspecting the wealth-to-income ratio, however, it is evident that the downward pressure is stronger for the primary graduates than for the business economist counterparts. The results, however, are generally robust to higher and lower values of β in either direction.

Risk-Free Rate

As with the discount factor, the risk-free rate is rarely a topic of discussion. While historically it has been best practice to use a risk-free rate of 2%, we chose to calibrate our benchmark model with a

rate of 1% given the recent low interest rate environment. A lower risk-free rate lowers the return on investment for all groups. As the business economist graduates maintain their numerical edge in excess returns, however, this makes the consumption equivalent higher for the primary graduates. A natural implication is that the wealth-to-income ratios decrease for both groups, as their effective returns are lower. Once again, however, we see that this effect is not symmetrical, as the primary graduates' wealth at retirement decreases more, creating more inequality as measured by the WIR ratio.

Elasticity of Consumption

Finally, we address the elasticity of consumption with respect to financial wealth. Viceira (2001) shows that $b_1 = 1$ holds when a household faces retirement, but also argues that $0 < b_1 < 1$ before retirement and that it rises monotonically throughout working life, arguing that as human capital decreases and financial wealth becomes a larger part of total resources, a positive shock to financial wealth should translate directly into higher current consumption. As the excess returns are tied to b_1 , it is natural that the consumption equivalents and WIR ratios are affected by changing it. For the more extreme case of $b_1 = 0.75$ we see an impact on the consumption equivalent of 1.57 percentage points. This is due to the fact that the excess returns of the economist graduates are no longer as attractive as they were before. Even for an elasticity of $b_1 = 0.75$, however, the WIR ratio is quite stable, illustrating a limited impact on the inequality at retirement.

9 Counterfactual Policy Analysis

The results until now establish a clear relationship between financial knowledge, inequality, and welfare implications for the households: if a social planner were to endow the lowest literacy household with the financial knowledge of the highest literacy household, the inequality at retirement in our model would decrease by 52.2% and the lowest literacy household would enjoy a 3.71% welfare gain corresponding to 7,500 DKK per year. While this is an interesting finding, it is difficult to specify how such a policy would be implemented in practice.

Replacement Rates

Considering the high mandated contribution rates to private pensions and the public pension transfers from the government that we described in section 4.5, it seems natural to consider the implications of lowering the replacement rates of the households. As we do not have pension savings as a model parameter in itself, we acknowledge that the analysis is rather one-sided - it would be natural to suggest that lowering contribution rates for liquidity-constrained households may be welfare improving despite lowering replacement rates. Or even, as Schlafmann et al. (2024) find, it may be

possible to design a contribution plan that is welfare improving without lowering replacement rates.

We instead focus on the crowding-out effects of private wealth accumulation due to high income during retirement: high replacement rates soften the sharp income drop at the time of retirement and therefore reduce the incentive to save during working life. We distinguish between a *symmetric* and *asymmetric* policy. While symmetric policies set a joint level of replacement rates, asymmetric policies set different rates across the educational groups.

Table 16: Simulation Results of Policy Experiments

Replacement Rate		Mean CE	Mean WIR		WIR Ratio
(0, 0)	(2, 2)	CE	WIR (0,0)	WIR (2,2)	
<i>Baseline</i>					
89%	91%	3.71%	7.29	10.62	1.457
<i>Symmetric Reduction</i>					
80%	80%	3.77%	8.52	11.59	1.362
70%	70%	3.81%	9.63	12.68	1.317
60%	60%	3.86%	10.77	13.80	1.281
<i>Asymmetric Reduction (Primary only)</i>					
80%	90%	3.77%	8.52	10.53	1.237
70%	90%	3.81%	9.63	10.53	1.094
60%	90%	3.86%	10.77	10.53	0.978

Notes. This table reports the consumption-equivalent welfare gain (CE) and wealth-to-income ratios under alternative pension replacement rate policy experiments. WIR Ratio = WIR(2,2) / WIR(0,0). Baseline replacement rates are 89% for type (0,0) and 91% for type (2,2).

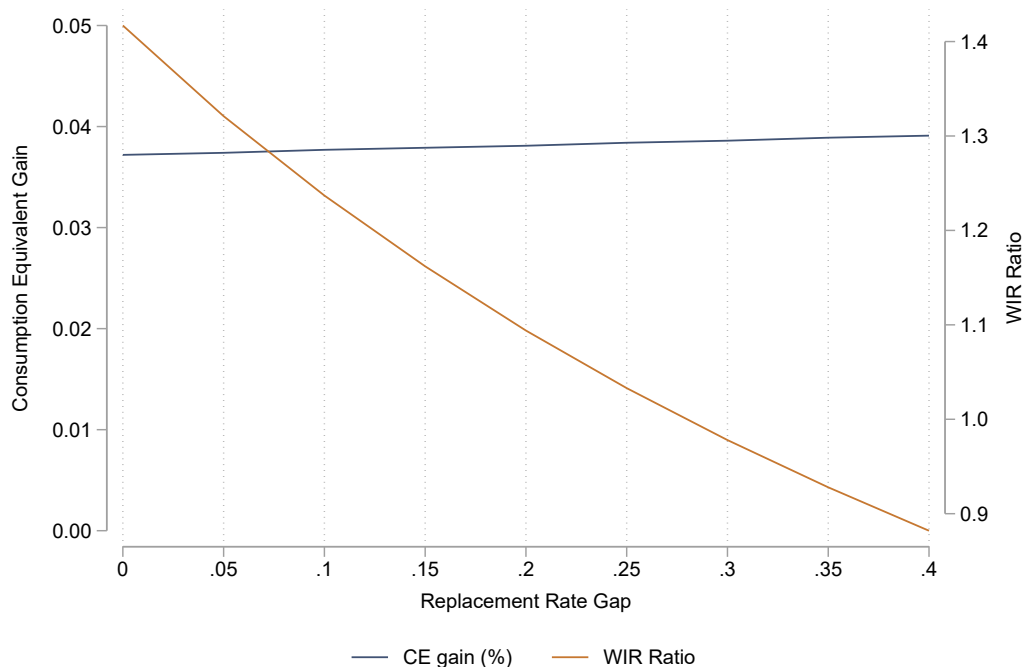
As presented in Table 16, a symmetric reduction in replacement rates increases the wealth-to-income ratios for both the lowest and highest literacy households. This illustrates the intuition described above: as the replacement rates decrease, the level of wealth that a household must have accumulated to live through retirement while maintaining a smooth consumption path increases. Notably, the symmetric decrease lowers the wealth inequality at retirement as measured by the WIR ratio. This is the case because the lowest literacy household's wealth-to-income ratio is more sensitive to replacement rates drops due to lower labor income uncertainty. To understand this, imagine constant labor income with a replacement rate of 100%, that is, with no life-cycle motive. The precautionary savings motive, however, leads an agent with a high labor income variance to have a higher wealth-to-income ratio at retirement compared to an agent with a low labor income variance. Dropping the replacement rate substantially makes the low-variance agent increase her assets at retirement by more than the high-variance agent, since the high-variance agents already has a large asset stock due to the precautionary savings motive. Hence, the savings rate of the low-variance increases more sharply than that of the high-variance agents yielding a lower inequality according to WIR ratio. For example, decreasing the replacement rate from 80% to 70% increases the wealth-to-income ratio by 13.02% for the lowest literacy household but only 9.49% for the highest

literacy household, which lowers the inequality compared to the benchmark scenario.

Perhaps slightly counterintuitively, lowering the replacement rates of the primary households asymmetrically reduces the wealth inequality at retirement. Once again, this is the case because the increased savings motive is one-sided. With a lower replacement rate, the primary group is encouraged to save for old age, meaning their wealth-to-income ratio increases substantially. At replacement rates of 60% and 90%, we observe a WIR ratio close to one. Furthermore, we also observe an increasing consumption equivalent as the replacement rate decreases for the primary group. This is a trivial result: as the primary group now holds more wealth, they would be even better off if they could access the excess returns of the highest literacy households. Figure 14 illustrates the asymmetric case, plotting the WIR ratio and consumption equivalents as a function of the gap in replacement rates ($\mu_{\text{Business}} - \mu_{\text{Primary}}$), where a larger value on the x-axis corresponds to greater asymmetry.

The wealth inequality decreases in the replacement rate gap and turns negative around a gap of 27 percentage points. The consumption equivalent stays rather constant with a maximum range of 9 percentage points. Hence, the model predicts that lower replacement rates for primaries compared to business economics graduates can tackle wealth inequalities while the relevance of financial knowledge stays roughly the same. This finding is similar to that of Lusardi et al. (2017).

Figure 14: WIR Ratio & Consumption Equivalent for Varying Primary Replacement Rates



Notes. This figure illustrates the consumption equivalent gain of a (0,0) household obtaining the excess returns of a (2,2) household and the WIR ratios for different replacement rate gaps.

It is important to note that the WIR ratio captures a purely mechanical notion of equality: the

ratio of accumulated wealth at retirement. Convergence of this ratio toward unity under asymmetric reductions reflects a forced increase in precautionary savings by primary households, not an improvement in their welfare. In fact, our consumption-equivalent metric rises as primary replacement rates fall. This is the model's way of signaling that lower replacement rates push primaries into a larger self-insurance problem, in a domain where their lack of financial literacy is most costly. Thus, the accounting metric of inequality decreases, but it comes at a welfare cost for the primaries. Furthermore, as mentioned above, our framework excludes the contribution side altogether.

10 Discussion

10.1 Interpretation of Findings

In this paper, we have extended the classical buffer-stock model with portfolio choice to allow for heterogeneity in attainable excess returns dependent on financial knowledge and intergenerational transfers of that knowledge. By calibrating the model with Danish register data, we account for roughly 40% of the observed inequality in wealth-to-income ratios at retirement between the lowest and the highest financial-literacy groups. A counterfactual analysis in which the lowest literacy household is endowed with the excess returns of the highest literacy household closes 52.2% of the simulated retirement gap and delivers a 3.71% consumption-equivalent welfare gain. Thus, the mechanism we identify can rationalize approximately $52.2\% \times 39.81\% = 20.7\%$ of the empirical wealth gap.

Two interpretations of this number deserve to be made explicit. Firstly, although the 20.7% is significant, the remaining 79.3% must reflect channels that lie outside our model, such as heterogeneity in preferences, dispersion in realized returns conditional on participation, or inheritances. Therefore, the 39.81% model fit should be read as a lower bound on the importance of financial knowledge once these alternative channels are controlled for inside a structural framework, but not as a claim that financial knowledge is the dominant driver of household wealth dispersion in Denmark.

The second caveat concerns the institutional setting in which our households live. Our model framework deviates from the Danish environment in a handful of ways. There is no pension, no housing and no explicit consumption floor. As such, our simulated wealth-to-income ratios are shifted upwards as households must self-insure against income risk that would otherwise have been covered by the government.

Taken together, these caveats imply that our results should be read as a benchmark for the optimal life-cycle response to financial knowledge heterogeneity in an otherwise neutral institutional environment, rather than as a forecast of actual Danish wealth dynamics. The strength of the framework lies in isolating the financial-knowledge channel while the cost is that several quantitatively important features of the Danish economy are held fixed.

Notably, we take no stance on the appropriate policy instrument for raising the financial knowledge of the lowest educational group. One may suggest that politicians should intervene in the educational sector in order to incentivize households to pursue longer educations and promote the teaching of financial literacy in primary schools. On the other hand, one might speculate that improving the financial knowledge of households through informational campaigns outside the educational sector could have even greater welfare improvements: after all, it seems counterintuitive to incentivize a passionate nurse to pursue a degree in business or economics if it is possible to teach the nurse about financial literacy while they stay in their field. We shall not pretend to know what the best intervention policy in the educational sector is. However, our results quantify the value of closing the financial literacy gap. The implementation question is therefore left for empirical work that can compare the welfare gains of improving financial literacy to the cost thereof.

10.2 Limitations

While our findings are robust to reasonable variations in parameters as documented in Section 8, certain modeling choices warrant further discussion.

Assets & Participation

Quantifying wealth and financial-market participation raises two challenging choices. First, one must decide whether to measure wealth as liquid financial assets, financial assets plus housing, or under an even broader definition that includes illiquid assets such as pension wealth and privately held businesses. Second, one must decide on the relevant population for the equity share calibration: the entire population, the subset of households that participate in financial markets, or those that hold a specific asset class such as stocks (Campbell et al., 2003, pp. 200–201). In this paper, we measure wealth as liquid financial assets, thus excluding housing, pension wealth and business equity, and we calibrate participation rates and risky shares across the entire population.

As we noted in Section 4.5, the first choice puts downward pressure on our wealth-to-income measures at retirement, since many Danish households accumulate a substantial part of their net wealth in illiquid pension accounts and real estate and live as wealthy hand-to-mouth households as measured by liquid-wealth (Kaplan et al., 2014). Replacing our liquid-asset measure with total net assets would shift the simulated wealth-to-income ratios upward substantially. Nonetheless, we maintain the liquid asset definition because our model allows for allocation between a risky and a risk-less financial instrument. Therefore, we believe a wider measure would be inappropriate and make the results less interpretable, unnecessarily.

Secondly, because we use CRRA preferences and do not introduce explicit participation costs, the model cannot mechanically match the low empirical participation rates given a strictly positive equity premium. A calibration restricted to financial-market participants would neglect the fact

that participating itself is part of decision-making based on financial literacy. Since the welfare cost of low financial literacy is, in our framework, primarily the welfare cost of non-participation, restricting the calibration to participants would remove the channel we set out to quantify.

Demographics

Three modeling assumptions on the spillover and categorization of financial knowledge across household members deserve discussion.

First, we treat two adults registered as married, partners or cohabiting as a single household in which wealth, income and financial decisions are joint. This is a simplifying assumption. It seems reasonable, however, to assume that a substantial fraction of registered couples maintain separate accounts and make financial decisions independently of each other. Because we count a household as participating if either adult holds a positive risky financial position, our measure likely overstates the true participation rate at the household level while understating the conditional risky share among those who do participate.

Second, we categorize households by the longest education observed among their adult members. Under this rule, a household consisting of a business economist and a primary graduate is classified as more literate than a household of two master's graduates. The simplification has two consequences: it likely understates the average income of the highest literacy group, since the partner's lower earnings pull down the head's. Furthermore, it forces the implicit assumption that the more educated partner either makes all financial decisions or transfers financial knowledge to the partner at zero cost. An alternative would be to use the average years of schooling within a household, with arbitrary cut-offs chosen ex-ante. We leave this as a specification check for future work.

Third, parental financial literacy is categorized in the same way, which may understate the parental channel. A household with four highly-educated parents has the same parental-literacy category as a household with one. Controlling for the number of literate parents, or using the parental-average length of education, would in principle allow the marginal effect of an additional literate parent to be identified.

Parental Effect on Income

As argued in Section 4.5, we assume that parental education has no direct effect on the income process of the household. Parental literacy therefore enters the model through attainable returns, own education and starting wealth. This is a strong assumption, especially if one were to consider the network effects of the corporate world. While the descriptive evidence in Figure B.1 is broadly consistent with the assumption, one might argue that the income densities are noisy proxies for the true income process. The decision is driven primarily by data availability: parental data, particularly for highly-educated parents in old households, are sparse in the early years of our sample. Therefore, including an effect of parental education in income would have biased the income-age

profiles toward younger cohorts.

Merton Inversion

The identification of the effects of financial literacy on attainable expected excess returns, rests on the Merton-inversion procedure described in Section 6.2. Although our returns to education line up closely with Fagereng et al. (2020), two limitations deserve to be mentioned.

First, while our risky-share regression in Equation 56 includes the 9th-grade GPA to absorb the effect of innate ability, and financial measures for the household, it does not include controls for parental income, parental wealth or parental risky shares. To the extent that the parental literacy categories are positively correlated with these omitted variables, the coefficient for parental education may be upward biased. The parental channel we estimate should therefore be read as an upper bound on the pure educational effect, since some of the estimated effect may be a result of other omitted variables.

Second, the Merton inversion does not yield the excess return that households actually receive in equity markets. It recovers the excess return that would rationalize their observed risky share under the model. If a household's low participation rate reflects a fixed cost, a more conservative preference parameter θ or any other behavioral bias outside our framework, our procedure attributes the behavior to a lower implied excess return. As such, when we estimate a higher risky share for business economist households than for primary households, our model interprets the gap as a difference in excess returns. An equally plausible interpretation is that business economist households have a higher coefficient of relative risk aversion. The assumption of homogeneous preference parameters across literacy groups is of course a simplifying assumption. Vissing-Jørgensen (2002) and Vissing-Jørgensen and Attanasio (2003) show that preferences differ substantially between stock-market participants and non-participants and thus across educational groups.

10.3 Further Research

The points above suggest four natural areas that we believe warrant further investigation but lie beyond the scope of this thesis.

Model Extensions

As described in Section 2, a handful of model extensions are well documented to have an impact on the predictions of the classical buffer-stock models. We therefore review several of these extensions that would have been desirable to include in our model framework.

First, it is well documented that fixed costs of investing, whether monetary, time-related or psychological, can rationalize non-participation in the financial markets (Haliassos & Michaelides, 2003; Vissing-Jørgensen, 2002). The introduction of a fixed cost of participation, even if small, may have

allowed us to better match the data on risky shares, as the low-wealth households would have had a larger incentive to stay out of the market.

Second, we do not have an explicit consumption floor in our model. Gourinchas and Parker (2002) and Hubbard et al. (1995) find that a consumption floor acts as a safety net for the low-income group, decreasing the precautionary savings motive for them, which may rationalize the lack of savings that they exhibit.

Finally, we do not allow for the income process and return process to be correlated. This is a standard feature of model extensions of Cocco et al. (2005). Furthermore, it would have complicated the identification of our expected attainable excess returns due to the hedging term in the optimal risky shares, cf. equation (54) (Viceira, 2001). However, assuming a positive correlation between the returns and income process would have decreased the optimal risky share, which may have allowed us to match the data more closely.

Heterogeneous Starting Ages

All educational groups in our model enter working life at age 25. This assumption is generous for the highly educated households but unreasonably late for the least educated. Although the descriptive wealth-to-income ratios at age 25 in Section 6.3 suggest that primary-educated households accumulate little wealth before that age, a meaningful share of their working life takes place between graduation at age 15 and 25 and is unrepresented in our framework. Campbell et al. (2003) adopt age 22 as the starting age for their low-education group, which, in a Danish context for primary graduates, may even be too late. Therefore, a proper cost-benefit analysis of educational policy would require this extension: primary graduates avoid the public cost of higher education and begin contributing tax revenue years before their more educated counterparts. Both effects accrue to society but are absent from our welfare calculation.

Epstein–Zin Preferences

Under the CRRA preferences adopted in Section 3.2, the coefficient of relative risk aversion θ is tied to the elasticity of intertemporal substitution. This is convenient for tractability but generates a tension that runs through the empirical literature. Portfolio-choice models typically require θ in the range of 5–10 to match observed risky shares and the equity premium, while consumption-Euler-equation estimates of the intertemporal elasticity are around 0.3–1 (Gourinchas & Parker, 2002; Vissing-Jørgensen, 2002). Under CRRA, the two cannot be simultaneously satisfied. Epstein–Zin preferences disentangle the two parameters and would allow our framework to match the joint distribution of consumption and portfolio choice more flexibly, at the cost of a higher-dimensional state space and a more demanding solution algorithm.

Simulated Method of Moments

We calibrate the model by setting preference parameters to values commonly used, yielding return differences that match prior literature, although market data suggests higher excess returns. An alternative, often done in the literature, would be to estimate the parameters (θ, β, b_1) , possibly separately by literacy group, via the simulated method of moments (see, for example, Cagetti (2003)). This would generate parameter estimates that match the empirical wealth-to-income profiles, the participation profiles and the conditional risky shares simultaneously, would deliver formal standard errors on the recovered parameters, and would allow us to perform specification tests of the model against the data. Given more time and computational resources, simulated method of moments would be the natural way to test whether the residual 60% of the empirical inequality that our calibration leaves unexplained reflects parameter misspecification or genuine economic mechanisms outside our framework.

11 Conclusion

This thesis quantifies the extent to which heterogeneity in financial knowledge, and its intergenerational transfers, accounts for the observed wealth inequality at retirement across Danish households. We embed financial knowledge into the canonical buffer-stock model with portfolio choice and calibrate the resulting framework using the Danish administrative registers. The model reproduces approximately 40% of the empirical wealth-to-income gap between the lowest and highest financial-literacy groups at retirement. A counterfactual analysis in which the lowest literacy households are endowed with the excess returns of highest literacy households closes 52.2% of the simulated gap and delivers a 3.71% consumption-equivalent welfare gain for the lowest group corresponding to 7,500 DKK per period. Thus, the financial-knowledge channel rationalizes 20.7% of the empirical inequality. We further document a synergy effect between own and parental literacy: the welfare cost of low financial knowledge is substantially larger when parental literacy is also low.

These results contribute to the literature by extending Cocco et al. (2005) with heterogeneous financial knowledge, including an intergenerational dimension and by providing the first quantification of the parental-literacy channel in a life-cycle portfolio-choice framework calibrated on Danish register data. Our welfare estimates are silent on the optimal policy instrument for raising financial knowledge in the lowest educational group, but they establish a meaningful upper bound on the welfare gain that any such instrument could in principle deliver. Whether that gain is best captured through educational reform, through targeted informational campaigns, or through some combination of the two is a question we leave to subsequent empirical work.

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A Tables

Table A.1: Descriptive Statistics of the Household–Year Sample

	Mean	SD	p10 [†]	p50 [†]	p90 [†]
<i>Panel A: Household demographics</i>					
Age (head)	55.23	19.38	30.00	54.00	82.78
Household type (ie_type)	1.16	0.40	1.00	1.00	2.00
Household size	1.79	1.14	1.00	1.00	4.00
Number of adults	1.46	0.66	1.00	1.00	2.00
Number of children	0.32	0.74	0.00	0.00	2.00
Male (0/1)	0.49	0.50	0.00	0.00	1.00
Married (0/1)	0.30	0.46	0.00	0.00	1.00
Widowed (0/1)	0.16	0.37	0.00	0.00	1.00
Divorced (0/1)	0.17	0.38	0.00	0.00	1.00
Age, youngest member	46.73	27.91	5.00	51.00	82.00
Age, oldest member	55.71	19.30	30.00	54.00	83.00
Equivalence factor	1.33	0.45	1.00	1.00	2.10
<i>Panel B: Education and financial literacy</i>					
Years of schooling, head	12.72	4.58	8.00	11.00	18.00
Years of schooling, partner	13.73	3.99	8.00	15.00	18.00
Years of schooling, parent	13.82	3.35	8.00	14.25	18.00
Education category	11.37	3.37	5.00	12.00	16.00
Folkeskole 9th-grade GPA	6.45	3.02	2.00	7.00	10.28
Financial literacy (0/1/2)	0.46	0.64	0.00	0.00	1.00
Financial literacy, parent (0/1/2)	0.09	0.32	0.00	0.00	0.00
<i>Panel C: Income (DKK)</i>					
Household income, gross	491,955	372,530	190,266	331,754	1,059,813
Equivalised gross income	351,228	195,789	175,391	279,126	617,705
Household disposable income	361,622	260,776	152,007	254,005	753,062
Individual income	258,517	128,832	139,102	213,494	439,096
<i>Panel D: Wealth and debt (DKK)</i>					
Financial assets (gross)	364,144	713,486	9,313	85,406	975,845
Financial assets, net of cash	138,672	473,341	0	0	321,864
Total assets	2,612,812	3,423,580	91,687	1,190,110	7,234,996
Total net assets	1,992,327	2,937,365	23,393	825,685	5,634,774
Total debt	607,420	1,023,787	0	70,142	2,140,177
Stock-market participation (0/1)	0.31	0.46	0.00	0.00	1.00

Notes. The table reports descriptive statistics for the final household–year sample (sample size 8,490,918, with the partner-, parent-, and Folkeskole-grade variables defined over reduced sub-samples). Monetary variables are reported in nominal DKK and have been winsorised at the bottom 5% and top 1% of the equivalised disposable income distribution.

[†] Reported percentiles are pseudo-percentiles (averages of ± 100 adjacent observations).

Table A.2: Income–Age Profiles by Financial Literacy: Cohort-Effects Specification

	Primary		Master's		Business	
	(1) Third	(2) Fifth	(3) Third	(4) Fifth	(5) Third	(6) Fifth
<i>Panel A: Age polynomial</i>						
Age	−0.018*** (0.001)	−0.273*** (0.028)	0.150*** (0.002)	0.255*** (0.047)	0.222*** (0.003)	0.841*** (0.079)
Age ² × 10 ²	0.101*** (0.003)	1.630*** (0.133)	−0.254*** (0.005)	−0.029 (0.224)	−0.416*** (0.008)	−2.580*** (0.378)
Age ³ × 10 ⁴	−0.080*** (0.002)	−4.350*** (0.309)	0.165*** (0.004)	−2.050*** (0.523)	0.281*** (0.006)	3.460*** (0.889)
Age ⁴ × 10 ⁶	— —	5.630*** (0.350)	— —	4.400*** (0.597)	— —	−1.480 (1.020)
Age ⁵ × 10 ⁸	— —	−2.820*** (0.155)	— —	−2.760*** (0.267)	— —	−0.270 (0.462)
Constant	11.500*** (0.017)	13.080*** (0.228)	9.493*** (0.027)	7.396*** (0.387)	8.646*** (0.047)	2.280*** (0.643)
<i>Panel B: Specification and fit</i>						
Time effects	No	No	No	No	No	No
Cohort effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,861,609	2,861,609	2,160,761	2,160,761	609,816	609,816
R ²	0.103	0.103	0.095	0.097	0.119	0.121
Adj. R ²	0.103	0.103	0.095	0.097	0.119	0.121

Notes. The dependent variable is log labor income. Each column reports pooled OLS estimates from a separate regression of log income on a polynomial in age and a full set of cohort dummies, with 2019 as the omitted base year. Errors are clustered. Cohort-dummy coefficients are estimated but suppressed for compactness. “Third” columns use a third-order polynomial in age; “Fifth” columns use a fifth-order polynomial. Higher-order age coefficients and their standard errors are reported on the scale indicated in the row label. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

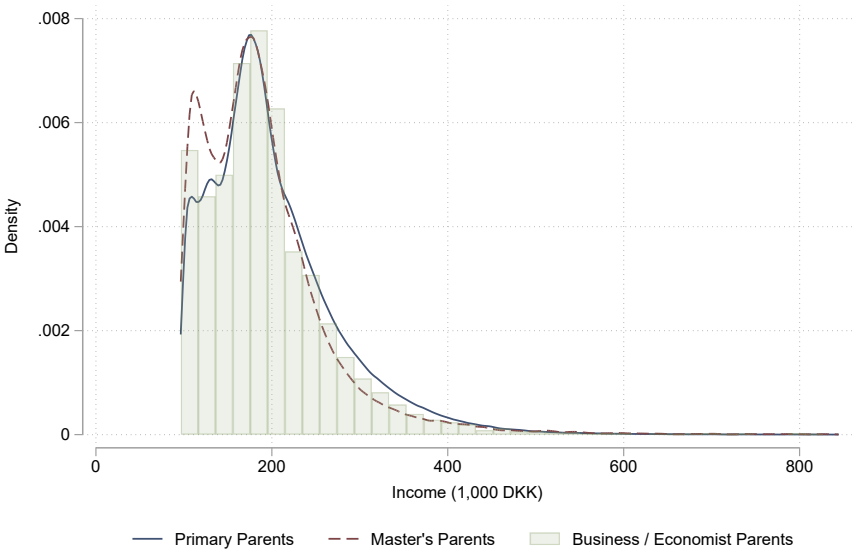
Table A.3: Full: Determinants of the Household Risky Asset Share

	(1) $\hat{\alpha}$
<i>Panel A: Financial Literacy</i>	
Finlit = 1	0.0162*** (0.0018)
Finlit = 2	0.0410*** (0.0028)
Parental finlit = 1	0.0076*** (0.0016)
Parental finlit = 2	0.0241*** (0.0042)
Finlit = 1 $\times$ Parental = 1	0.0190*** (0.0023)
Finlit = 1 $\times$ Parental = 2	0.0170*** (0.0047)
Finlit = 2 $\times$ Parental = 1	0.0179*** (0.0033)
Finlit = 2 $\times$ Parental = 2	0.0200*** (0.0053)
<i>Panel B: Resources and Ability</i>	
Folkeskole Math 9th-grade GPA	0.0050*** (0.0002)
HH gross income $\times 10^9$	-4.85*** (1.64)
HH financial assets $\times 10^7$	1.81*** (0.014)
Constant	-0.0021 (0.0039)
<i>Panel C: Controls and Specification</i>	
Demographic controls	Yes
Year fixed effects	Yes
Observations	362,750
R^2	0.192
Adj. R^2	0.192

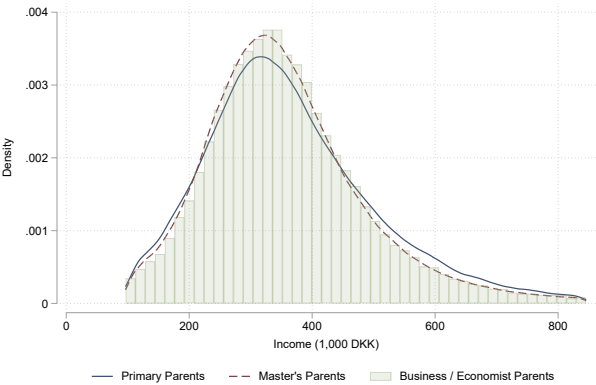
Notes. The dependent variable is the household risky asset share. Column (1) reports OLS estimates with robust standard errors in parentheses. Demographic controls include gender, marital status, number of children, and age. Year fixed effects (2014-2024) are included with 2019 as the base year. The omitted categories are Finlit = 0 and Parental finlit = 0. Coefficients on income and financial assets are scaled for readability. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Figures

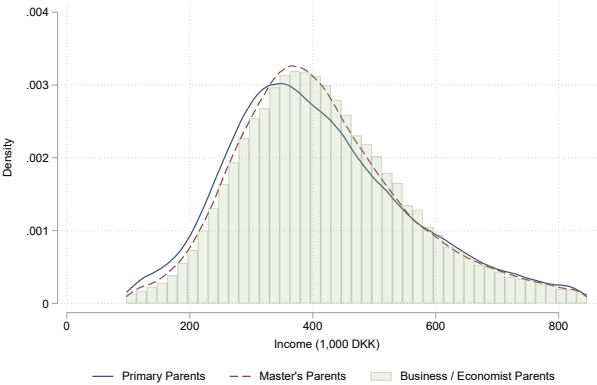
Figure B.1: Income Densities by Financial Literacy and Parental Financial Literacy



(a) Income densities, Primary

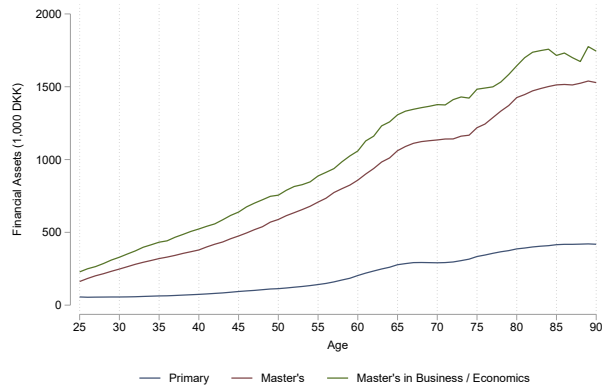
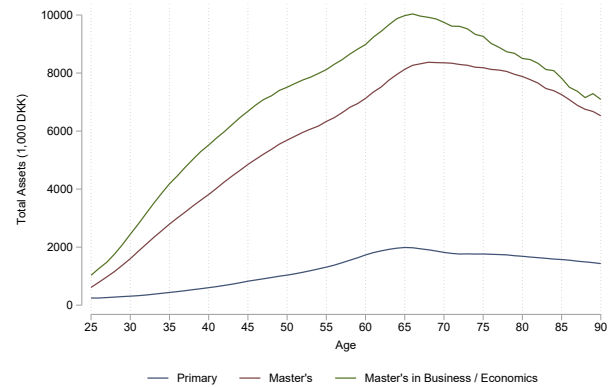


(b) Income densities, Master's

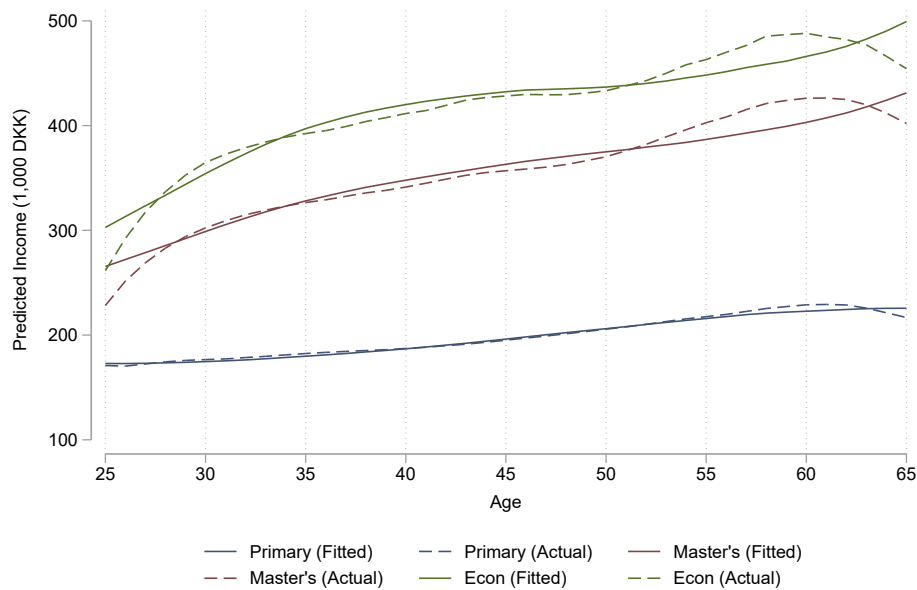


(c) Income densities, Business

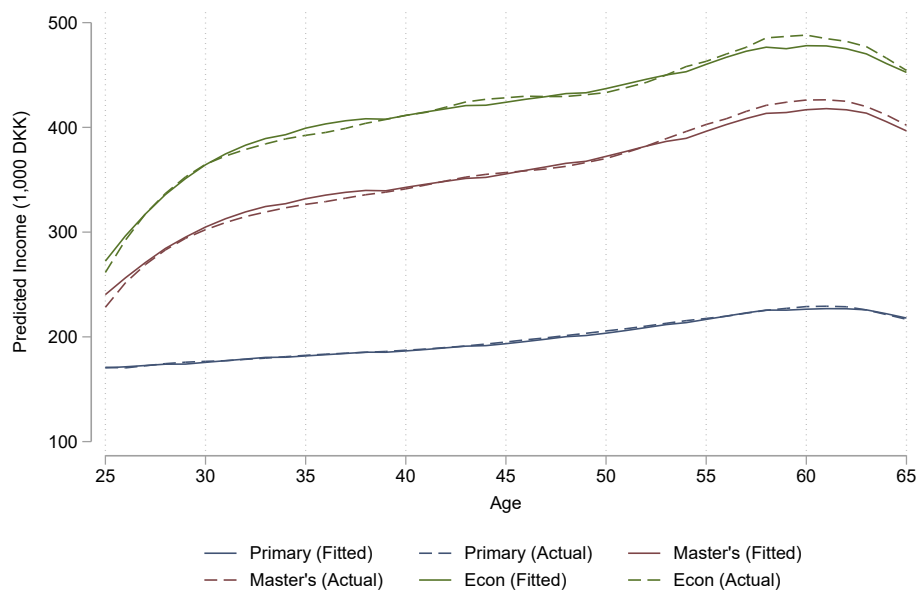
Notes. Income densities of Primary, Master's and Business graduates for different parental backgrounds.

Figure B.2: Financial and Total Net Assets of the Households**(a) Financial assets of the households****(b) Total net assets of the households**

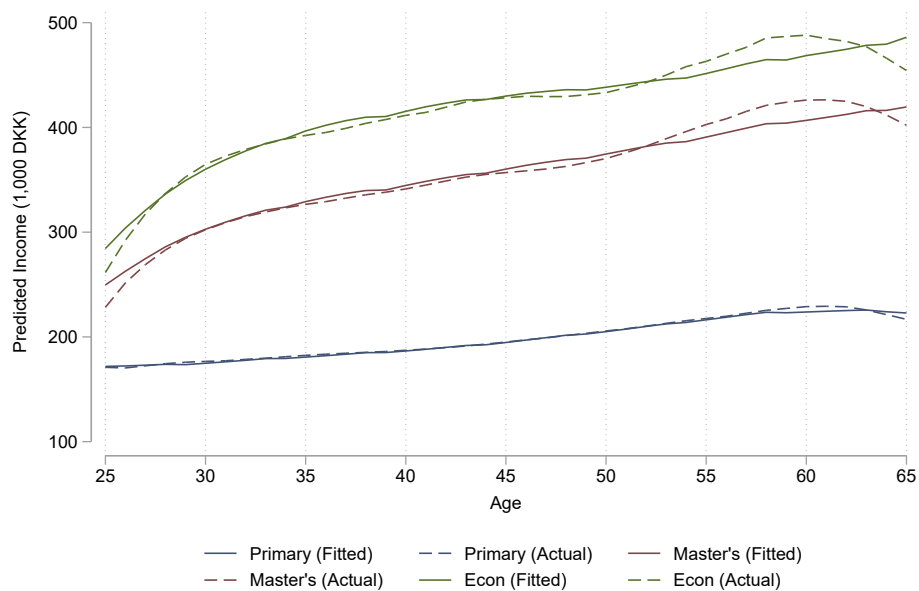
Notes. Panel (a) plots financial assets of the household (in 1,000 DKK) by age. Panel (b) Plots total net assets of the household (in 1,000 DKK) by age.

Figure B.3: Predicted Income-Age Profile, Third-Degree Time-Effects Specification

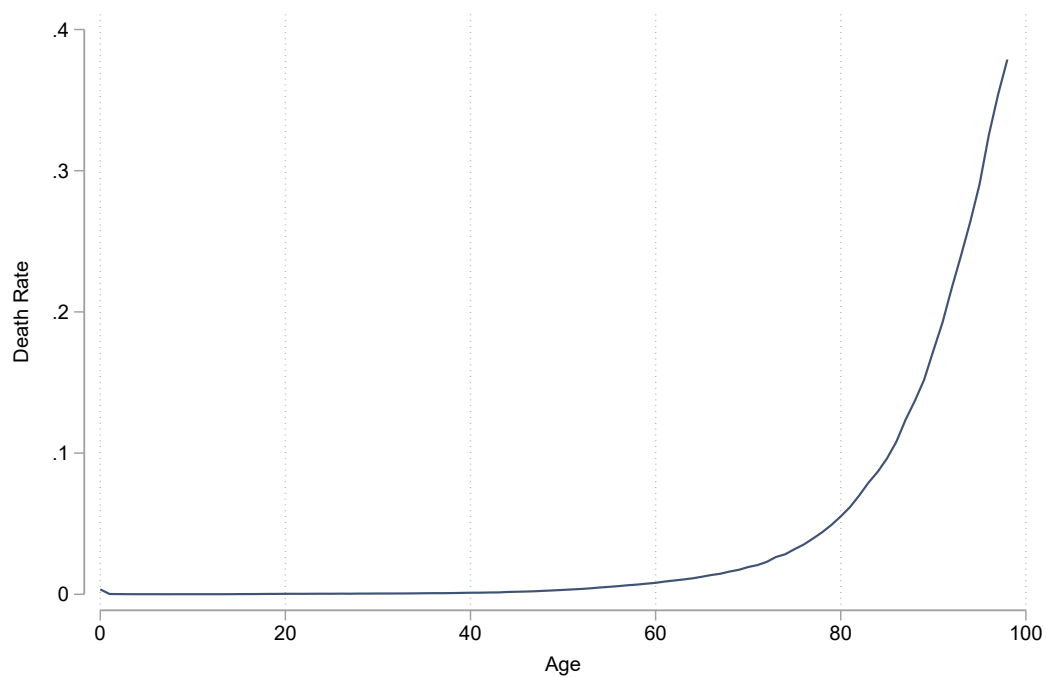
Notes. Predicted income-age profiles from using coefficients presented in Table 6 with a fifth-degree polynomial and time effects against their actual empirical observations.

Figure B.4: Predicted Income-Age Profile, Fifth-Degree Cohort-Effects Specification

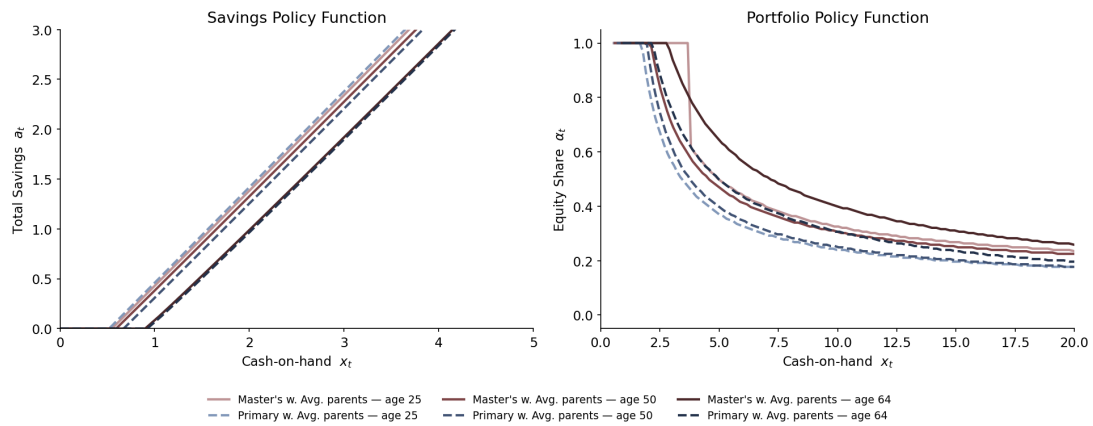
Notes. Predicted income-age profiles from using coefficients presented in Table A.2 with a fifth-degree polynomial and cohort effects against their actual empirical observations.

Figure B.5: Predicted Income-Age Profile, Third-Degree Cohort-Effects Specification

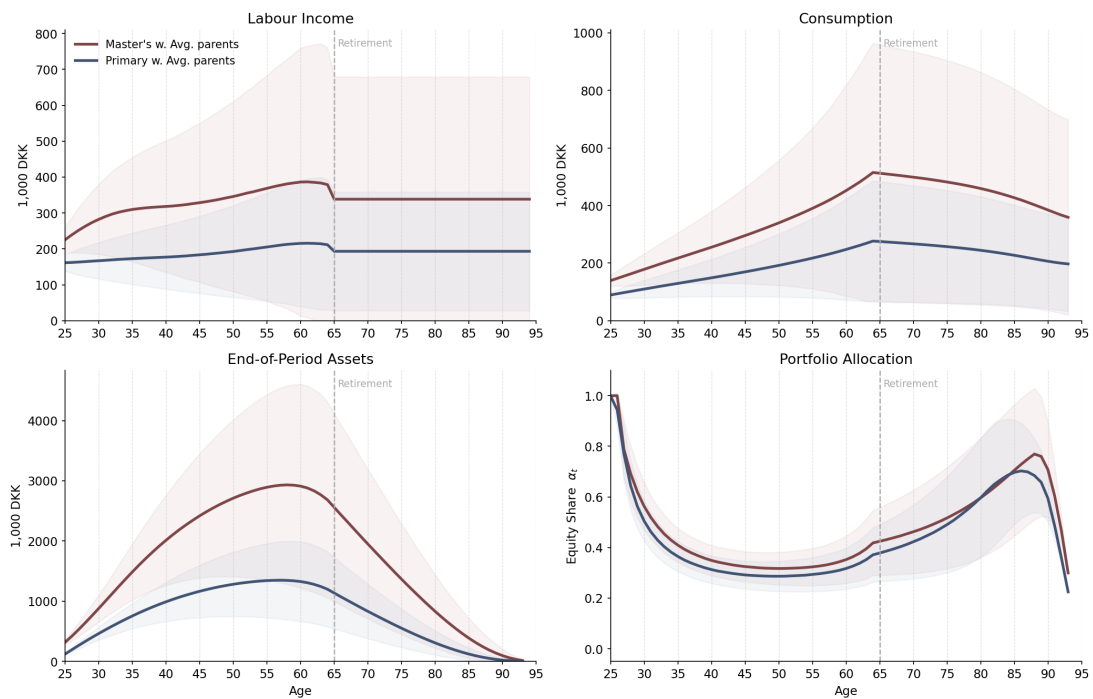
Notes. Predicted income-age profiles from using coefficients presented in Table A.2 with a third-degree polynomial and cohort effects against their actual empirical observations.

Figure B.6: Death Rate Calibration

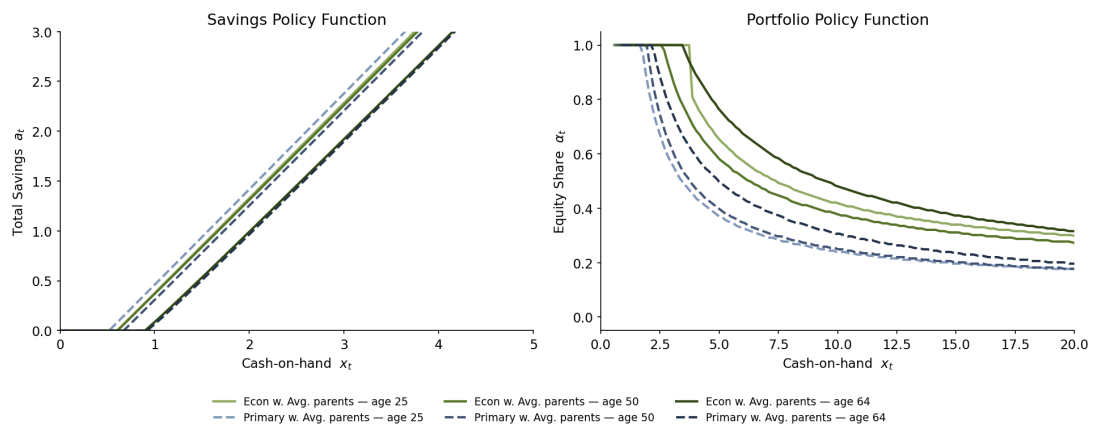
Notes. Danish Death Rates calibrated using Statistics Denmark's, StatBank. Tables BEF5 & DOD

Figure B.7: Policy Functions: Primary & Master's with Average Parents

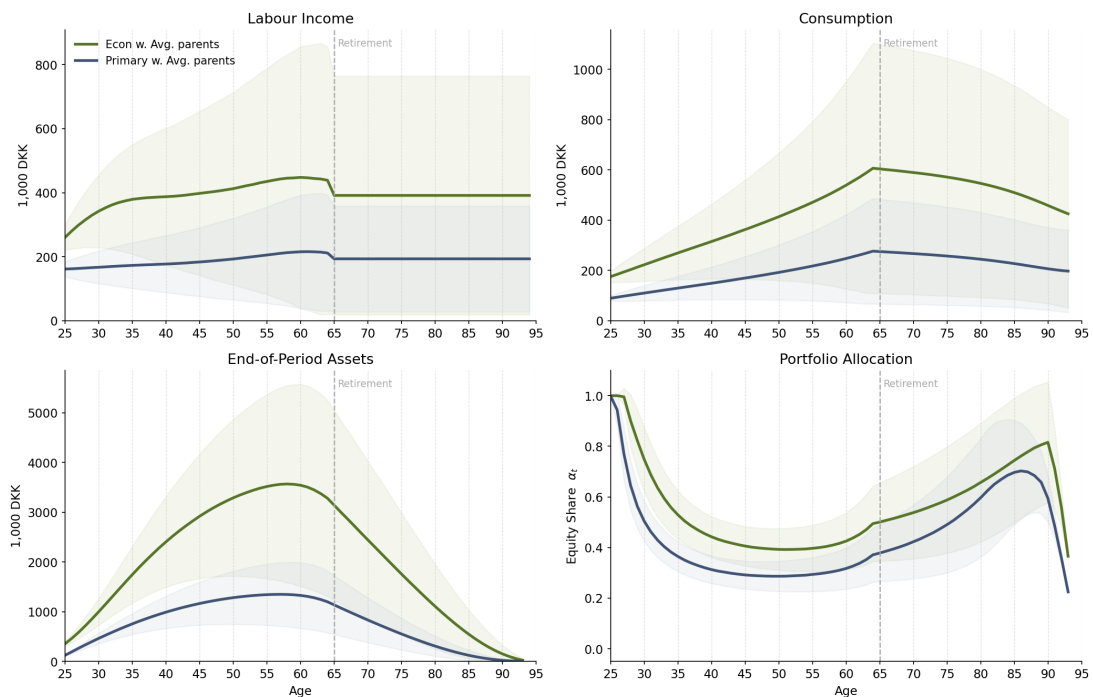
Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for households with average parents.

Figure B.8: Life-Cycle Profiles: Primary & Master's with Average Parents

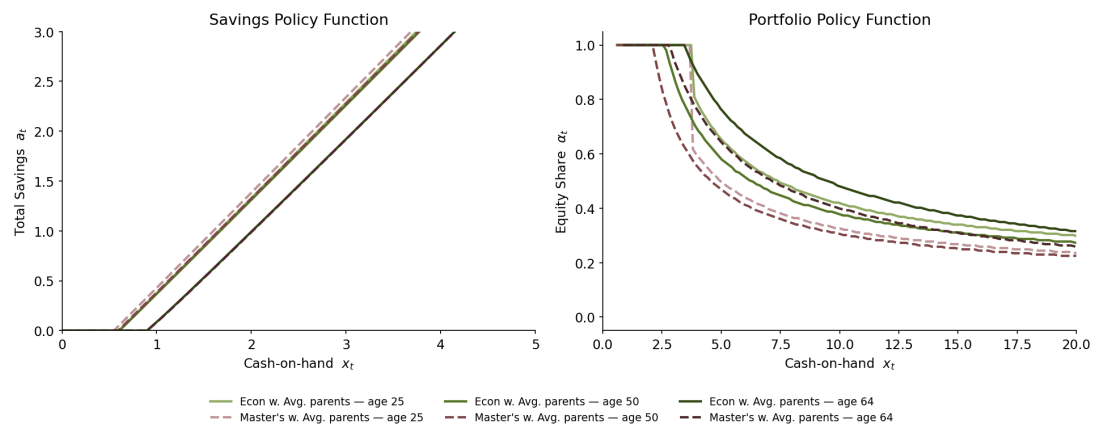
Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for households with average parents.

Figure B.9: Policy Functions: Primary & Business Economics with Average Parents

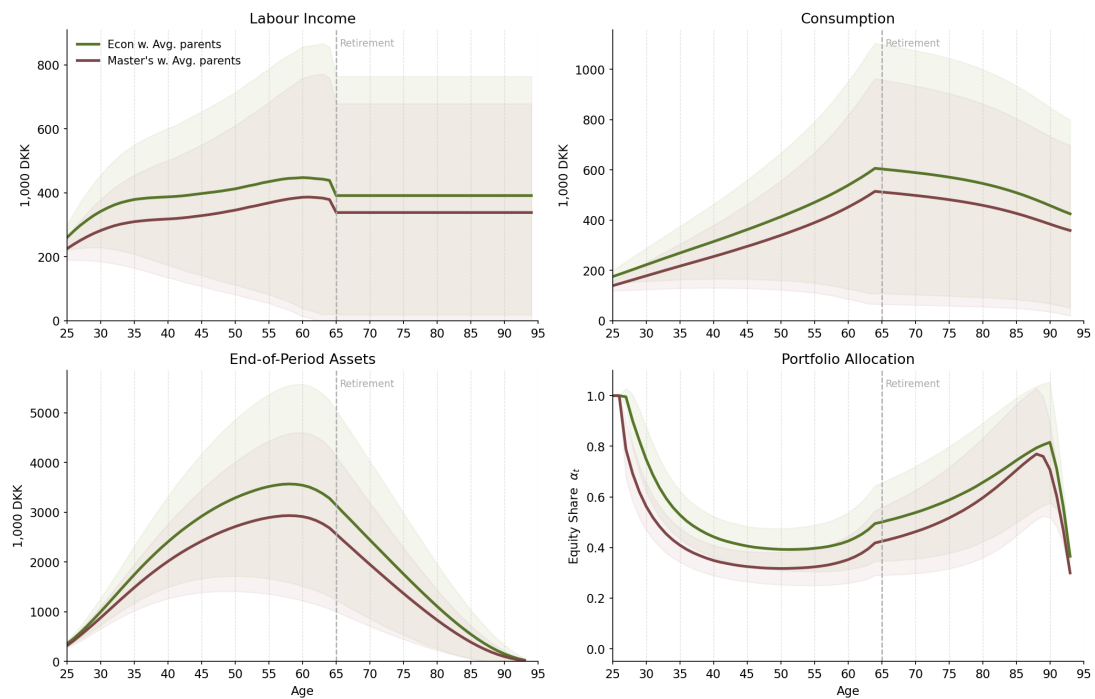
Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for households with average parents.

Figure B.10: Life-Cycle Profiles: Primary & Business Economics with Average Parents

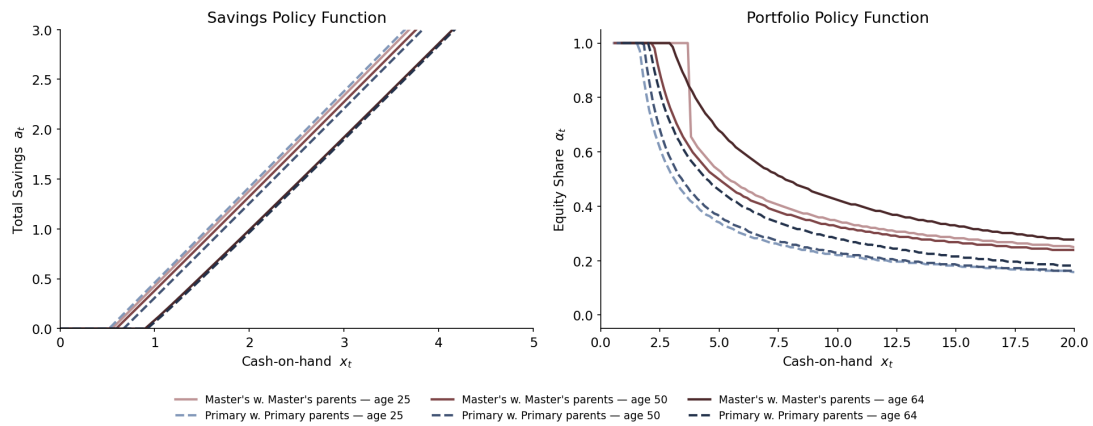
Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for households with average parents.

Figure B.11: Policy Functions: Master's & Business Economics with Master's Parents

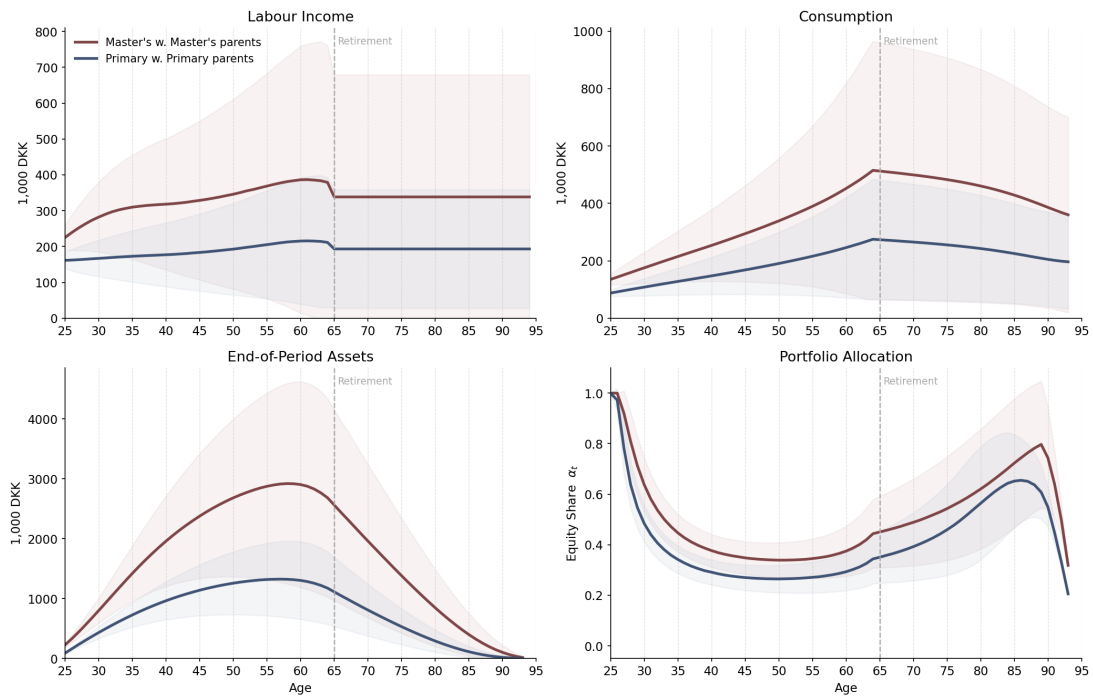
Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for households with Master's parents.

Figure B.12: Life-Cycle Profiles: Master's & Business Economics with Master's Parents

Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for households with Master's parents.

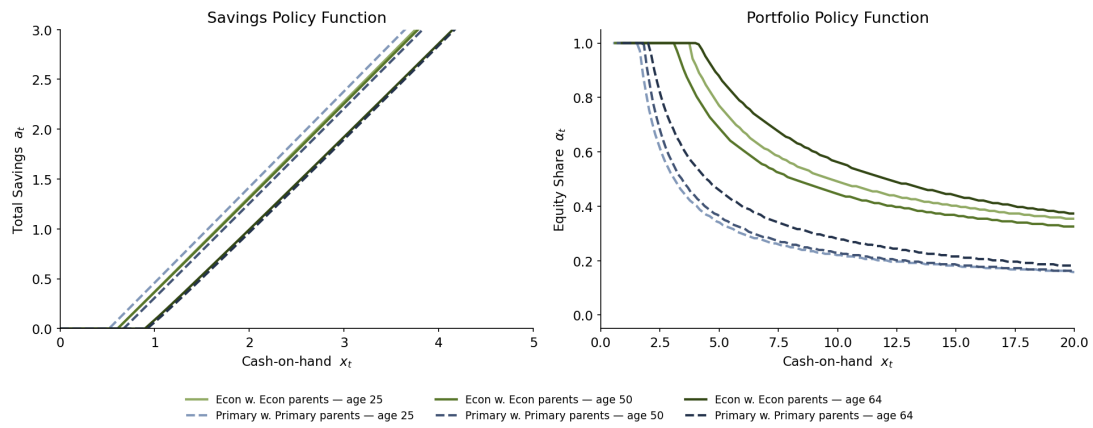
Figure B.13: Policy Functions: Master's with Master's Parents vs. Primary with Primary Parents

Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for a (1,1) and a (0,0) household.

Figure B.14: Life-Cycle Profiles: Master's with Master's Parents vs. Primary with Primary Parents

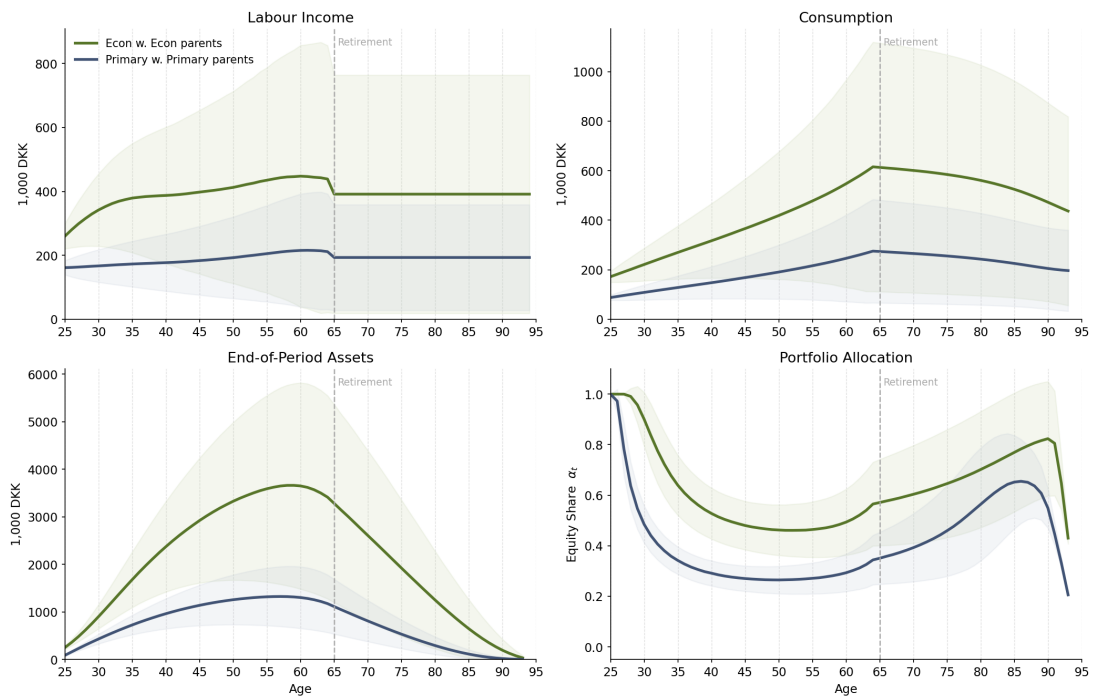
Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for a (1,1) and a (0,0) household.

Figure B.15: Policy Functions: Business Economist with Business Economics Parents vs. Primary with Primary Parents



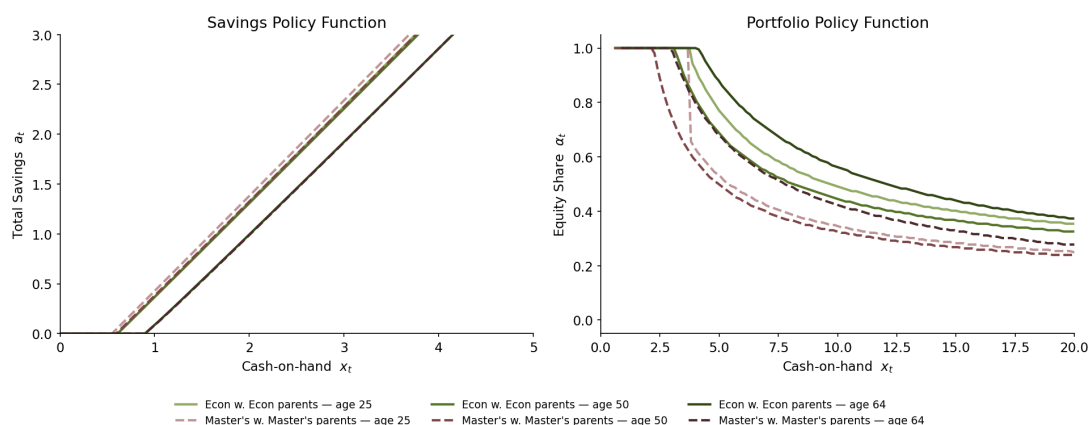
Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for a (2,2) and a (0,0) household.

Figure B.16: Life-Cycle Profiles: Business Economist with Business Economics Parents vs. Primary with Primary Parents



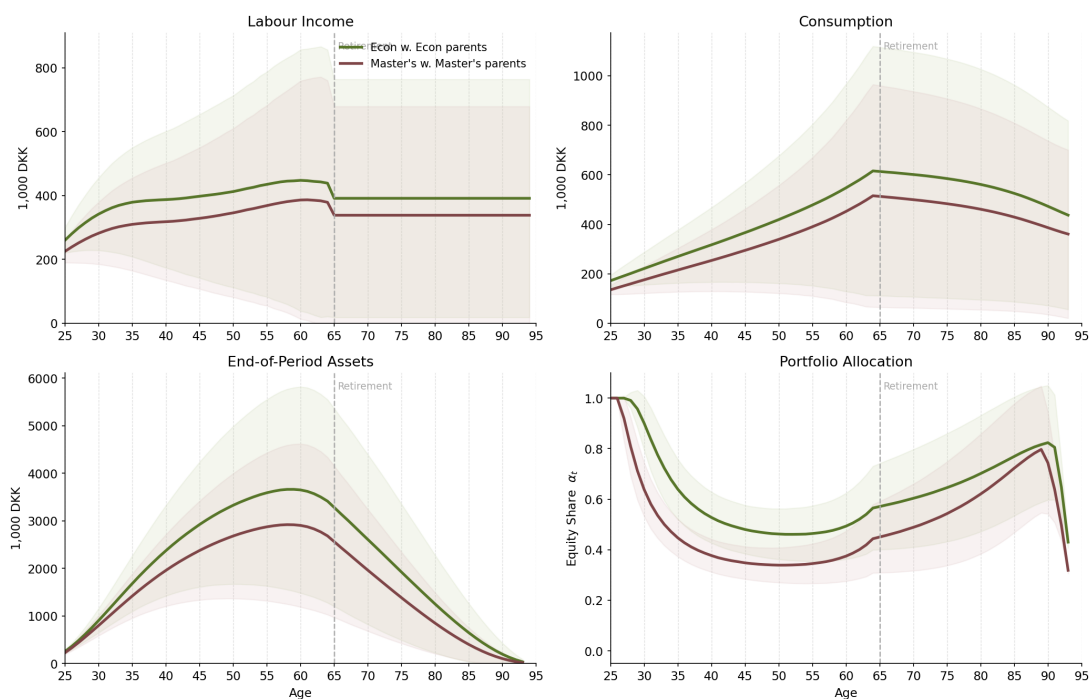
Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for a (2,2) and a (0,0) household.

Figure B.17: Policy Functions: Business Economist with Business Economics Parents vs. Master's with Master's Parents



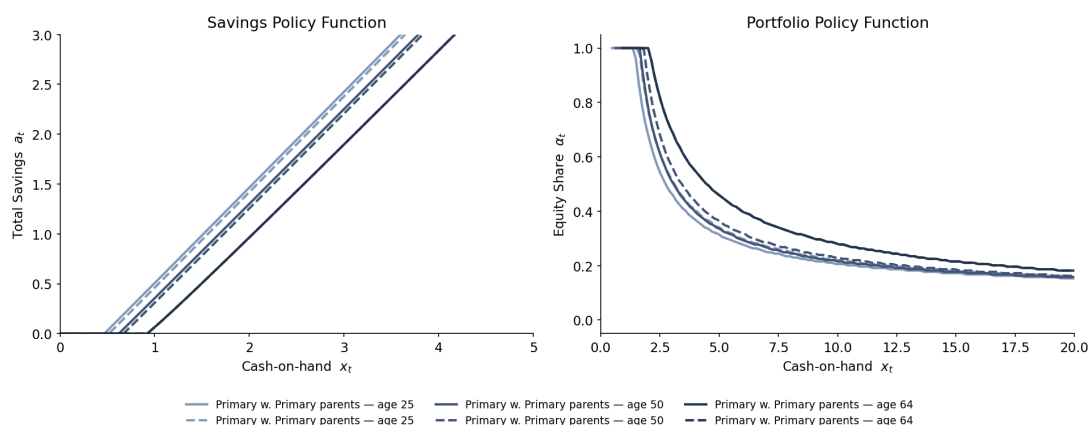
Notes. Savings policy $a_t(x_t)$ and portfolio policy $\alpha_t(x_t)$ as a function of cash-on-hand x_t at ages 25, 50, and 64 for a (2,2) and a (1,1) household.

Figure B.18: Life-Cycle Profiles: Business Economist with Business Economics Parents vs. Master's with Master's Parents



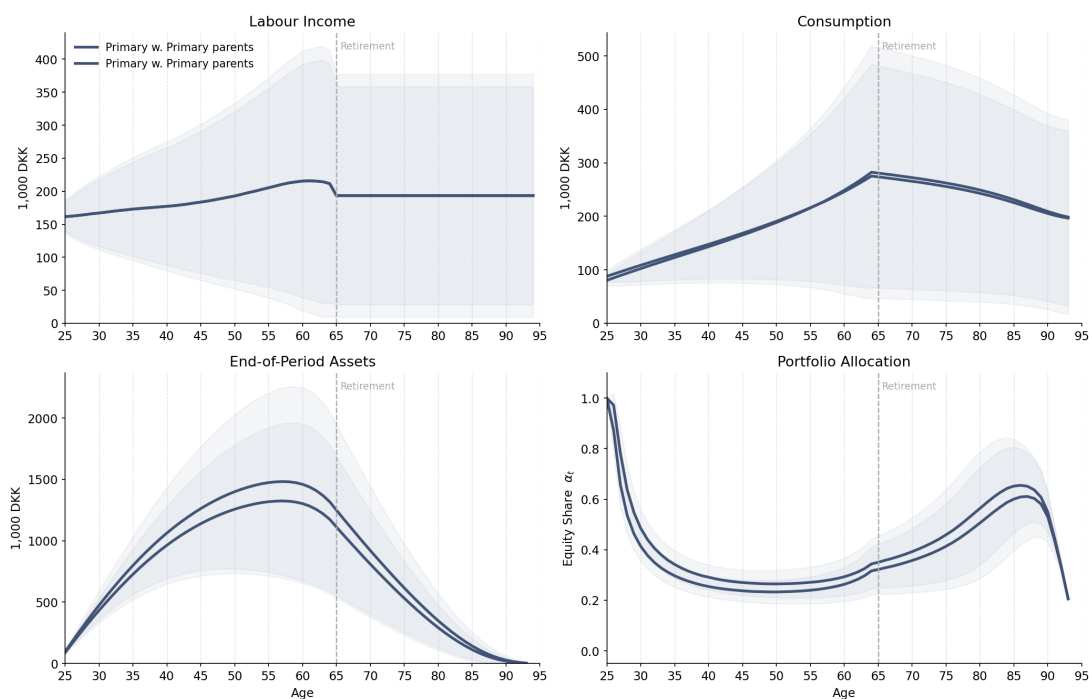
Notes. Simulated life-cycle profiles of labor income, consumption, end-of-period assets, and portfolio allocation for a (2,2) and a (1,1) household.

Figure B.19: Decomposition: Income Uncertainty Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



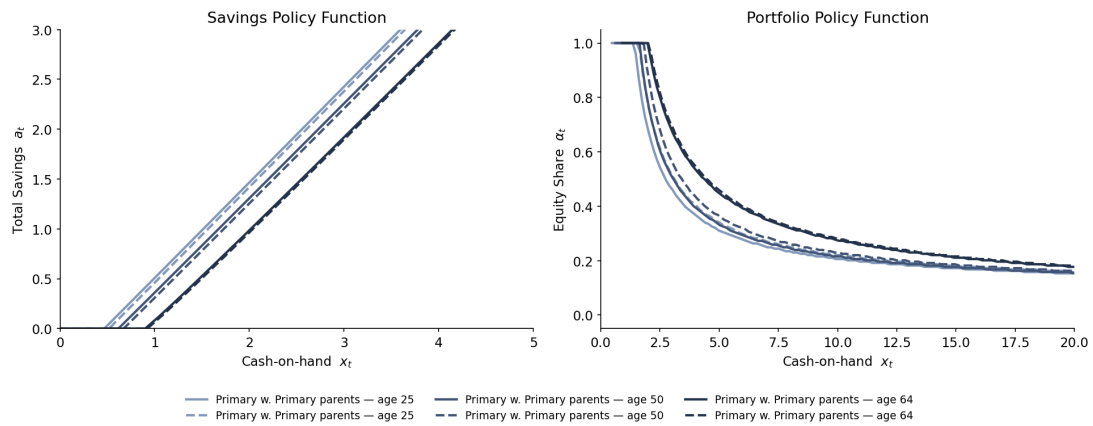
Notes. Policy functions for a (0,0) and (2,2) household when only the income uncertainty channel differs between groups.

Figure B.20: Decomposition: Income Uncertainty Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



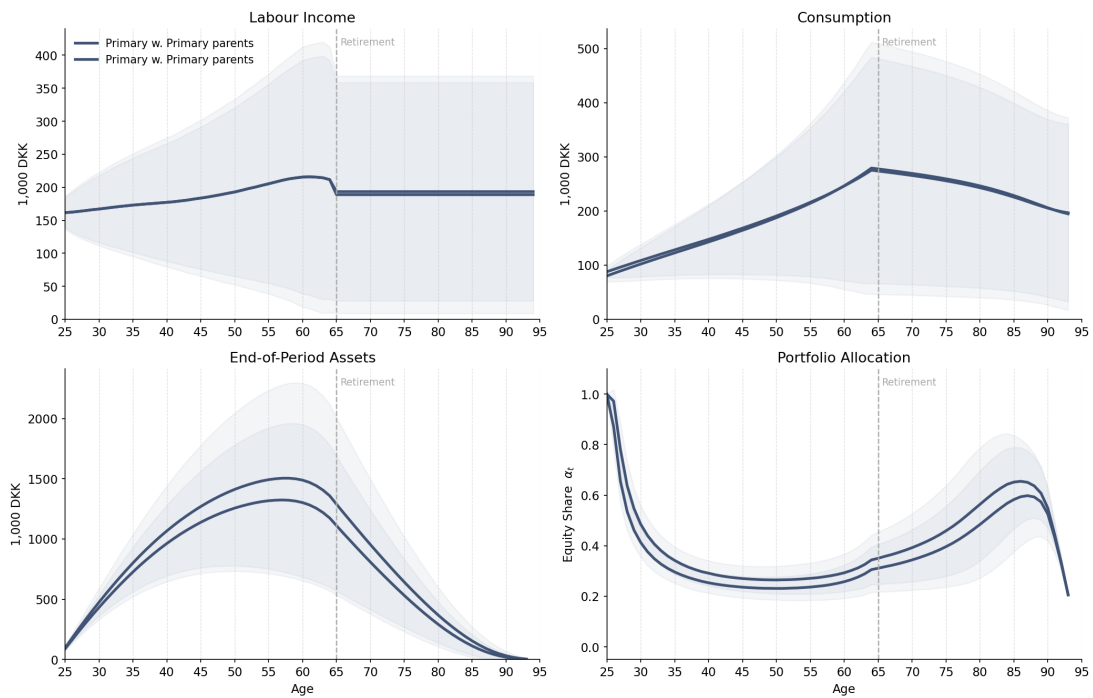
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the income uncertainty channel differs between groups.

Figure B.21: Decomposition: Replacement Rate Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



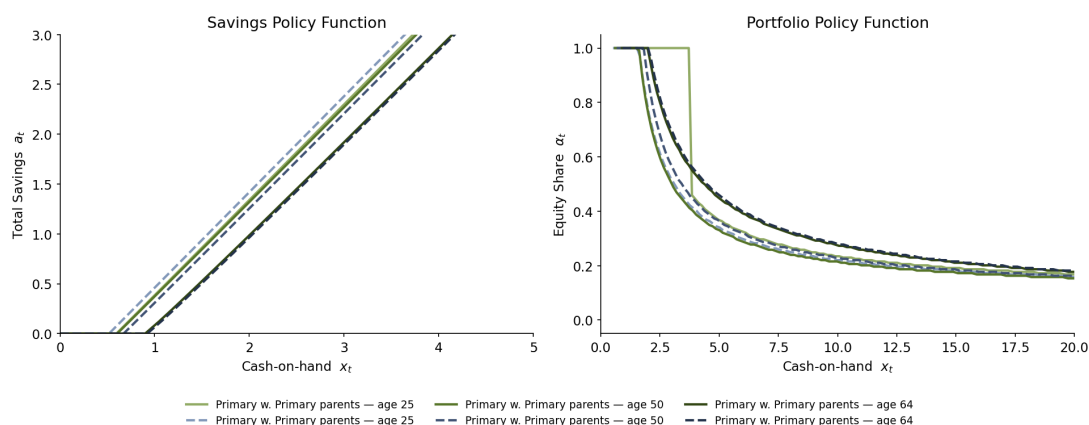
Notes. Policy functions for a (0,0) and (2,2) household when only the replacement rate channel differs between groups.

Figure B.22: Decomposition: Replacement Rate Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



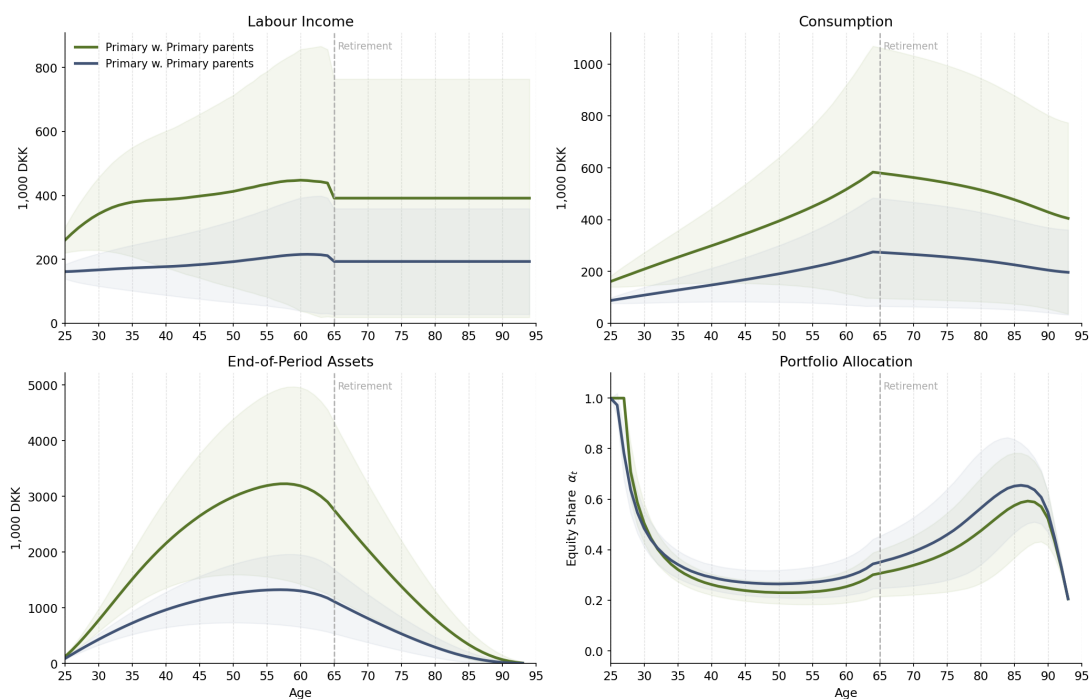
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the replacement rate channel differs between groups.

Figure B.23: Decomposition: Income Path Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



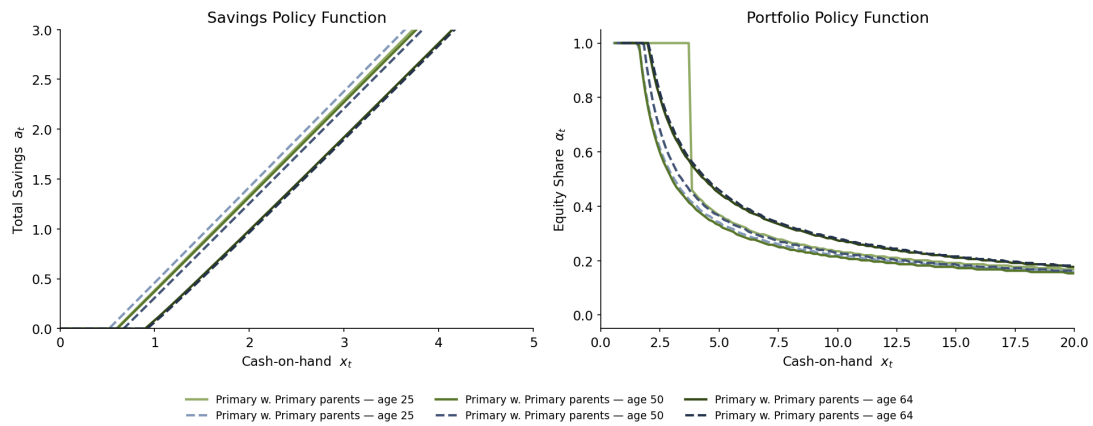
Notes. Policy functions for a (0,0) and (2,2) household when only the income path channel differs between groups.

Figure B.24: Decomposition: Income Path Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



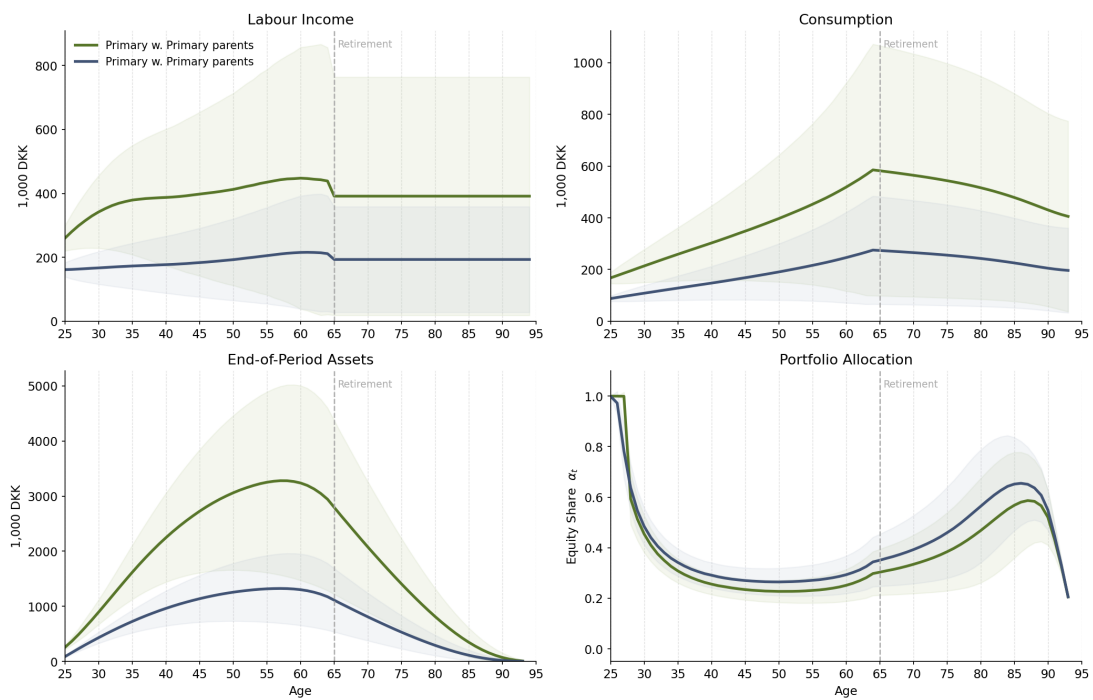
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the income path channel differs between groups.

Figure B.25: Decomposition: Initial Wealth Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



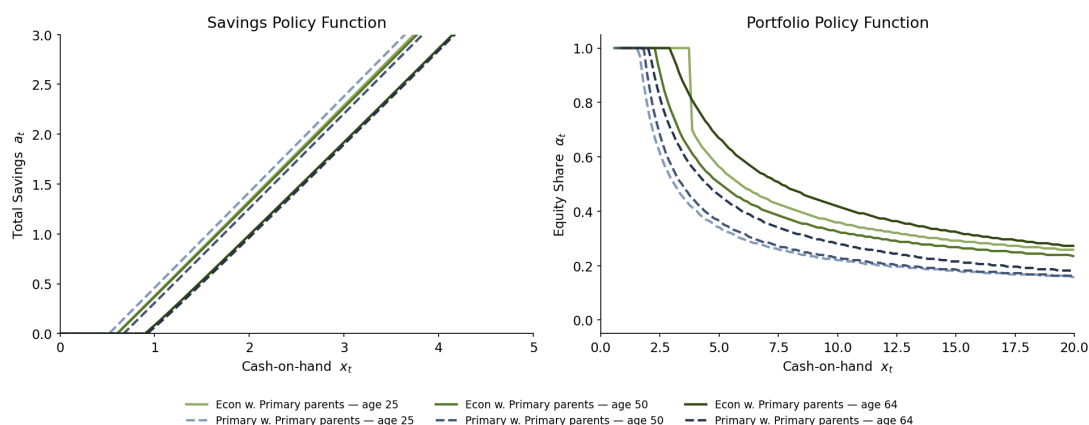
Notes. Policy functions for a (0,0) and (2,2) household when only the initial wealth endowment channel differs between groups.

Figure B.26: Decomposition: Initial Wealth Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



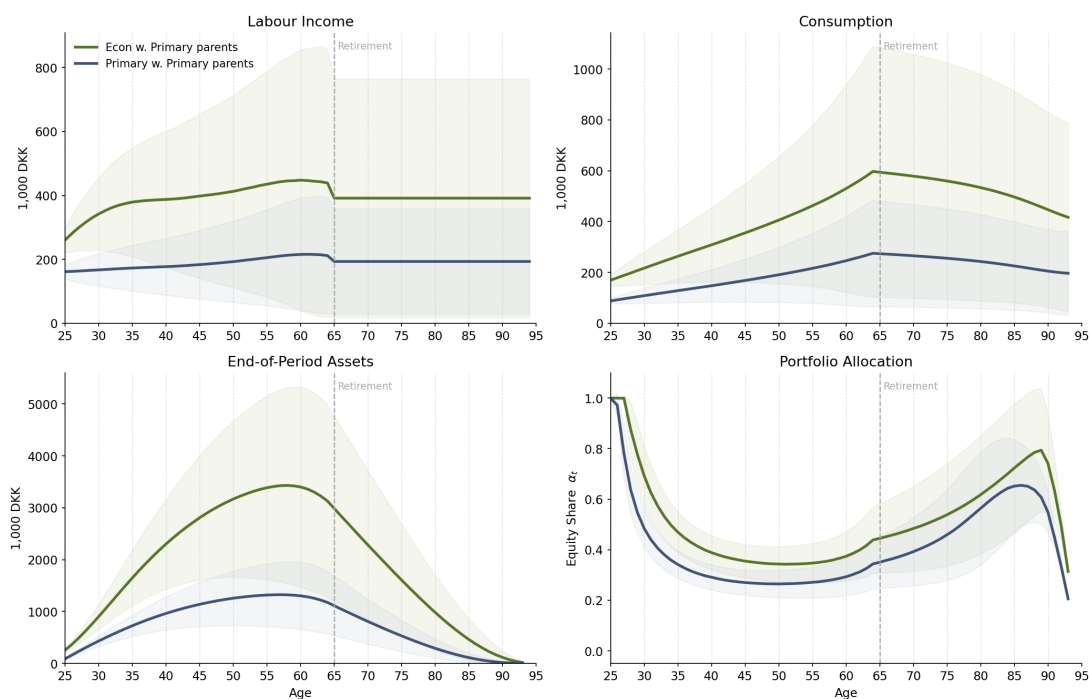
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the initial wealth endowment channel differs between groups.

Figure B.27: Decomposition: Own Education Return Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



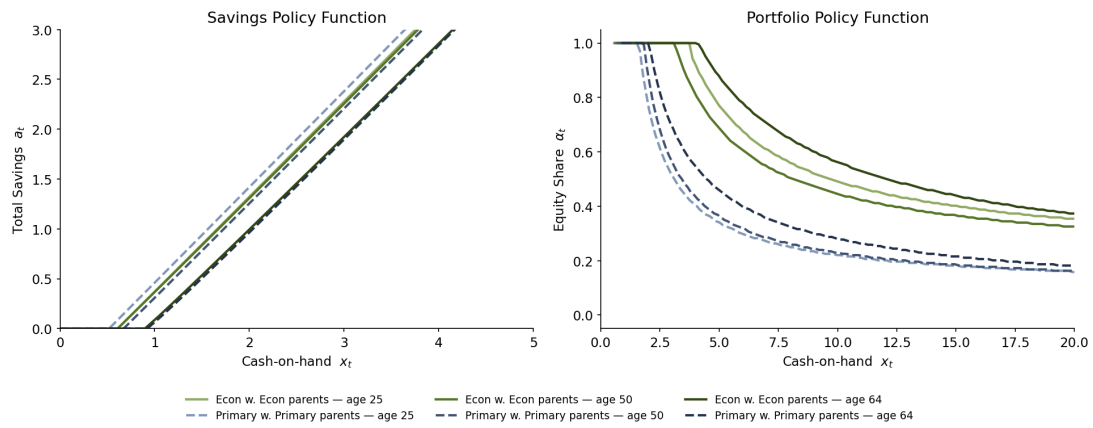
Notes. Policy functions for a (0,0) and (2,2) household when only the own-education excess return channel differs between groups.

Figure B.28: Decomposition: Own Education Return Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



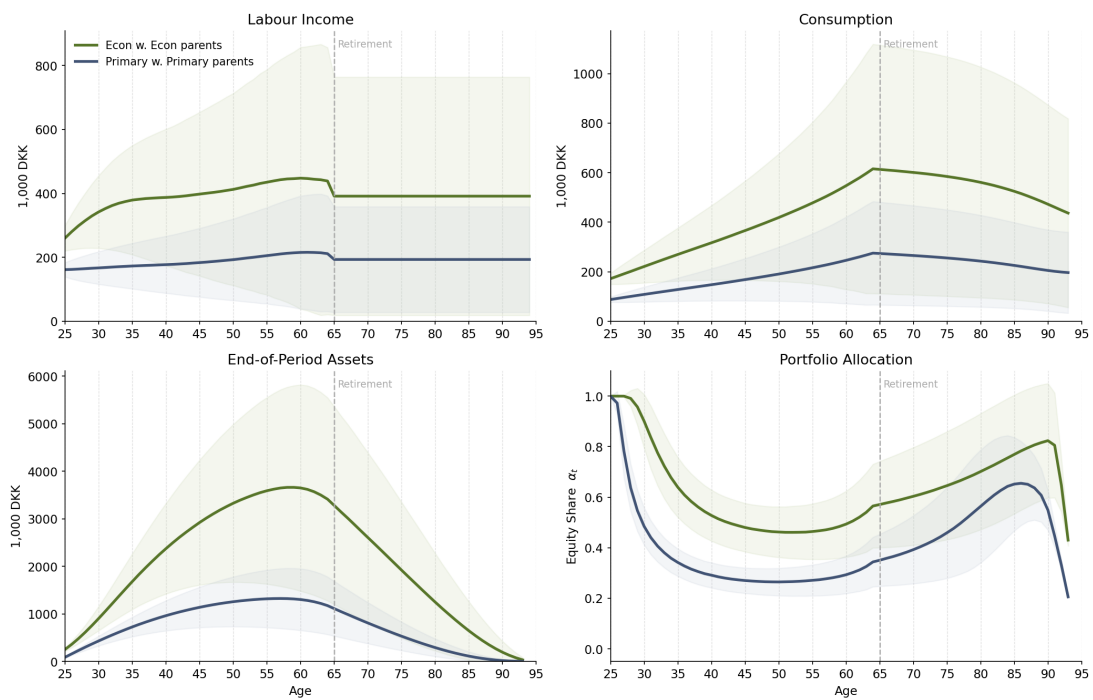
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the own-education excess return channel differs between groups.

Figure B.29: Decomposition: Parental Financial Literacy Return Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



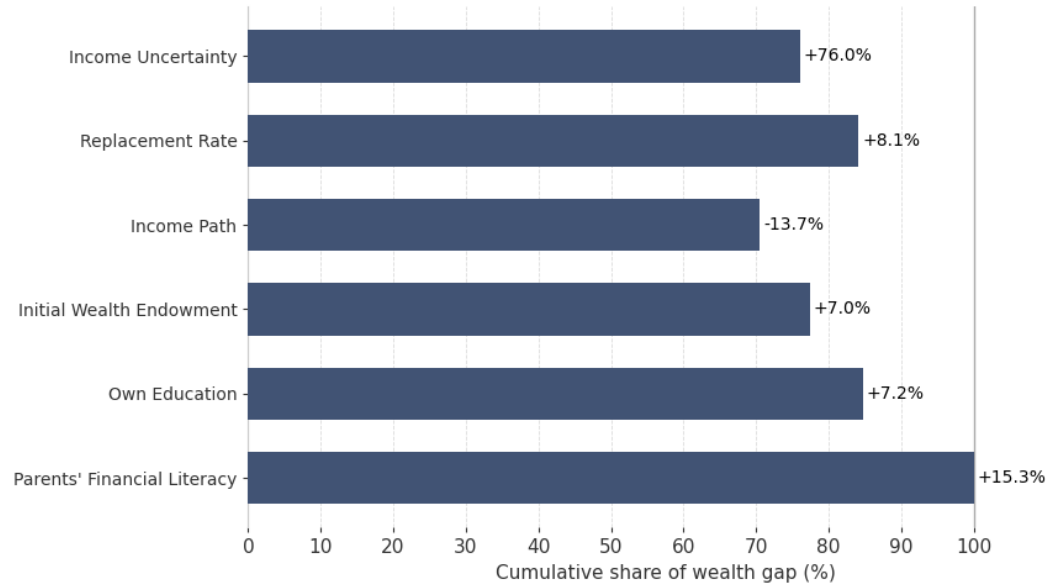
Notes. Policy functions for a (0,0) and (2,2) household when only the parental financial literacy excess return channel differs between groups.

Figure B.30: Decomposition: Parental Financial Literacy Return Channel, Business Economist with Business Economics Parents vs. Primary with Primary Parents



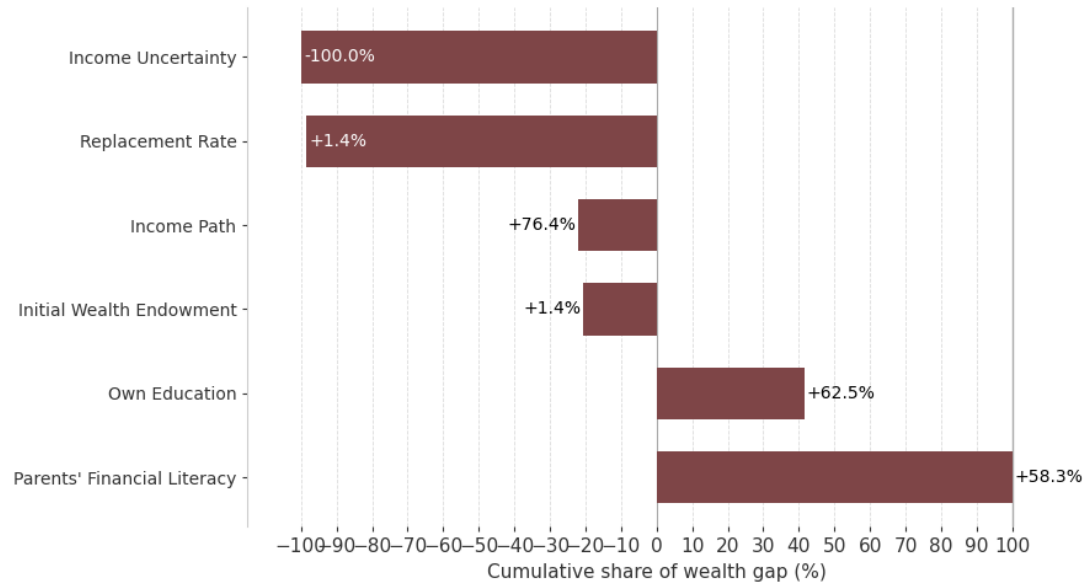
Notes. Life-cycle profiles for a (0,0) and (2,2) household when only the parental financial literacy excess return channel differs between groups.

Figure B.31: Wealth Inequality Decomposition: (0,0) vs. (1,1)



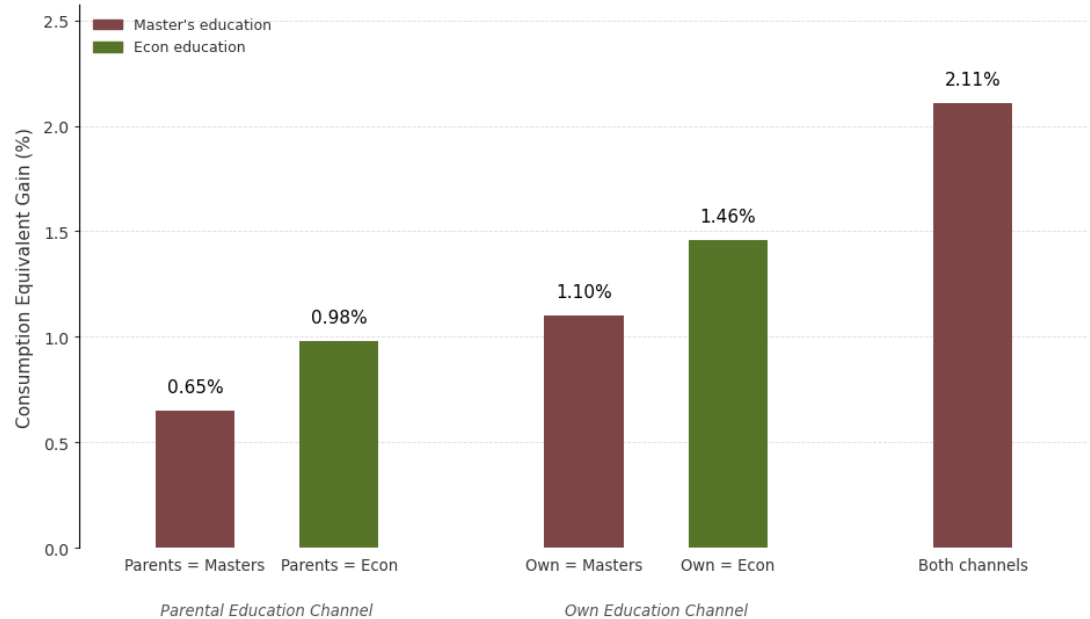
Notes. Decomposition of the simulated (1,1)/(0,0) wealth-to-income ratio at retirement into its constituent channels, analogous to Figure 11.

Figure B.32: Wealth Inequality Decomposition: (1,1) vs. (2,2)



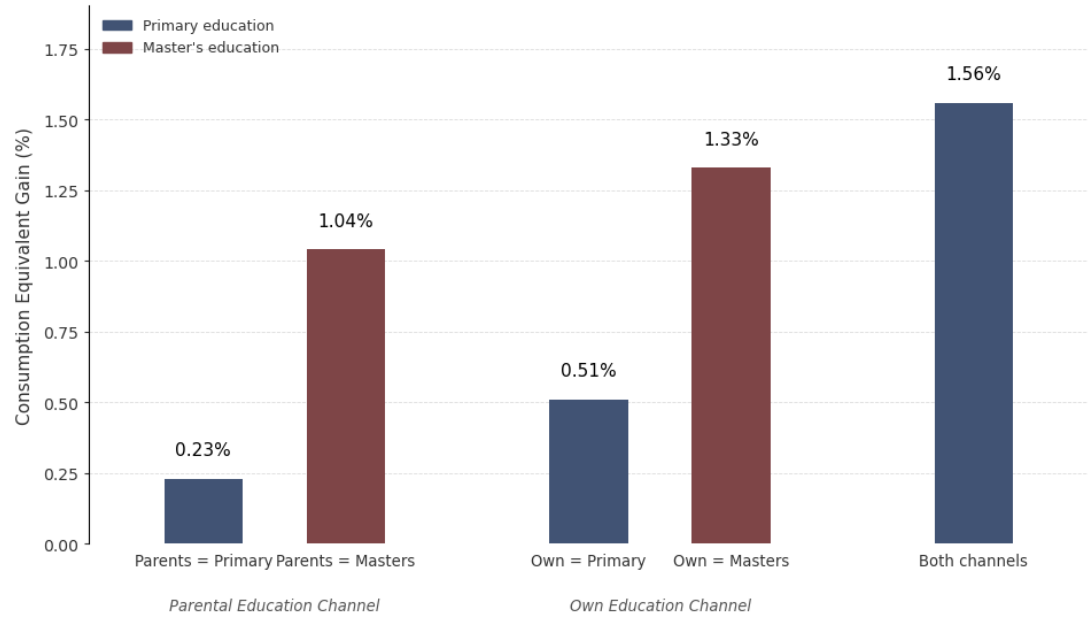
Notes. Decomposition of the simulated (2,2)/(1,1) wealth-to-income ratio at retirement into its constituent channels, analogous to Figure 11.

Figure B.33: Consumption Equivalent Decomposition: (0,0) vs. (1,1)

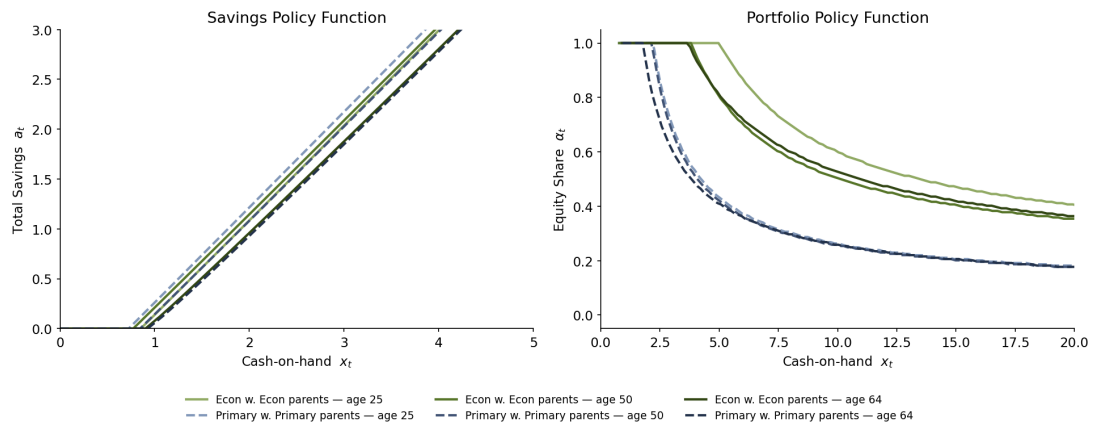


Notes. Consumption-equivalent welfare gains of moving a (0,0) household to (1,1) excess returns, decomposed into the parental education channel and the own education channel, analogous to Figure 12.

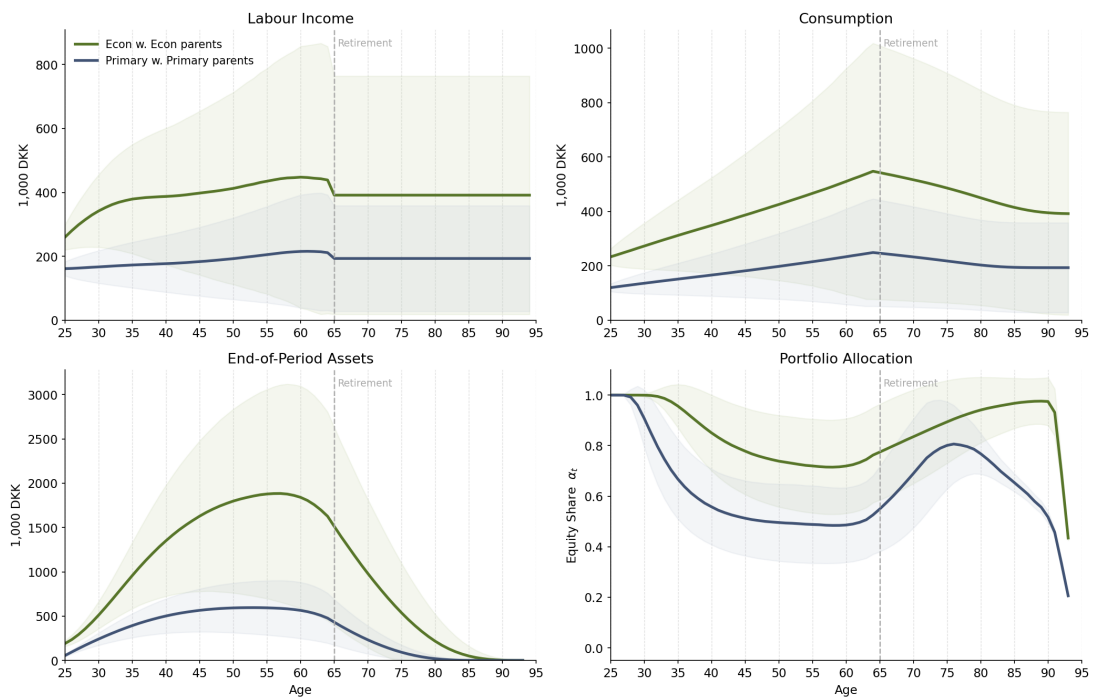
Figure B.34: Consumption Equivalent Decomposition: (1,1) vs. (2,2)



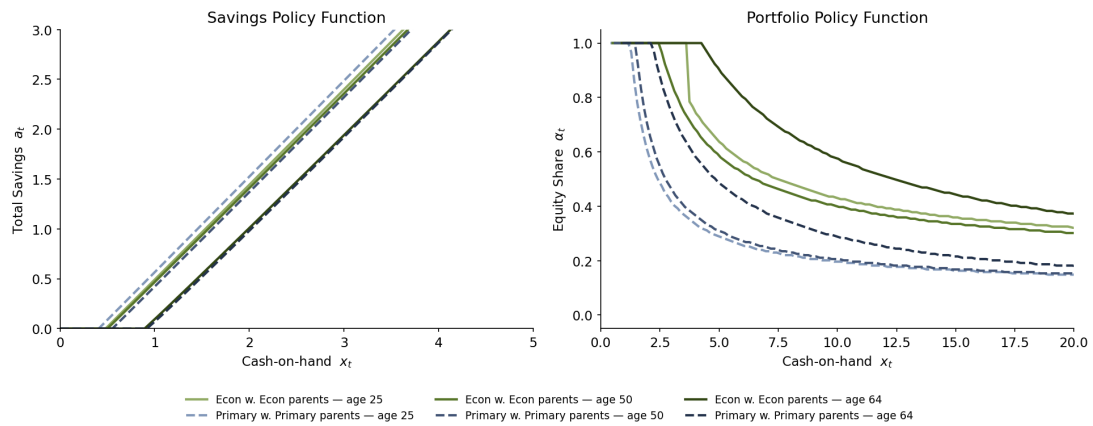
Notes. Consumption-equivalent welfare gains of moving a (1,1) household to (2,2) excess returns, decomposed into the parental education channel and the own education channel, analogous to Figure 12.

Figure B.35: Robustness: Policy Functions, $\theta = 4$ 

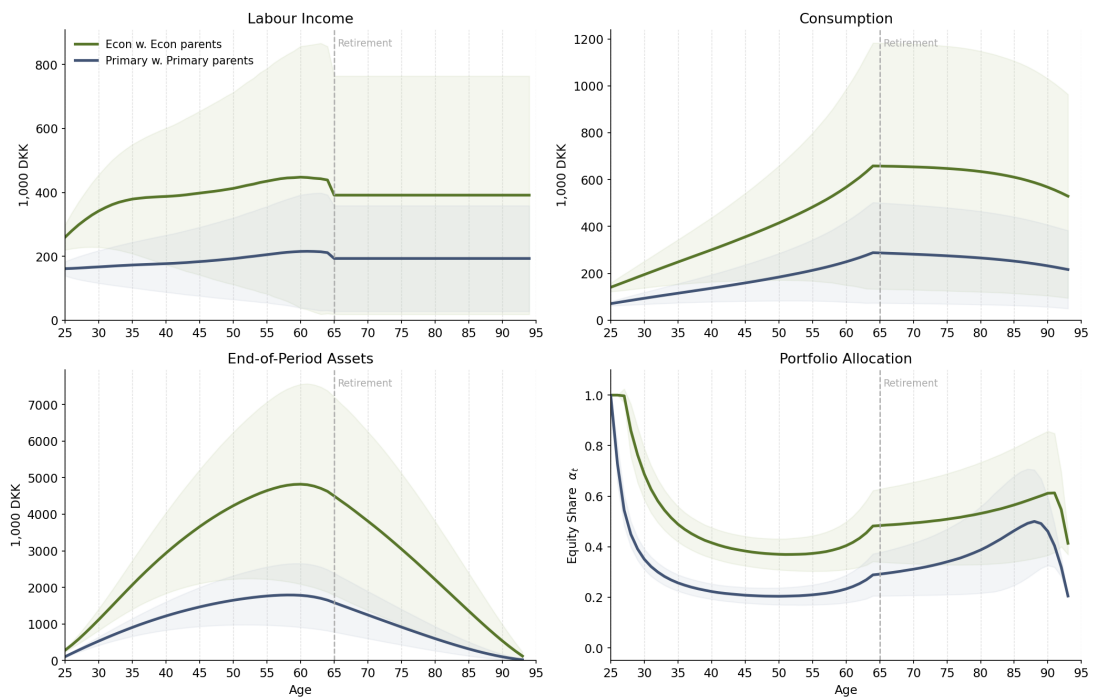
Notes. Policy functions for a (0,0) and (2,2) household under $\theta = 4$.

Figure B.36: Robustness: Life-Cycle Profiles, $\theta = 4$ 

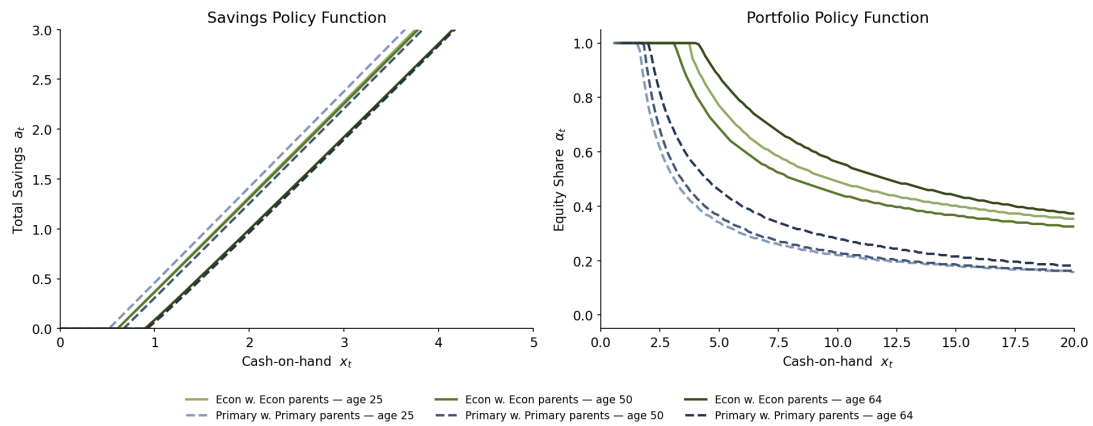
Notes. Life-cycle profiles for a (0,0) and (2,2) household under $\theta = 4$.

Figure B.37: Robustness: Policy Functions, $\theta = 10$ 

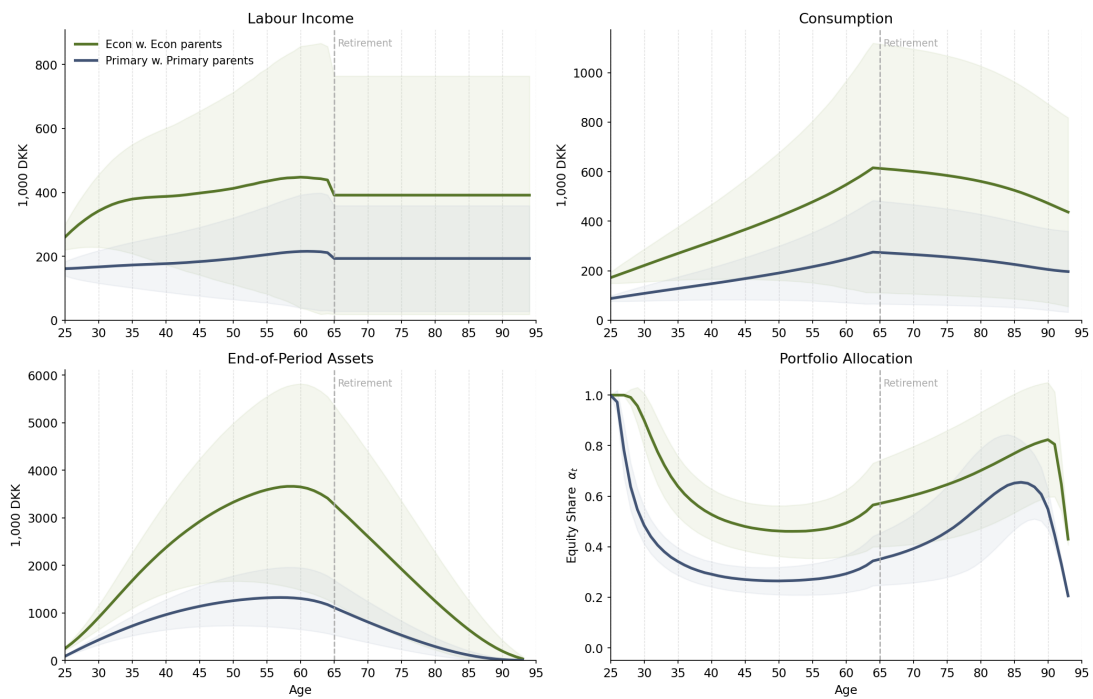
Notes. Policy functions for a (0,0) and (2,2) household under $\theta = 10$.

Figure B.38: Robustness: Life-Cycle Profiles, $\theta = 10$ 

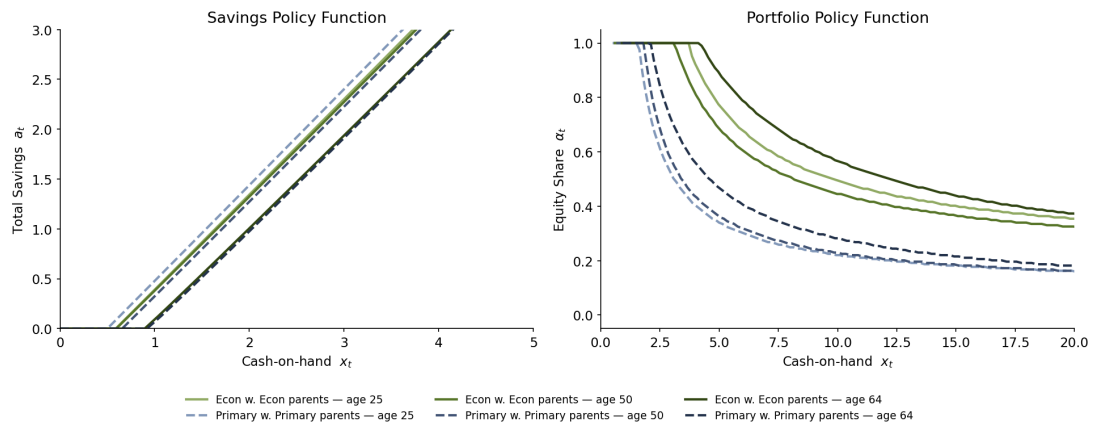
Notes. Life-cycle profiles for a (0,0) and (2,2) household under $\theta = 10$.

Figure B.39: Robustness: Policy Functions, $\beta = 0.95$ 

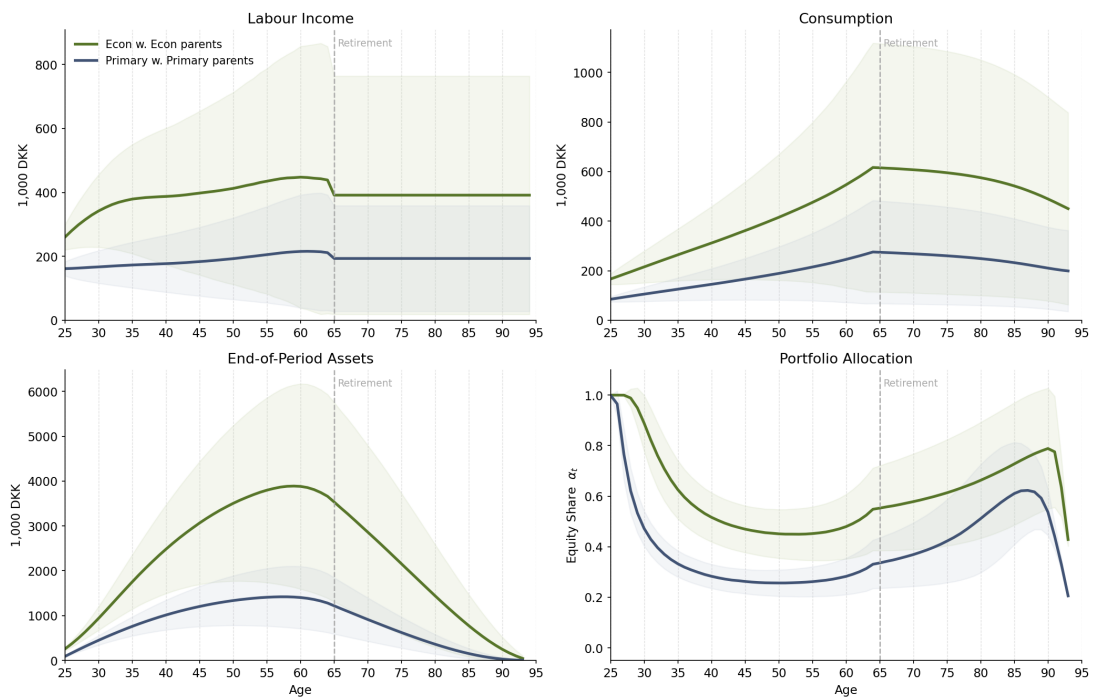
Notes. Policy functions for a (0,0) and (2,2) household under $\beta = 0.95$.

Figure B.40: Robustness: Life-Cycle Profiles, $\beta = 0.95$ 

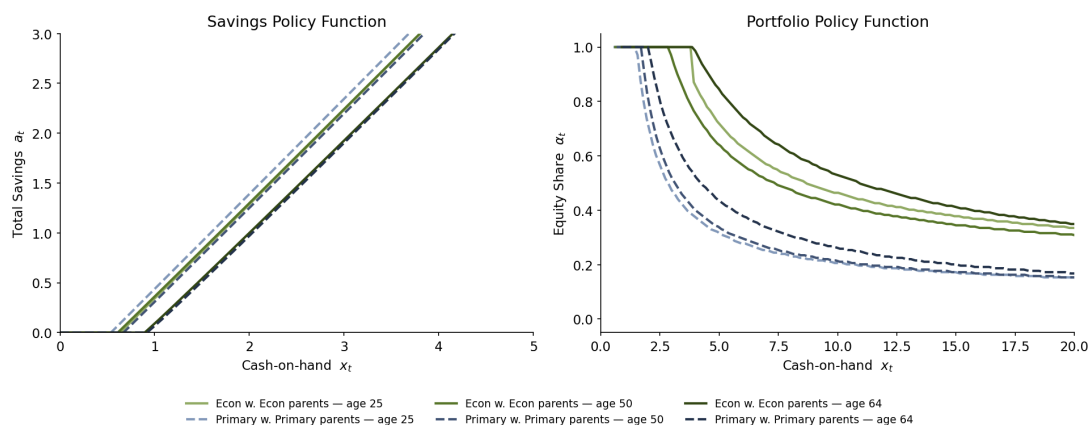
Notes. Life-cycle profiles for a (0,0) and (2,2) household under $\beta = 0.95$.

Figure B.41: Robustness: Policy Functions, $\beta = 0.97$ 

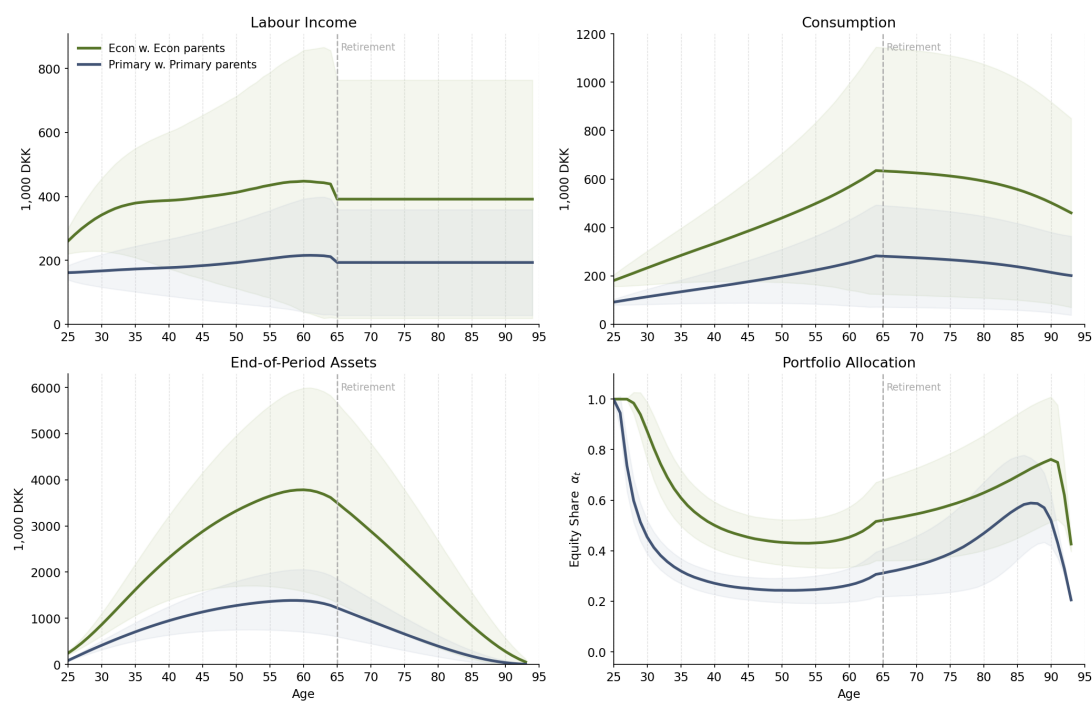
Notes. Policy functions for a (0,0) and (2,2) household under $\beta = 0.97$.

Figure B.42: Robustness: Life-Cycle Profiles, $\beta = 0.97$ 

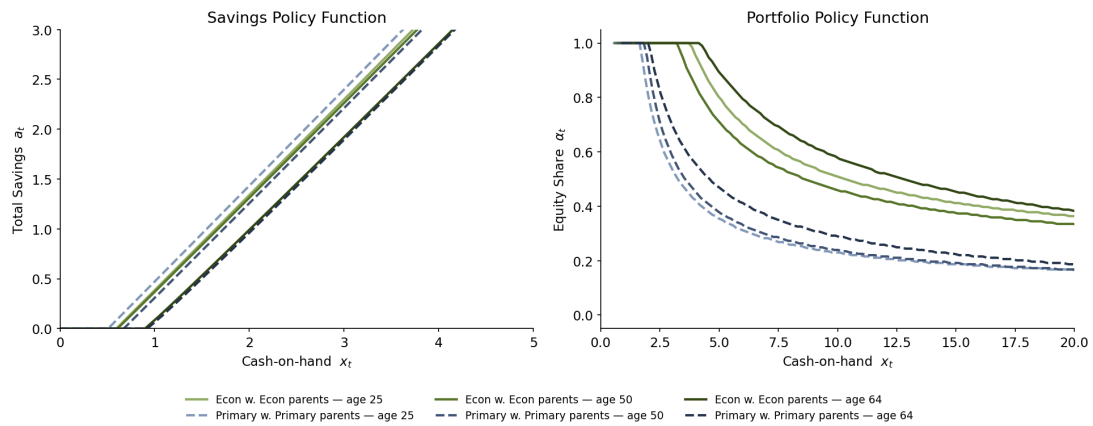
Notes. Life-cycle profiles for a (0,0) and (2,2) household under $\beta = 0.97$.

Figure B.43: Robustness: Policy Functions, $r_f = 0.02$ 

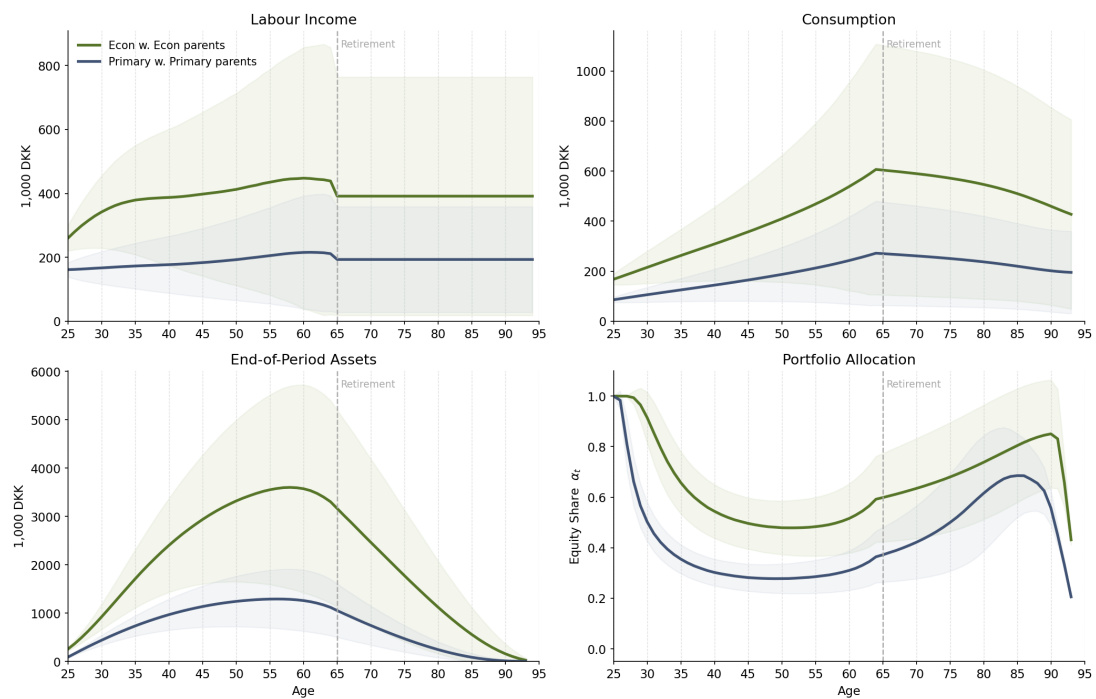
Notes. Policy functions for a (0,0) and (2,2) household under $r_f = 0.02$.

Figure B.44: Robustness: Life-Cycle Profiles, $r_f = 0.02$ 

Notes. Life-cycle profiles for a (0,0) and (2,2) household under $r_f = 0.02$.

Figure B.45: Robustness: Policy Functions, $r_f = 0.005$ 

Notes. Policy functions for a (0,0) and (2,2) household under $r_f = 0.005$.

Figure B.46: Robustness: Life-Cycle Profiles, $r_f = 0.005$ 

Notes. Life-cycle profiles for a (0,0) and (2,2) household under $r_f = 0.005$.

C Derivations

Variance Decomposition

We follow the variance decomposition approach of Carroll and Samwick (1997) and decompose log income into a deterministic life-cycle component, a permanent component, and a transitory component:

$$\log y_{i,e,t} = g_{e,t} + \mu_{i,e,t} + v_{i,e,t}, \quad (\text{C.1})$$

where $g_{e,t}$ is the deterministic life-cycle component, $\mu_{i,e,t}$ is the permanent component, and $v_{i,e,t} \sim N(0, \sigma_v^2)$ is an i.i.d. transitory shock. The deterministic component is parameterised as an age polynomial, estimated separately for each educational group with calendar year fixed effects as in equation (51)

The permanent component follows a unit-root process,

$$\mu_{i,e,t} = \mu_{i,e,t-1} + \varepsilon_{i,e,t}, \quad (\text{C.2})$$

where $\varepsilon_{i,e,t} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. innovation to permanent income, mutually independent of the transitory shock $v_{i,e,t}$ and of its own past. The unit-root specification implies that permanent income shocks are never forgotten, in contrast to transitory shocks which die out in a single period. This corresponds to the standard formulation of MaCurdy (1982) and is the special case of a more general AR(1) specification in which shocks are fully persistent. The residuals of equation (51) relative to the deterministic profile is

$$r_{i,e,t} \equiv \log y_{i,e,t} - \hat{g}_{e,t} = \mu_{i,e,t} + v_{i,e,t}. \quad (\text{C.3})$$

Taking first differences eliminates the non-stationary permanent level:

$$\Delta r_{i,e,t} = r_{i,e,t} - r_{i,e,t-1} = (\mu_{i,e,t} - \mu_{i,e,t-1}) + (v_{i,e,t} - v_{i,e,t-1}) = \varepsilon_{i,e,t} + v_{i,e,t} - v_{i,e,t-1}. \quad (\text{C.4})$$

Since $\varepsilon_{i,e,t}$, $v_{i,e,t}$, and $v_{i,e,t-1}$ are mutually independent,

$$\text{Var}(\Delta r_{i,e,t}) = \sigma_\varepsilon^2 + 2\sigma_v^2. \quad (\text{C.5})$$

The autocovariance at lag one is

$$\begin{aligned} \text{Cov}(\Delta r_{i,e,t}, \Delta r_{i,e,t-1}) &= \text{Cov}(\varepsilon_{i,e,t} + v_{i,e,t} - v_{i,e,t-1}, \varepsilon_{i,e,t-1} + v_{i,e,t-1} - v_{i,e,t-2}) \\ &= -\text{Var}(v_{i,e,t-1}) \\ &= -\sigma_v^2. \end{aligned} \quad (\text{C.6})$$

These two moments are sufficient to identify the variances of the permanent and transitory shocks:

$$\sigma_v^2 = -\text{Cov}(\Delta r_{i,e,t}, \Delta r_{i,e,t-1}), \quad (\text{C.7})$$

$$\sigma_\varepsilon^2 = \text{Var}(\Delta r_{i,e,t}) + 2\text{Cov}(\Delta r_{i,e,t}, \Delta r_{i,e,t-1}). \quad (\text{C.8})$$

More generally, the same logic applied to K -period differences yields

$$\text{Var}(\Delta^K r_{i,e,t}) = K\sigma_\varepsilon^2 + 2\sigma_v^2, \quad (\text{C.9})$$

so that σ_ε^2 can be recovered as the slope, and $2\sigma_v^2$ as the intercept, of a regression of $\text{Var}(\Delta^K r)$ on the differencing horizon K . We estimate the decomposition separately for each educational group, allowing the permanent and transitory variances to differ across the population in a manner consistent with the heterogeneity in returns and participation documented elsewhere in the empirical literature.

Carroll and Samwick (1997) recommend identification from long differences (typically $K \geq 5$) primarily to remove the role of measurement error. Because our income data are drawn from Danish administrative tax records, third-party reported by employers and financial institutions and used directly by the tax authority, classical measurement error of the kind that motivates long-horizon identification is plausibly small. We therefore rely on the one-period moment conditions described above and treat σ_v^2 as the variance of genuine transitory income shocks rather than as a composite of transitory shocks and measurement error.

The Envelope Theorem

The Envelope theorem is based on differentiating the value function with respect to the state variable, answering the question: how much additional lifetime utility does the agent get with one more unit of the state variable? Differentiating the Bellman equation with respect to the state variable x yields

$$\begin{aligned} v'(x) = & F_x(x, g(x)) + F_d(x, g(x)) \cdot g'(x) \\ & + \beta v'(h(x, g(x))) \cdot [h_x(x, g(x)) + h_d(x, g(x)) \cdot g'(x)], \end{aligned} \quad (\text{C.10})$$

and collecting terms in $g'(x)$:

$$g'(x) \cdot [F_d(x, g(x)) + \beta v'(h(x, g(x))) \cdot h_d(x, g(x))]. \quad (\text{C.11})$$

The expression in brackets is exactly the optimality condition in equation (8): the marginal benefit today equals the marginal cost of all discounted future periods. The indirect effect through the

optimal policy channel therefore vanishes, and equation (C.10) reduces to

$$v'(x) = F_x(x, g(x)) + \beta v'(h(x, g(x))) \cdot h_x(x, g(x)). \quad (\text{C.12})$$

In life-cycle models, the return function is typically a utility function $F(x, c) = u(c)$, so it depends on the control variable only and the direct effect of the state variable disappears: $F_x = 0$. We are left with

$$v'(x) = \beta v'(h(x, g(x))) \cdot h_x(x, g(x)). \quad (\text{C.13})$$

Dividing by the optimality condition $F_d(x, g(x)) = -\beta v'(h(x, g(x))) \cdot h_d(x, g(x))$, and using the ratio $h_x/h_d = -1$, gives

$$v'(x) = F_d(x, g(x)), \quad (\text{C.14})$$

or in a life-cycle scenario

$$v'(x) = u'(c^*(x)). \quad (\text{C.15})$$

Consumption Equivalent Welfare Measure

Consumption equivalent (CE) is an intuitive way to express differences in welfare. The general interpretation is that an agent is indifferent between moving from the baseline scenario to the counterfactual (in our case that is a higher level of financial literacy) and receiving a consumption increase of $(\lambda - 1)\%$ every period.

As shown in equation (25), CRRA has the convenient characteristic of being homogeneous. In this case, rather than normalizing consumption by permanent income, we factorize consumption by λ :

$$u(\lambda c) = \frac{(\lambda c)^{1-\sigma}}{1-\sigma} = \lambda^{1-\sigma} \cdot u(c), \quad (\text{C.16})$$

and as discussed in section 3.4, we can apply that characteristic to the value function. Here, the consumption equivalent λ is defined as the scalar such that scaling all baseline consumption by λ makes the agent indifferent between the baseline and the counterfactual:

$$\lambda^{1-\sigma} \cdot V_{\text{base}}(x_0) = V_{\text{cf}}(x_0). \quad (\text{C.17})$$

Isolating λ , we obtain a closed-form expression based on the two value functions:

$$\lambda = \left(\frac{V_{\text{cf}}(x_0)}{V_{\text{base}}(x_0)} \right)^{1/(1-\sigma)}. \quad (\text{C.18})$$

This is a typical way to compare welfare across model specifications, expressed in the equivalence of the counterfactual characteristic and an annual consumption increase.

Gauss-Hermite Quadrature

We use Gauss-Hermite shocks in order to model the stochasticity of the income path and market returns. The following section describes what Gauss-Hermite quadratures are and how to implement them. The goal is to obtain lognormal transitory and lognormal permanent shocks for the income path. The initial problem is that the continuous lognormal distribution needs to be discretely approximated to guarantee computational feasibility. Gauss-Hermite provides the following solution:

$$\int_{-\infty}^{+\infty} e^{-x^2} f(x) dx \approx \sum_{i=1}^n w_i f(x_i), \quad (\text{C.19})$$

where the nodes x_i and weights w_i are a closed form based on the level of approximation n only. It is important to understand that once n is chosen, nodes and weights are automatically pinned down.¹⁴ In our case, we want to generate a discrete approximation of the income shocks N and V , where both are log-normally distributed.

So far this is just a mathematical scaling, but in order to translate this into the economic income shock that we need, we calculate the corresponding *lognormal shock realizations* based on the corresponding mean and variance:

$$\int \phi(N) dN = \int \frac{1}{N\sigma_N\sqrt{2\pi}} \exp\left(-\frac{(\ln N - \mu_N)^2}{2\sigma_N^2}\right) dN \quad (\text{C.20})$$

$$\text{Set} \quad -x^2 = -\frac{(\ln N - \mu_N)^2}{2\sigma_N^2} \quad (\text{C.21})$$

$$x = \frac{\ln N - \mu_N}{\sqrt{2}\sigma_N} \quad (\text{C.22})$$

$$N = \exp\left(\mu_N + \sqrt{2}\sigma_N x\right), \quad dN = \sqrt{2}\sigma_N N dx \quad (\text{C.23})$$

$$\int \phi(N) dN = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-x^2} dx \approx \sum_{i=1}^n \frac{w_i}{\sqrt{\pi}}. \quad (\text{C.24})$$

Therefore, n fully determines x_i and w_i . The Gauss-Hermite quadrature rule is defined for integrals of the form $\int e^{-x^2} f(x) dx$, so to approximate an expectation over a lognormally distributed shock $N \sim \log \mathcal{N}(-\frac{1}{2}\sigma_N^2, \sigma_N^2)$, a change of variables is applied to the lognormal density. This yields the node transformation $N_i = \exp(\mu_N + \sqrt{2}\sigma_N x_i)$, which maps the standard Gauss-Hermite nodes into shock realizations consistent with the target distribution. The weights are divided by $\sqrt{\pi}$ to

¹⁴Python provides a package that generates nodes and weights with a simple function:
`nodes_h, weights_h = hermgauass(par["Nyy"])`

ensure they sum to one, consistent with approximating an expectation. The mean and variance of N thus enter exclusively through this node transformation.